DAY 41
LEVEL 41

BIOLOGY | CHAPTER 9 | PART 1 ASTRA SERIES

LIFE PROCESSES & PHOTOSYNTHESIS

Goal: life processes and how plants make food.



by Nitesh Prakash

01 Life Processes | 02 Nutrition Types | 03 Photosynthesis | 04 Stomata | 05 NCERT Activities | +140 XP today | Level 41 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT ARE LIFE PROCESSES? ★★★ MOST ASKED

DEFINITION

- **Life processes:** the basic processes that together carry out the **maintenance job** of an organism and keep it alive, even when it is sleeping or sitting still
- Molecules in living bodies keep moving all the time, so repair never stops

The four life processes ★★★

Process	What it does
Nutrition	takes in food (energy source) from outside the body
Respiration	breaks down food to release energy (mostly using oxygen)
Transportation	carries food, oxygen and wastes to every cell
Excretion	removes harmful by-products from the body

Why do we need energy all the time?

- Maintenance and repair go on even in sleep, so the body needs a **constant supply of energy**
- Energy comes from **food** taken from outside the body (carbon-based, mostly)
- Growth needs raw materials from outside too, since the body must make new molecules
- Food is broken down by **oxidising–reducing reactions** inside cells; many organisms use oxygen from outside for this

MNEMONIC: NARATE

"NaRaTE" = **N**utrition, **R**espiration, **T**ransportation, **E**xcretion. Jo NaRaTE karta hai, wahi zinda hai!



BUJHO *Main toh so raha tha, tab bhi energy kharch ho rahi thi?*

PAHELI *Haan Bujho! Sote waqt bhi heart, breathing aur cell repair chalte rehte hain. Maintenance kabhi chutti nahi leta, isliye energy roz chahiye.*





★★★ *NaRaTE ke chaar processes ka naam aur kaam: 2 marks pakke, Nitesh Bhaiya ki guarantee!*

Unicellular vs multicellular organisms

Unicellular	Multicellular
whole body surface touches the surroundings	most cells are far from the surroundings
simple **diffusion** meets the need for food, gases, waste	diffusion is too slow, so **special tissues and organs** do the job
e.g. Amoeba, Paramecium	e.g. humans, plants

2 NUTRITION AND ITS TYPES ★★★

DEFINITION

- **Nutrition:** the process of taking in food and using it to get energy and raw materials for growth and repair

Autotrophic vs heterotrophic nutrition ★★★

Autotrophic	Heterotrophic
makes its own food from simple **inorganic** substances (CO_2, water)	depends on food made by other organisms, directly or indirectly
uses sunlight and chlorophyll	breaks down complex food with **enzymes**
e.g. green plants, some bacteria	e.g. animals, fungi, most bacteria
called **autotrophs** (producers)	called **heterotrophs** (consumers)



MEME BREAK

*Biology mein pehla pyaar: **life processes**. Ye chaar samajh liye toh poora chapter apna lagega!*



★★★ *Autotrophic vs heterotrophic ka difference har saal 1–2 marks mein aata hai, Nitesh Bhaiya ki guarantee!*

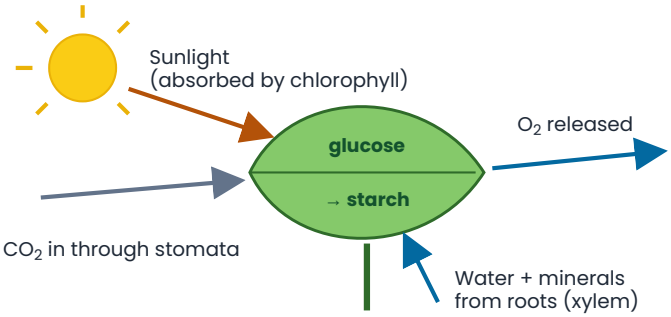
3 PHOTOSYNTHESIS ★★★ MOST ASKED

DEFINITION

- **Photosynthesis:** the process by which autotrophs take in **CO_2** and **water** and convert them into **carbohydrates** in the presence of **sunlight** and **chlorophyll**, releasing oxygen

BALANCED EQUATION

$$6CO_2 + 12H_2O \xrightarrow{\text{(sunlight, chlorophyll)}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$



Three events of photosynthesis ★★★ MOST ASKED

1. **Absorption** of light energy by chlorophyll
2. **Conversion** of light energy into chemical energy and **splitting of water** into hydrogen and oxygen
3. **Reduction** of carbon dioxide to carbohydrates

MNEMONIC: THREE EVENTS

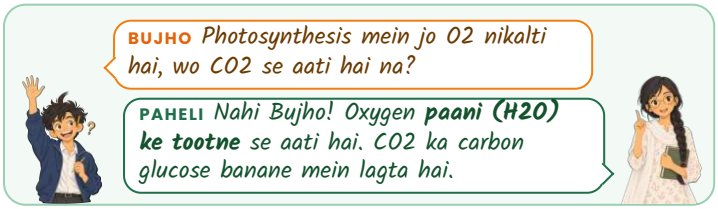
"A–C–R: Aaj Chai Ready" = **A**bsorb light, **C**onvert energy + split water, **R**educe CO_2. Chai ki tarah order fix!

Desert plants: a clever timing trick ★★☆

- Their stomata stay **closed in the day** to save water
- They take up **CO_2 at night** and store it as an intermediate
- In the day, energy absorbed by chlorophyll acts on this intermediate to make food

Chlorophyll, chloroplasts and starch

- Leaf cells contain green dots called **chloroplasts**; they hold the pigment **chlorophyll**
- Glucose not used at once is stored as **starch**, the plant's energy reserve

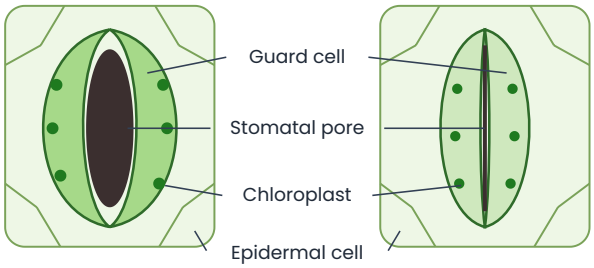


DO NOT MIX UP

- **Chloroplast** = the green organelle (the kitchen)
- **Chlorophyll** = the green pigment inside it that traps light (the stove)

4 STOMATA: DOORS OF THE LEAF ★★★ MOST ASKED

Stomata are tiny pores on the surface of the leaf. Most gas exchange for photosynthesis happens through them (gases also pass through stems and roots).



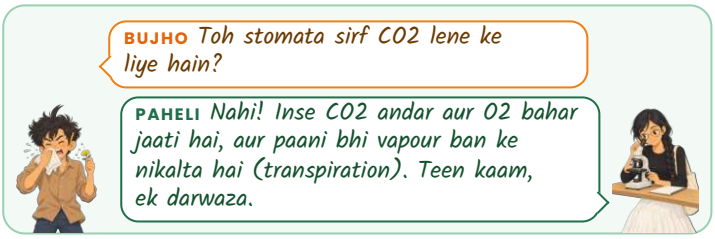
(a) Open stomatal pore (b) Closed stomatal pore

How the pore opens and closes ★★★

- Water **flows into** the guard cells → they swell → the pore **opens**
- Water **flows out** → guard cells shrink → the pore **closes**
- A lot of water is lost through open stomata, so the plant closes them when CO_2 is not needed

MNEMONIC: GUARD CELLS

"Paani aaya, darwaza khula; paani gaya, darwaza band." Swollen guard cells = open stoma.





MEME BREAK

*Plant bole: delivery app nahi chahiye, **sunlight + CO_2 + paani** se apna khana khud bana leta hoon!*

Why photosynthesis matters ★☆☆

- It is the **starting point of almost every food chain**: all heterotrophs depend on it
- It releases the **oxygen** that most organisms use for respiration
- It removes CO_2 from the air

NCERT in-text questions

Q How do autotrophs get carbon dioxide and nitrogen?

- CO_2 from air through stomata; nitrogen from soil as nitrates or nitrites

Q Name the raw materials of photosynthesis.

- CO_2, water, sunlight and chlorophyll (plus minerals for other compounds)

5 RAW MATERIALS FOR PHOTOSYNTHESIS ★★☆

Where each raw material comes from

Raw material	Source and route
Carbon dioxide	air, enters through **stomata**
Water	soil, absorbed by **roots** and carried up
Minerals (N, P, Fe, Mg)	soil, taken up by roots
Sunlight	absorbed by chlorophyll

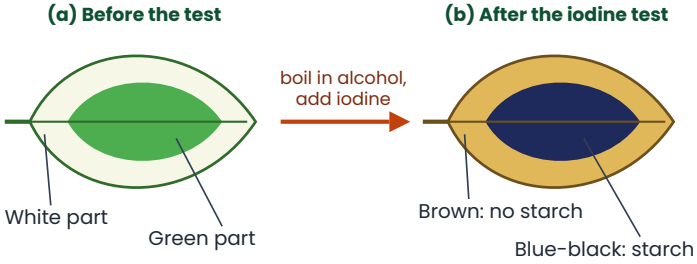
Nitrogen: why and how

- Nitrogen is essential to make **proteins** and other compounds
- Plants take it as **inorganic nitrates or nitrites**
- Or as **organic compounds** made by bacteria from atmospheric nitrogen

6 NCERT ACTIVITIES ★★★ MOST ASKED

Activity 1: is chlorophyll needed? (variegated leaf)

1. Keep a potted plant with variegated leaves (green and white parts) in the **dark for three days**, so its starch is used up (**destarching**)
2. Keep it in sunlight for about six hours; pluck a leaf and trace its green parts
3. Dip the leaf in boiling water, then boil it in **alcohol** on a water bath to remove chlorophyll
4. Dip it in dilute **iodine solution** and compare with the tracing



Result: only the parts that were green turn blue-black, so **chlorophyll is needed** for photosynthesis (starch is formed only there).



MEME BREAK
*Cat ne bhi photosynthesis try kiya! Par bina **chlorophyll** ke koi starch nahi banta, iodine test fail.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Using water with a heavy oxygen isotope (^{18}O), scientists showed that the oxygen released in photosynthesis comes from **water**, not from carbon dioxide.

Activity 2: is carbon dioxide needed? ★★★

1. Take two healthy potted plants of the same size; keep both in the **dark for three days**
2. Place each on a glass plate; put a watch-glass of **potassium hydroxide (KOH)** beside one plant
3. Cover each with a bell jar and seal the bottom with **vaseline** (air-tight)
4. Keep both in sunlight for two hours, then test one leaf from each for starch

Set-up	Starch test	Why
with KOH	**no** blue-black	KOH absorbs CO_2
without KOH	**blue-black**	CO_2 available

Conclusion: carbon dioxide is essential for photosynthesis.

WHY ALCOHOL, WHY IODINE?

- Alcohol dissolves **chlorophyll**, so the colour change is easy to see
- Iodine turns **starch blue-black**; boil alcohol on a **water bath**, never on a direct flame

Exam tips and common mistakes

- Write the equation with **"sunlight" and "chlorophyll"** on the arrow; both carry marks
- Photosynthesis releases **O_2** that comes from **water**, not from CO_2
- Destarching is done **first**, so that any starch found later is new
- Do not say "plants respire only at night": they respire all the time
- The plant without KOH is the **control**; vaseline keeps outside air (CO_2) from entering



MEME BREAK
*Bhookh lagne ka tension hi nahi, jab khana khud banta ho. **Autotrophs** ki smart life!*

7 PRACTICE MCQS

1. Which of these is NOT a life process? ★★☆

(a) nutrition (b) excretion
(c) photosynthesis in animals (d) respiration

2. The correct sequence of events in photosynthesis is best described as: ★★★

(a) reduction of CO_2, light absorption, splitting of water
(b) light absorption, splitting of water, reduction of CO_2
(c) splitting of water, reduction, light absorption
(d) none of these

3. Opening and closing of the stomatal pore is controlled by: ★★★

(a) epidermal cells (b) guard cells
(c) xylem (d) chloroplasts

4. In the KOH activity, KOH is used to: ★★★

(a) provide oxygen (b) absorb water
(c) absorb carbon dioxide (d) kill bacteria

5. In the iodine test, a starch-containing part of a leaf turns: ★★☆

(a) red (b) blue-black
(c) yellow (d) white

6. Desert plants take up CO_2: ★★☆

(a) only in the day (b) at night
(c) through roots (d) never

7. Plants absorb nitrogen mostly in the form of: ★★☆

(a) N_2 gas (b) nitrates or nitrites
(c) ammonia gas (d) proteins

8. Excess glucose in a plant is stored as: ★★☆

(a) glycogen (b) starch
(c) cellulose (d) sucrose

9. **Assertion (A):** The parts of a variegated leaf that were white do not turn blue-black in the iodine test.
Reason (R): White parts lack chlorophyll, so no starch is made there. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Guard cells close the stomatal pore when they lose water.
Reason (R): Plants lose a large amount of water through open stomata. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Animals do not photosynthesise.
2	(b)	Absorb, convert and split, reduce.
3	(b)	Guard cells swell or shrink.
4	(c)	KOH absorbs CO_2 inside the jar.
5	(b)	Iodine + starch gives blue-black.
6	(b)	Stomata open at night in deserts.
7	(b)	Inorganic nitrates or nitrites.
8	(b)	Starch is the plant's reserve.
9	(a)	No chlorophyll, no starch: R explains A.
10	(b)	Both true; R is why plants close them, not how.

Day 41 hit-list: Eduspark PYQ analysis

Topic	Rating
Photosynthesis equation and events	★★★
Stomata diagram and opening	★★★
Starch test activities	★★★
Autotrophic vs heterotrophic	★★☆
Desert plants and CO_2	★☆☆

Short and long answer questions

1. What are life processes? Why is energy needed even when we are asleep? **(2 marks)**
2. Write the balanced equation of photosynthesis and list its three events. NCERT **(3 marks)**
3. Draw a labelled diagram of an open and a closed stomatal pore. How do guard cells open the pore? **(3 marks)**
4. Describe an activity to show that CO_2 is essential for photosynthesis. NCERT **(5 marks)**

Model answers

1 Life processes
- Processes that maintain the body: nutrition, respiration, transport, excretion
- Repair and molecular movement never stop, so energy is needed all the time

2 Equation and events
- Equation with sunlight, chlorophyll on the arrow
- Absorption of light; conversion and splitting of water; reduction of CO_2

3 Stomata
- Diagram: two guard cells, pore, chloroplasts (2 labels each view)
- Water enters guard cells, they swell and the pore opens

4 KOH activity
- Destarch two plants, KOH beside one, seal, sunlight 2 h
- Leaf with KOH shows no starch: CO_2 is essential

Day 41 checklist: tick before you sleep

- ☐ I can name the four life processes and why energy is needed
- ☐ I can write the photosynthesis equation and its three events
- ☐ I can draw open and closed stomata
- ☐ I can explain both NCERT starch-test activities
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 41 COMPLETE +140 XP earned. Next: Level 42



BUJHO *Samajh gaya! Plant khud chef hai: CO_2, paani aur dhoop se glucose.*

PAHELI *Shabaash! Kal Level 42: hum khana kaise digest karte hain.*



NITESH SIR SAYS
Photosynthesis ka equation aur stomata ka diagram roz ek baar likho. ***Kal milte hai Day 42 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 42
LEVEL 42

BIOLOGY | CHAPTER 9 | PART 2 ASTRA SERIES

HETEROTROPHIC NUTRITION & DIGESTION

Goal: how animals take in food and digest it.



by Nitesh Prakash

01 Nutrition Types | 02 Amoeba | 03 Digestive System | 04 Enzymes | 05 Villi and Caries | +140 XP today | Level 42 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HETEROTROPHIC NUTRITION ★★★ MOST ASKED

DEFINITION

- **Heterotrophic nutrition:** obtaining ready-made food (made by other organisms) and breaking it down with **enzymes**
- How an organism eats depends on the **form of food** available and **how it gets it**

Three types of heterotrophic nutrition ★★★ MOST ASKED

Type	How	Examples
Holozoic	takes in whole food and digests it inside the body	Amoeba, cow, humans
Saprotrophic	breaks down food **outside** the body (on dead matter) and absorbs it	fungi: bread mould, yeast, mushrooms
Parasitic	takes food from a living host **without killing it**	Cuscuta (amarbel), ticks, lice, leeches, tapeworms

MNEMONIC: THREE TYPES

"HSP": **H**olozoic = haath se poora khana; **S**aprotrophic = sadi-gali cheez pe; **P**arasitic = parayi thali, host zinda rehta hai.

Examples that confuse students

- **Cuscuta** (dodder) is a **plant** parasite: it has no chlorophyll
- **Yeast and mushrooms** are saprotrophs, not parasites
- Bacteria can be saprotrophic or parasitic, depending on the species
- Some organisms (like grass for a cow) are food; the cow is holozoic



BUJHO *Parasite apne host ko maar deta hai na?*

PAHELI *Nahi Bujho! Parasite host se khana leta hai par usse **turant nahi maarta**. Host mar gaya toh khana bhi khatam!*

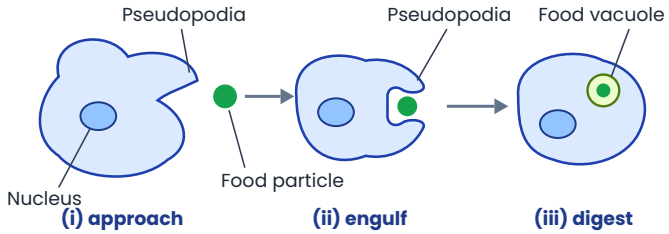


Saprotrophs such as fungi and bacteria act as nature's **cleaners**: they break down dead plants and animals and return nutrients to the soil.

2 HOW DO ORGANISMS OBTAIN FOOD? ★★★

Single-celled organisms take food in through the **whole cell surface**; there is no separate digestive system.

Amoeba: food through pseudopodia ★★★ MOST ASKED



1. Temporary finger-like extensions of the cell, the **pseudopodia**, surround the food particle
2. They fuse to form a **food vacuole** around it
3. Inside the vacuole, complex food is broken into simpler substances that **diffuse** into the cytoplasm
4. Undigested material moves to the surface and is thrown out

Paramecium: food at a fixed spot ★★☆

- Its cell has a **definite shape**
- Food enters at a **specific spot** (oral groove)
- Hair-like **cilia** covering the cell move the food to this spot

Steps of holozoic nutrition ★★☆

Step	What happens
Ingestion	taking food into the body
Digestion	breaking complex food into simple soluble forms
Absorption	digested food passes into blood
Assimilation	cells use the food for energy, growth and repair
Egestion	throwing out undigested waste

MNEMONIC: HOLOZOIC STEPS

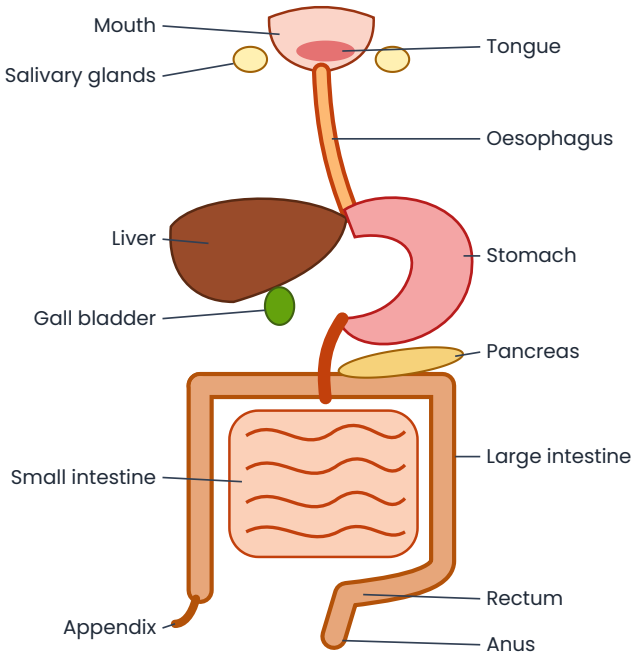
"I Do Always Amaze Everyone" = **I**ngestion, **D**igestion, **A**bsorption, **A**ssimilation, **E**gestion.



★★★ *Amoeba ka labelled diagram (pseudopodia, food vacuole) board ka favourite hai, Nitesh Bhaiya ki guarantee!*

3 HUMAN DIGESTIVE SYSTEM ★★★ MOST ASKED

The **alimentary canal** is a long tube from the mouth to the anus. Different parts do different jobs, helped by **glands** that release digestive juices.



Human alimentary canal with associated glands (schematic)

MNEMONIC: ORDER OF THE CANAL

"Mummy Ordered Samosa, Sister Loved Aloo" = **M**outh, **O**esophagus, **S**tomach, **S**mall intestine, **L**arge intestine, **A**nus.

Glands that join the canal

- **Salivary glands** → mouth
- **Gastric glands** → stomach wall
- **Liver** and **pancreas** → small intestine
- **Intestinal glands** → wall of small intestine

Enzyme summary table ★★★ MOST ASKED

Juice	Enzyme	Acts on	Product
Saliva	Salivary amylase	Starch	Maltose (sugar)
Gastric juice	Pepsin (+ HCl)	Proteins	Peptides
Pancreatic juice	Trypsin	Proteins	Peptides
Pancreatic juice	Lipase	Emulsified fats	Fatty acids, glycerol
Pancreatic juice	Amylase	Starch	Sugars
Intestinal juice	Several enzymes	All three	Amino acids, glucose, fatty acids + glycerol



MEME BREAK

*Mouth shuru karta hai, stomach madad karta hai, par **small intestine** asli hero hai. Change my mind!*

4 MOUTH, OESOPHAGUS AND STOMACH ★★★

Mouth: the start ★★★

- **Teeth** chew and grind food; the **tongue** mixes it with saliva
- Food is made **small and soft** so enzymes can act on it
- **Salivary amylase** in saliva breaks **starch** into a simple sugar (maltose)

Oesophagus: food pipe ★★☆

- Carries food from the mouth to the stomach
- Rhythmic contraction of its muscles, called **peristalsis**, pushes food forward (it happens all along the gut)

Stomach: the acid tank ★★★ MOST ASKED

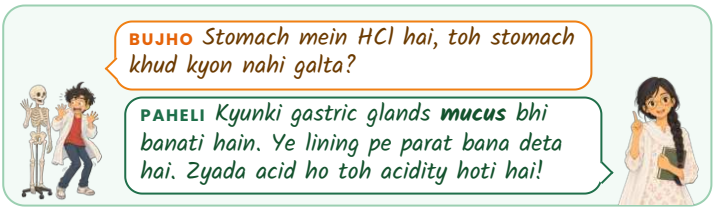
Gastric juice part	Role
Hydrochloric acid (HCl)	makes the medium **acidic** for pepsin; kills germs
Pepsin	digests **proteins**
Mucus	protects the stomach lining from the acid

Muscular walls churn the food. A **sphincter muscle** lets food leave the stomach into the small intestine little by little.



MEME BREAK

*HCl sab kuch gala dena chahta hai, par **mucus** stomach wall ko bachata hai. Mucus = stomach ka bodyguard!*



MNEMONIC: WHERE DIGESTION STARTS

"S–P–F" in order of the canal: **S**tarch starts in the **mouth**, **P**rotein in the **stomach**, **F**at in the **small intestine**.

Exam tips and common mistakes

- **Pepsin** works in an **acidic** medium; **trypsin** works in an **alkaline** medium
- Bile is made by the **liver** (stored in the gall bladder), not by the gall bladder
- Write **salivary amylase** (not just "saliva") as the enzyme
- Peristalsis happens **all along the gut**, not only in the oesophagus
- Fat digestion needs **both** bile (emulsify) and lipase (digest)

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Adult humans have **32 teeth** of four kinds: incisors (cut), canines (tear), premolars and molars (grind). Enamel is the **hardest** substance in the body.

5 SMALL INTESTINE ★★★ MOST ASKED

Longest part of the canal ★★☆

- Tightly coiled; the site of **complete digestion** of carbohydrates, proteins and fats
- Its length differs with diet: **herbivores** eating grass need a **longer** small intestine to digest cellulose; **carnivores** have a shorter one

Juices that work here ★★★ MOST ASKED

Juice (source)	What it does
Bile (liver)	makes the acidic food from the stomach **alkaline**; bile salts **emulsify** fats (big globules → tiny droplets)
Pancreatic juice (pancreas)	**amylase** for starch, **trypsin** for proteins, **lipase** for emulsified fats
Intestinal juice (intestinal wall)	finishes the job: proteins → **amino acids**, complex carbohydrates → **glucose**, fats → **fatty acids + glycerol**

MNEMONIC: PANCREATIC ENZYMES

"Pancreas ka TAL" = **T**rypsin (protein), **A**mylase (starch), **L**ipase (fat). Yaad rakho: bile mein koi enzyme nahi!



MEME BREAK

*Bile ki deal: bade fat globules do, chhote droplets lo. Ise kehte hain **emulsification**, aur lipase khush!*

BUJHO *Bile fat ko digest karta hai na?*

PAHELI *Nahi! Bile mein **enzyme hi nahi** hota. Wo sirf fat ko chhote droplets mein todta hai (emulsify). Digest toh **lipase** karta hai.*

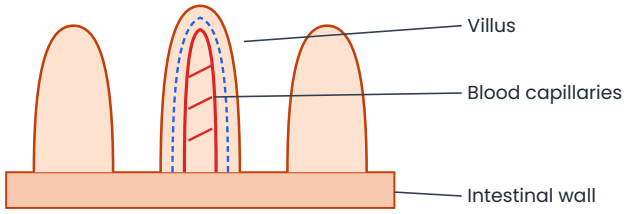
Absorption and assimilation

- **Absorption:** digested food passes through the walls of the villi into the **blood**
- **Assimilation:** body cells use the absorbed food for **energy, growth and repair**

Large intestine and anus ★★☆

- Unabsorbed food enters the **large intestine**, where more villi absorb **water**
- The rest leaves the body through the **anus**
- The exit of waste is controlled by the **anal sphincter**

Villi: built for absorption ★★★ MOST ASKED



- The inner wall has millions of finger-like folds called **villi**
- They **increase the surface area** for absorption
- Each villus is richly supplied with **blood vessels**, which carry absorbed food to every cell, where it gives energy, builds new tissue and repairs old tissue



MEME BREAK

*Muh wala amylase ho ya pancreas wala, kaam ek hi: **starch** → **sugar**. Bas jagah alag hai!*

Dental caries (tooth decay) ★★☆

- Bacteria act on **sugars** and make **acids** that soften (demineralise) the **enamel**
- Bacterial cells and food particles stick to teeth as **dental plaque**; saliva cannot reach the enamel to neutralise the acid
- Brushing after eating removes plaque before acid forms
- If untreated, germs reach the **pulp**, causing inflammation and infection

6 PRACTICE MCQS

1. Cuscuta (amarbel) shows which mode of nutrition? ★★★

(a) saprotrophic (b) parasitic
(c) holozoic (d) autotrophic

2. In Amoeba, digestion of food takes place in the: ★★☆

(a) nucleus (b) pseudopodia
(c) food vacuole (d) cell membrane

3. Which enzyme of the pancreas digests proteins? ★★★

(a) amylase (b) trypsin
(c) lipase (d) pepsin

4. The function of HCl in the stomach is to: ★★★

(a) digest fats (b) make the medium acidic for pepsin
(c) emulsify fats (d) absorb water

5. Emulsification of fats is done by: ★★★

(a) pancreatic juice (b) bile salts
(c) saliva (d) mucus

6. Water is mainly absorbed from undigested food in the: ★★☆

(a) stomach (b) small intestine
(c) large intestine (d) oesophagus

7. Villi in the small intestine help mainly by: ★★★

(a) secreting bile (b) increasing the surface area for absorption
(c) killing germs (d) churning food

8. Herbivores have a longer small intestine than carnivores because: ★★☆

(a) they eat more
(b) cellulose takes longer to digest
(c) they drink more water
(d) they have no stomach

9. **Assertion (A):** The inner lining of the stomach is not damaged by its own hydrochloric acid.
Reason (R): Mucus secreted by gastric glands protects the stomach lining. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Bile juice is called a digestive juice although it has no enzyme.
Reason (R): Bile makes the food alkaline and emulsifies fats, helping digestion. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(b)	A plant that feeds on a living host.
2	(c)	Food vacuole holds and digests food.
3	(b)	Trypsin digests proteins.
4	(b)	Acidic medium for pepsin; also kills germs.
5	(b)	Bile salts break fat into droplets.
6	(c)	Large intestine absorbs water.
7	(b)	More surface area, faster absorption.
8	(b)	Cellulose needs a long gut.
9	(a)	Mucus explains why the lining is safe.
10	(a)	R explains why bile still counts.

Day 42 hit-list: Eduspark PYQ analysis

Topic	Rating
Human digestive system diagram	★★★
Enzymes and their juices	★★★
Role of HCl, mucus, bile	★★★
Amoeba nutrition diagram	★★☆
Villi, absorption, dental caries	★★☆

Short and long answer questions

1. Name the three types of heterotrophic nutrition with one example each. **(3 marks)**
2. Draw a labelled diagram of Amoeba taking in food. NCERT **(2 marks)**
3. Draw the human alimentary canal and label the liver, stomach, pancreas and small intestine. Give the role of each gland. **(5 marks)**
4. What is dental caries? How can it be prevented? NCERT **(2 marks)**

Model answers

1 Types
- Holozoic (human), saprotrophic (mushroom), parasitic (Cuscuta)

2 Amoeba
- Pseudopodia around the food, food vacuole formed (Section 2 diagram)

3 Digestive system
- Liver: bile; pancreas: trypsin, lipase, amylase; gastric glands: HCl, pepsin, mucus

4 Dental caries
- Acid from bacteria on sugar softens enamel; brush after meals, less sugar

Day 42 checklist: tick before you sleep

- ☐ I can name and give examples of the three heterotrophic types
- ☐ I can draw Amoeba taking in food
- ☐ I can draw and label the human digestive system
- ☐ I know every enzyme, its juice and its product
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.

LEVEL 42 COMPLETE +140 XP earned. Next: Level 43



BUJHO *Samajh gaya! Muh se amylase, stomach mein pepsin, aur intestine mein TAL.*

PAHELI *Shabaash! Kal Level 43: respiration aur energy ka khel.*



NITESH SIR SAYS
Digestive system ka diagram 3 baar bana lo, labels khud aa jayenge. ***Kal milte hai Day 43 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 43
LEVEL 43

BIOLOGY | CHAPTER 9 | PART 3 ASTRA SERIES

RESPIRATION & BREATHING

Goal: learn how cells release energy and how we breathe.



by Nitesh Prakash

01 Glucose Paths | 02 ATP | 03 Anaerobic | 04 Lungs | 05 Breathing

+140 XP today | Level 43 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 RESPIRATION: GETTING ENERGY FROM FOOD ★★☆

DEFINITION

- **Respiration:** the process in which food (glucose) is broken down in cells to release **energy**; this energy runs every life process
- Two parts: **breathing** (taking in O_2, giving out CO_2) and **cellular respiration** (breaking glucose inside cells)
- Respiration is a **catabolic**, energy-releasing process: the opposite of photosynthesis

ATP: the energy currency of the cell ★★★

- Energy released in respiration is stored as **ATP** (adenosine triphosphate)
- When a cell needs energy, ATP breaks into **ADP + phosphate** and releases about **30.5 kJ/mol**
- ATP powers muscle contraction, protein making, nerve impulses and active transport
- Like money in your pocket: earn once, spend anywhere in the cell

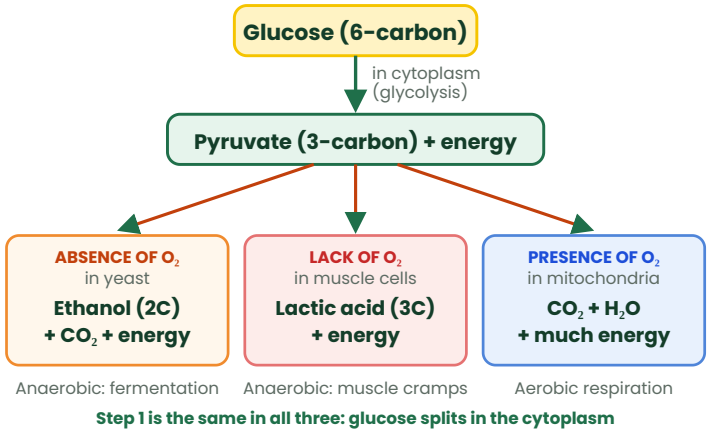


MEME BREAK
*Mitochondria ATP ko aise **sprinkle** karta hai jaise salt bae namak. Isliye mitochondria = **powerhouse of the cell**, aur ATP = energy currency.*

Breathing vs respiration ★☆☆

Breathing	Cellular respiration
physical process	chemical (biochemical) process
in the lungs	in every living cell
takes in O_2, gives out CO_2	breaks glucose, releases energy
no energy released	energy released as ATP

2 BREAKDOWN OF GLUCOSE: THREE PATHWAYS ★★★ MOST ASKED



Step 1 is the same in all three: glucose splits in the cytoplasm



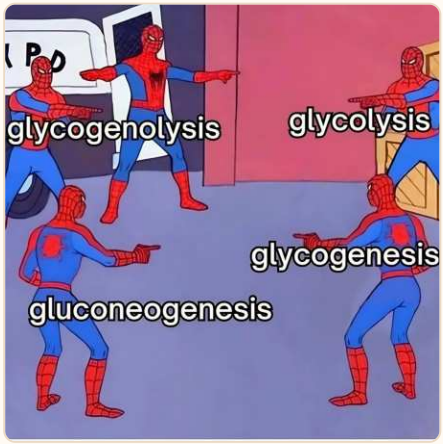
★★★ *Glucose → pyruvate hamesha cytoplasm mein; aage ka rasta O_2 decide karta hai. Ye flowchart har saal poochha jaata hai, Nitesh Bhaiya ki guarantee!*

MNEMONIC: THREE ENDS OF PYRUVATE

"Yeast ka Ethanol, Muscle ka Lactic, Mito ka CO_2–Paani." Y-E, M-L, M-CW: yeast gives ethanol + CO_2, muscles give lactic acid, mitochondria give CO_2 + water.

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

In the mitochondria, pyruvate enters the **Krebs cycle** and the electron transport chain; most of the ATP of aerobic respiration is made here. NCERT does not test these names, only the pathway flowchart.



MEME BREAK
*Glycogenolysis, glycolysis, glycogenesis, gluconeogenesis: naam sab ek jaise! Board ke liye bas ek yaad rakho: **glycolysis = glucose → pyruvate**, cytoplasm mein.*

Aerobic vs anaerobic respiration ★★★

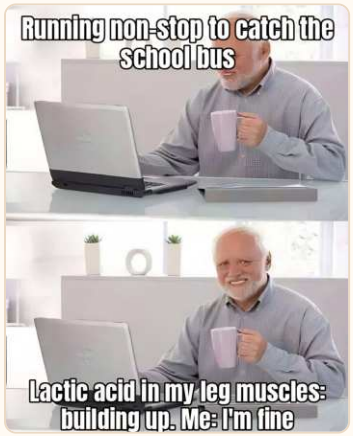
Point	Aerobic	Anaerobic
Oxygen	needed	not needed
Site	cytoplasm + mitochondria	cytoplasm only
Breakdown	complete	incomplete
Products	$CO_2 + H_2O$	ethanol + CO_2 (yeast); lactic acid (muscles)
Energy	much more (about 38 ATP per glucose)	very little (2 ATP per glucose)
Example	most plants and animals	yeast, some bacteria, our muscles in a sprint

MUSCLE CRAMPS: WHY?

During heavy exercise the muscles need energy faster than O_2 can reach them. They switch to anaerobic respiration and **lactic acid** builds up, which causes **cramps**. Rest and a massage bring blood and O_2 back.

EXAM TIPS

Pathway flowchart mein har arrow pe **site** (cytoplasm / mitochondria) aur **condition** (O_2 hai ya nahi) likho. Activity questions mein hamesha **observation + conclusion** dono.



MEME BREAK

*Lactic acid jam raha hai, phir bhi "main theek hoon". Ye hi hai **anaerobic respiration** in muscles.*



BUJHO *Anaerobic respiration sirf yeast mein hoti hai na? Hum toh oxygen lete hain!*

PAHELI *Nahi Bujho! Daudte waqt tumhare **muscle cells** bhi anaerobic chalate hain aur **lactic acid** banta hai. Yeast se ethanol banta hai, muscles se lactic acid.*



3 NCERT ACTIVITIES ★★☆

Activity 1: limewater test for exhaled air ★★☆

- Pump ordinary air through limewater in one test tube; blow **exhaled air** through limewater in another
- Limewater turns **milky faster** with exhaled air
- **Conclusion:** exhaled air has more CO_2 than ordinary air; $CO_2 + Ca(OH)_2 \rightarrow CaCO_3$ (milky) + H_2O

Activity 2: yeast and sugar ★★☆

- Add yeast to a sugar solution in a stoppered test tube; pass the gas formed into limewater
- Limewater turns milky: yeast gives out **CO_2**; the solution smells of **alcohol**
- **Conclusion:** yeast respires anaerobically (fermentation)

4 EXCHANGE OF GASES IN PLANTS ★★☆

Plants breathe by diffusion ★★☆

- Gases enter and leave through **stomata** (leaves), **lenticels** (stems) and the surface of roots, by **diffusion**
- Plants have no special breathing organs; every cell is close to the air

Time	What happens	Gas out
Day	photosynthesis faster than respiration; CO_2 used up	O_2
Night	only respiration	CO_2



BUJHO *Plants din mein sirf photosynthesis karte hain, respiration raat ko na?*

PAHELI *Galat Bujho! Plants **din-raat respiration** karte hain. Din mein photosynthesis tez hoti hai, isliye net O_2 bahar aati hai; raat ko sirf CO_2.*



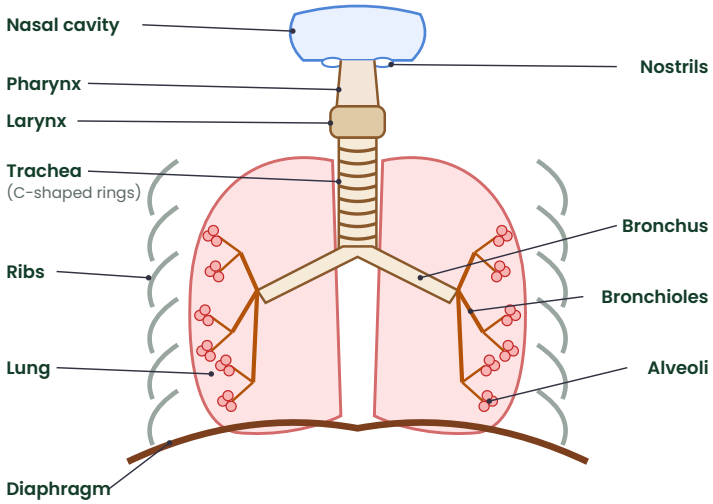
Breathing rate: water vs land ★★☆

- Water holds far **less dissolved O_2** than air
- So aquatic animals such as fish breathe **much faster** than land animals
- Fish take water in through the mouth and push it over the **gills**, where dissolved oxygen is absorbed

SMALL ORGANISMS

In **unicellular** organisms such as Amoeba, gases diffuse across the whole body surface; large animals need lungs or gills.

5 HUMAN RESPIRATORY SYSTEM ★★★ MOST ASKED



Path of air (draw it as a flow)

Nostrils → nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli

- Fine **hair** and **mucus** in the nose filter dust and germs and warm the air
- A flap, the **epiglottis**, closes the trachea while we swallow food
- **Rings of cartilage** keep the trachea open so it does not collapse when there is little air in it
- Alveoli are balloon-like sacs with thin walls and a dense network of **blood capillaries**

Mechanism of breathing ★★★

Step	Inhalation	Exhalation
Ribs	lifted up and out	move down and in
Diaphragm	contracts, flattens	relaxes, domes up
Chest cavity	becomes larger	becomes smaller
Air	rushes into the lungs	pushed out of the lungs

MNEMONIC: INHALE VS EXHALE

"Andar saans: Ribs Upar, Diaphragm Neeche (flat)." Inhale = chest bada, air andar. Exhale mein sab ulta: ribs neeche, diaphragm upar (dome).

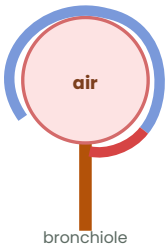
Residual volume and alveoli ★★☆

- The lungs always hold some air even after breathing out: the **residual volume**
- So there is always time for O_2 to be absorbed and CO_2 to be released
- The alveoli spread out would cover about **80 m^2**: a huge surface for gas exchange



MEME BREAK

*Saans rok li? Fikar not! Lungs mein **residual volume** hamesha rehta hai, isliye O_2 absorb hona chalta rehta hai.*

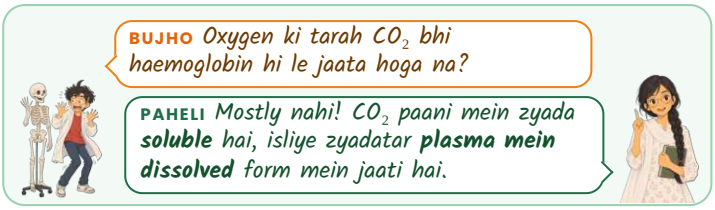


- O_2: alveolus → blood
- CO_2: blood → alveolus

Walls one cell thick, wrapped in capillaries: gases cross by diffusion in a fraction of a second

How O_2 and CO_2 travel in blood ★★★

- **Haemoglobin**, the respiratory pigment in red blood cells, has a very high affinity for O_2 and carries it from the lungs to the tissues
- CO_2 is more **soluble in water** than O_2, so it is mostly carried **dissolved in the blood plasma**
- In large animals diffusion alone is too slow, so a pigment is needed



MNEMONIC: PATH OF AIR

"Nani Ne Pakode Laye, Tea Bina Biscuit Aayi." Nostrils, Nasal cavity, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli.



MEME BREAK

*Anaerobic respiration se bahut ATP? Nahi bhai, sirf **2 ATP** per glucose. Bahut energy chahiye toh **oxygen + mitochondria**.*

COMMON MISTAKES

- Writing that glycolysis happens in mitochondria: it happens in the **cytoplasm**
- Saying plants give out only O_2: at night they give out **CO_2**
- Calling breathing and respiration the same: breathing is physical, respiration is chemical
- Forgetting the **arrows** and site names in the pathway flowchart

Case-based question ★★☆

- Riya runs a 400 m race and feels a sharp pain in her calf muscles. (i) Which type of respiration took place in her muscles? (ii) Name the product that caused the pain. (iii) Where in the cell does the first step occur? **(1 + 1 + 1)**
- **Answers:** (i) anaerobic respiration (ii) lactic acid (iii) cytoplasm

6 PRACTICE MCQS

1. The first step of glucose breakdown, forming pyruvate, takes place in the: ★★★

(a) mitochondria (b) cytoplasm
(c) nucleus (d) chloroplast

2. In yeast, anaerobic respiration produces: ★★★

(a) lactic acid + energy (b) $CO_2 + H_2O$ + energy
(c) ethanol + CO_2 + energy (d) pyruvate only

3. Muscle cramps after a long run are due to the build-up of: ★★★

(a) ethanol (b) lactic acid
(c) pyruvic acid (d) carbon dioxide

4. The energy currency of the cell is: ★★☆

(a) ADP (b) glucose
(c) ATP (d) DNA

5. Rings of cartilage are present in the: ★★☆

(a) oesophagus (b) trachea
(c) alveoli (d) bronchioles

6. Aquatic animals breathe faster than terrestrial animals because: ★★☆

(a) water is denser
(b) water has less dissolved O_2
(c) gills are small
(d) they move more

7. Most CO_2 is carried in human blood: ★★☆

(a) by haemoglobin only
(b) in dissolved form in plasma
(c) by platelets
(d) by white blood cells

8. During inhalation the diaphragm: ★★☆

(a) relaxes and domes up
(b) contracts and flattens
(c) does not move
(d) moves into the chest

9. **Assertion (A):** The lungs always contain some air even after a forceful exhalation.
Reason (R): The residual volume of air allows time for oxygen to be absorbed and carbon dioxide to be released. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Aerobic respiration releases more energy than anaerobic respiration.
Reason (R): In aerobic respiration glucose is broken down completely into carbon dioxide and water. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Why do muscles cramp after vigorous exercise? Name the substance responsible. **(2 marks)**
2. Write the three pathways of breakdown of glucose in a flowchart and name the site of each step. **(3 marks)**
3. Draw a labelled diagram of the human respiratory system. Explain how the alveoli are designed for maximum exchange of gases. **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Glycolysis happens in the cytoplasm.
2	(c)	Fermentation in yeast.
3	(b)	Anaerobic respiration in muscles.
4	(c)	ATP stores and supplies energy.
5	(b)	Rings stop the trachea collapsing.
6	(b)	Water holds less oxygen than air.
7	(b)	CO_2 is more soluble in water.
8	(b)	Diaphragm flattens, chest enlarges.
9	(a)	R explains A.
10	(a)	Complete breakdown gives more energy.

YAAD RAKHNE KI TRICK: ATP COUNT

"O_2 mila toh 38 ka jackpot, O_2 nahi toh sirf 2." Aerobic: about 38 ATP per glucose; anaerobic: 2 ATP.

Model answers

1 Cramps

> Lack of O_2 in a sprint makes muscles respire anaerobically; **lactic acid** builds up and causes cramps (1 + 1)

2 Pathways

> Use the flowchart on page 1: cytoplasm step, then yeast, muscle and mitochondria branches with products (1 mark each)

3 Alveoli

> Diagram with 6 labels (2); huge surface area (about 80 m^2), very thin walls, rich blood supply, residual air (3)

Day 43 checklist: tick before you sleep

- [ ] I can draw the three pathways of glucose breakdown with sites
- [ ] I know aerobic vs anaerobic in a table and why muscles cramp
- [ ] I can explain the limewater and yeast activities
- [ ] I can draw and label the human respiratory system
- [ ] I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 43 COMPLETE +140 XP earned. Next: Level 44



BUJHO *Samajh gaya! Glucose ka pehla step cytoplasm mein, aur mitochondria mein poori energy nikalti.*

PAHELI *Shabaash! Kal Level 44: heart, blood aur double circulation.*



NITESH SIR SAYS

*Flowchart mein **site** aur **product** dono likho, poore marks wahi se aate hain.* ***Kal milte hai Day 44 pe!***

Day 43 hit-list: Eduspark PYQ analysis

Topic	Rating
Pathways of glucose breakdown (flowchart)	★★★
Aerobic vs anaerobic respiration	★★★
Respiratory system diagram	★★★
Alveoli and exchange of gases	★★☆
Gas exchange in plants, day and night	★★☆
Haemoglobin and CO_2 transport	★☆☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 44 · LEVEL 44

BIOLOGY | CHAPTER 9 | PART 4

TRANSPORTATION IN HUMANS

Goal: master blood, the heart and double circulation.



by Nitesh Prakash

01 Blood | 02 Heart | 03 Double Circulation | 04 Vessels | 05 BP and Lymph | +140 XP today | Level 44 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

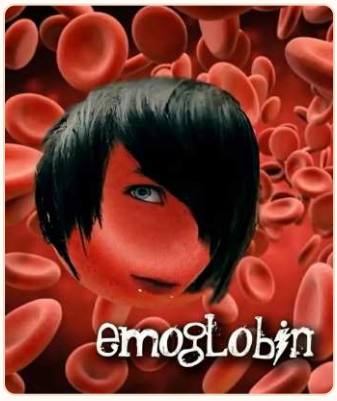
1 WHY TRANSPORT? BLOOD AS THE CARRIER ★★☆

DEFINITION

- Food, O_2, water and wastes must move to and from every cell; in big animals diffusion is far too slow
- **Blood** is a fluid **connective tissue**; the **heart** pumps it through **blood vessels**
- Humans have a closed circulatory system: blood always flows inside vessels

Components of blood ★★★

Component	Key facts
Plasma	fluid part (about 55%); carries food, CO_2, salts, hormones and nitrogenous wastes in dissolved form
RBCs	contain **haemoglobin**; carry O_2; no nucleus in humans
WBCs	fight germs (body defence); have a nucleus
Platelets	cell fragments that help in **clotting** of blood at a wound



MEME BREAK

*O_2 ko sabse zyada pyaar **haemoglobin** karta hai. Isliye RBC ke andar ye red pigment O_2 ko lungs se tissues tak le jaata hai.*

Clotting: sealing a wound ★★☆

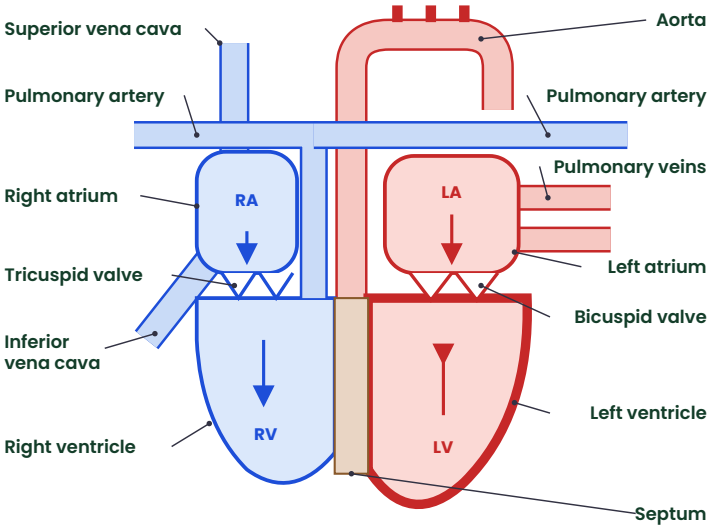
- At a cut, **platelets** collect and start a chain of reactions
- A mesh of **fibrin** threads traps blood cells and forms a **clot**
- The clot plugs the leak, so blood loss and pressure drop are stopped



MEME BREAK

*Wound pe jo crust banti hai na? Wo **platelets** ki mehnat hai: clot banake leak band. Usse khurachna mat!*

2 THE HUMAN HEART ★★★ MOST ASKED



Heart at a glance

- A muscular organ about the size of your **fist**, with **four chambers**: two atria on top, two ventricles below
- **Atria** receive blood (thin walls); **ventricles** pump it out (thick walls); the **left ventricle** is thickest as it pumps to the whole body
- **Valves** (tricuspid on the right, bicuspid on the left, semilunar at the arteries) stop **backflow**
- A wall called the **septum** keeps the two sides apart

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

At rest the heart beats about **72 times a minute**; each beat can be felt as the **pulse** at the wrist. The beat starts at a natural pacemaker in the wall of the right atrium (the SA node).



★★★ *Heart ka labelled diagram har board paper ka favourite hai: chambers, valves aur chaaron vessels. Nitesh Bhaiya ki guarantee!*

3 PATH OF BLOOD THROUGH THE HEART ★★★

Flow of blood (learn as a chain)

Body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left ventricle → aorta → body

- Right side carries **deoxygenated** blood; left side carries **oxygenated** blood
- Both atria contract together; then both ventricles contract together

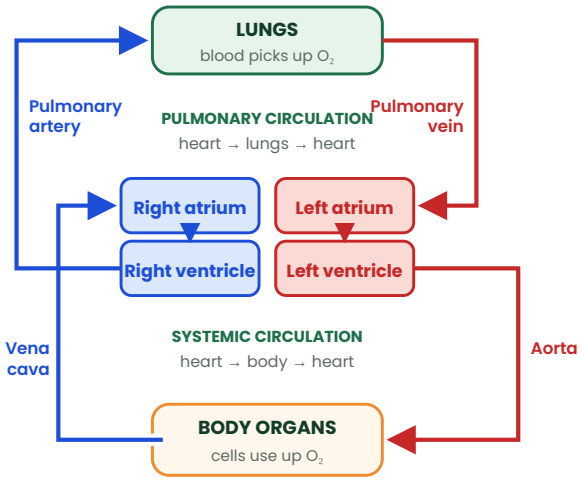
MNEMONIC: PATH OF BLOOD

"Vimal Ramu Ravi Pappu Lalu Pinku Lali Lata Aaye." Vena cava, R-atrium, R-ventricle, Pulmonary artery, Lungs, Pulmonary vein, L-atrium, L-ventricle, Aorta.

Four big vessels: quick revision ★★★

Vessel	From → To	Blood
Vena cava	body → right atrium	deoxygenated
Pulmonary artery	right ventricle → lungs	deoxygenated
Pulmonary veins	lungs → left atrium	oxygenated
Aorta	left ventricle → body	oxygenated

4 DOUBLE CIRCULATION ★★★ MOST ASKED



DEFINITION

- **Double circulation:** blood passes through the heart **twice** in one complete round of the body
- **Pulmonary circulation:** heart → lungs → heart; **systemic circulation:** heart → body → heart

EXAM TIPS

Heart diagram mein right side ko **blue** (deoxygenated) aur left side ko **red** (oxygenated) dikhao; valves aur septum label karna mat bhoolo. Viewer ka left = heart ka **right**!

Why four chambers? ★★★

- The septum keeps **oxygenated and deoxygenated** blood from mixing
- So the body gets a highly efficient supply of O_2: vital for **birds and mammals**, which need lots of energy to keep a constant body temperature

Animal	Heart	Circulation
Fish	2 chambers	single: heart → gills → body → heart
Amphibians, most reptiles	3 chambers	some mixing of blood; body temperature depends on surroundings
Birds, mammals	4 chambers	double, no mixing



BUJHO *Pulmonary artery mein toh oxygenated blood hoga, artery jo hai!*

PAHELI *Nahi Bujho!* ***Pulmonary artery*** *deoxygenated blood le jaati hai (heart → lungs), aur* ***pulmonary vein*** *oxygenated blood laati hai. Artery = heart se door, chahe blood koi bhi ho.*



5 BLOOD VESSELS ★★★

Arteries vs veins vs capillaries ★★★

Feature	Arteries	Veins	Capillaries
Carry blood	away from heart	to the heart	link arteries and veins
Blood	oxygenated (not the pulmonary artery)	deoxygenated (not the pulmonary vein)	exchange O_2, food, CO_2, wastes
Walls	thick, elastic	thin	one cell thick
Valves	absent	present	absent
Pressure	high	low	low

YAAD RAKHNE KI TRICK

"A for Away, V for Visit (heart)." Artery heart se door, Vein heart ko visit. Veins mein **valves** hote hain kyunki pressure kam hai.

Cross-sections of blood vessels



CLASS TRICK: VALVE NAMES

"Try Right, Be Left." **Tri**cuspid valve on the **right** side, **Bi**cuspid (mitral) valve on the **left**. Semilunar valves guard the aorta and pulmonary artery.



BUJHO *Capillaries itni patli kyun, bekaar nahi hai?*

PAHELI *Bilkul nahi! Capillary wall sirf* ***ek cell moti*** *hai, isliye O_2, food aur CO_2 ka exchange blood aur cells ke beech yahi hota hai.*





MEME BREAK
Cheers to the ***septum****! Isi deewar ki wajah se oxygenated aur deoxygenated blood kabhi mix nahi hote.*

6 BLOOD PRESSURE ★★★

Systolic and diastolic ★★★

- **Blood pressure:** the force that blood exerts on the walls of a vessel; much higher in arteries than in veins
- **Systolic** pressure: during ventricular contraction, normally about **120 mm Hg**
- **Diastolic** pressure: during ventricular relaxation, normally about **80 mm Hg**; normal BP is written **120/80**
- Measured with a **sphygmomanometer**
- **Hypertension** (high BP) is caused by constriction of **arterioles**, which raises resistance to blood flow; it can burst an artery and cause internal bleeding

BUJHO *120/80 mein 120 diastolic hai na?*



PAHELI *Ulta Bujho! Bada number* ***systolic*** *(ventricle contract) aur chhota number* ***diastolic*** *(ventricle relax). "****S = Squeeze = bada.****"*



7 LYMPH: THE SECOND TRANSPORT FLUID ★★☆

Lymph (tissue fluid) ★★☆

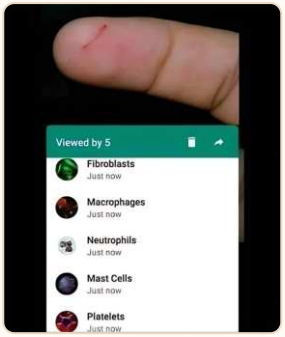
- Some plasma, proteins and blood cells escape through pores in capillary walls into the spaces between cells: this is **tissue fluid** or **lymph**
- Lymph is like plasma but **colourless** and has **less protein**
- It drains into **lymphatic capillaries** and then larger vessels that open into veins

Functions:

- ✔ carries **digested fat** absorbed in the intestine (lacteals)
- ✔ drains excess fluid from tissues back into the blood
- ✔ its lymphocytes help fight infection



MEME BREAK
Tissue fluid akela intezaar karta reh jaata agar ***lymph vessels*** *na hote. Yahi lymph use wapas veins tak le jaata hai.*



MEME BREAK
Ek chhota sa cut, aur "seen by 5": fibroblasts, macrophages, neutrophils, mast cells, platelets! ***Platelets*** *clot banate, WBCs germs se ladte.* ***Team blood****!*

COMMON MISTAKES

Writing that all arteries carry oxygenated blood (pulmonary artery does not). Mixing up systolic and diastolic. Calling lymph red: it is **colourless**. Forgetting valve names in the heart diagram.

NCERT in-text: quick answers

- **Components of the transport system:** heart (pumps), blood vessels (carry), blood (medium: plasma, RBCs, WBCs, platelets), lymph
- **Why separate oxygenated and deoxygenated blood in birds and mammals?** They need high energy to keep a constant body temperature, so an efficient O_2 supply is essential

MNEMONIC: FOUR PARTS OF BLOOD

"Paani, Rang, Wardi, Plaster." Plasma = paani (fluid), RBC = rang (red, O_2), WBC = wardi (soldier, defence), Platelets = plaster (seal the wound).

8 PRACTICE MCQS

1. Which blood component helps in clotting? ★★★
(a) RBCs (b) WBCs
(c) platelets (d) plasma

2. The chamber of the heart with the thickest muscular wall is the: ★★★
(a) right atrium (b) left atrium
(c) right ventricle (d) left ventricle

3. Oxygenated blood from the lungs enters the heart through the: ★★★
(a) pulmonary artery (b) pulmonary veins
(c) vena cava (d) aorta

4. Valves are present in: ★★☆
(a) arteries (b) veins
(c) capillaries (d) all vessels

5. Normal blood pressure of an adult is about: ★★★
(a) 80/120 mm Hg (b) 120/80 mm Hg
(c) 100/60 mm Hg (d) 140/100 mm Hg

6. The instrument used to measure blood pressure is: ★★☆
(a) sphygmomanometer (b) stethoscope
(c) barometer (d) thermometer

7. A fish heart has: ★★☆
(a) 2 chambers (b) 3 chambers
(c) 4 chambers (d) 1 chamber

8. Lymph differs from blood plasma as it: ★★☆

(a) is red (b) has more protein
(c) is colourless with less protein (d) has RBCs

9. **Assertion (A):** Birds and mammals have a four-chambered heart. **Reason (R):** Separation of oxygenated and deoxygenated blood gives them an efficient O_2 supply needed to maintain a constant body temperature. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** All arteries carry oxygenated blood. **Reason (R):** Arteries carry blood away from the heart to the organs. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. Why is double circulation necessary in human beings? **(2 marks)**
2. Differentiate between arteries and veins (any three points). **(3 marks)**
3. Draw a labelled sectional view of the human heart and trace the path of blood from the body back to the body. **(5 marks)**

Case-based question ★★☆

> A doctor records Rohan's BP as 150/100 mm Hg. (i) Which value is systolic? (ii) Name the condition. (iii) Give one cause. **(1 + 1 + 1)**

> **Answers:** (i) 150 mm Hg (ii) hypertension (iii) constriction of arterioles raises resistance to flow

DIAGRAM PRACTICE

Aaj raat 5 minute ka timer lagao: heart ka sectional view aur double circulation flowchart bina dekhe banao, phir page 1–2 se milao.

Answer key: MCQs

Q	Ans	Why
1	(c)	Platelets start clotting.
2	(d)	LV pumps blood to the whole body.
3	(b)	Pulmonary veins bring blood from the lungs.
4	(b)	Low pressure; valves stop backflow.
5	(b)	Systolic/diastolic = 120/80.
6	(a)	Sphygmomanometer.
7	(a)	Fish: 2 chambers, single circulation.
8	(c)	Lymph is colourless, less protein.
9	(a)	R explains A.
10	(d)	Pulmonary artery carries deoxygenated blood.

Model answers

1 Double circulation

> Keeps oxygenated and deoxygenated blood separate, so tissues get a rich O_2 supply for high energy needs (2)

2 Arteries vs veins

> Use the table on page 3: direction, wall thickness, valves (1 each)

3 Heart

> Diagram with chambers, valves and four vessels (3); path chain vena cava → RA → RV → pulmonary artery → lungs → pulmonary veins → LA → LV → aorta (2)

Day 44 checklist: tick before you sleep

- ☐ I can list the four components of blood and their jobs
- ☐ I can draw and label the heart with valves and vessels
- ☐ I can trace the path of blood and explain double circulation
- ☐ I know arteries vs veins vs capillaries and BP 120/80
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

LEVEL 44 COMPLETE +140 XP earned. Next: Level 45



BUJHO *Samajh gaya! Right side blue, left side red, aur blood heart se do baar guzarta.*

PAHELI *Shabaash! Kal Level 45: plants mein transport aur excretion.*



NITESH SIR SAYS

*Heart diagram roz ek baar banao, 7 din mein haath set ho jaayega. **Kal milte hai Day 45 pe!***

Day 44 hit-list: Eduspark PYQ analysis

Topic	Rating
Heart diagram with labels	★★★
Double circulation, why 4 chambers	★★★
Arteries vs veins vs capillaries	★★★
Blood pressure 120/80, hypertension	★★☆
Components of blood, clotting	★★☆
Lymph and its functions	★☆☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 45 | LEVEL 45

BIOLOGY | CHAPTER 9 | PART 5 | ASTRA SERIES

PLANT TRANSPORT & EXCRETION

Goal: move water and food in plants and filter wastes in us.



by Nitesh Prakash

01 Xylem | 02 Transpiration | 03 Phloem | 04 Nephron | 05 Dialysis

+140 XP today | Level 45 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

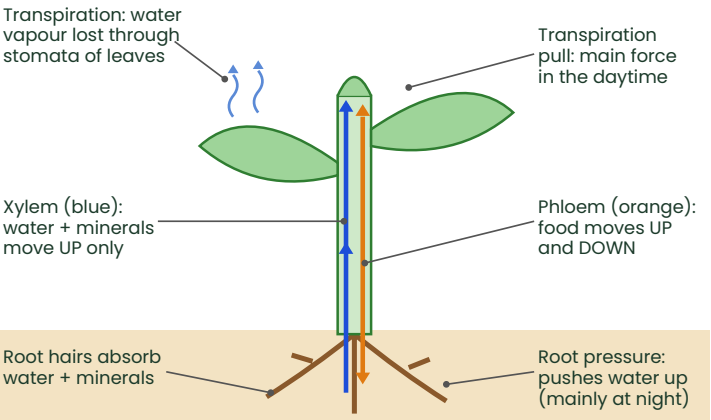
1 TRANSPORTATION IN PLANTS ★★★

Why do plants need a transport system?

1. Roots take **water and minerals** from the soil; leaves make **food**. Each must reach the other part.
2. Plants do not move and have many **dead cells** (e.g. wood), so they need **less energy** and can use a slow transport system.
3. Two conducting tissues do the job: **xylem** and **phloem** (together called vascular tissue).

DEFINITION

- **Xylem:** carries **water and minerals** from roots to leaves. Made of **vessels and tracheids** (dead, tube-like)
- **Phloem:** carries **food** (sucrose, amino acids) from leaves to other parts. Made of **sieve tubes and companion cells**



Root pressure: how water enters the xylem ★★☆

- Root cells in contact with soil **actively take up ions** (minerals)
- The ion concentration in the root becomes **higher** than in the soil
- So **water moves into the root** from the soil to remove this difference
- This makes a steady push of water up the xylem, called **root pressure**; it matters most **at night**

MNEMONIC: DAY VS NIGHT

"Din mein Dhoop, patte kheenchte hain = transpiration pull. Raat mein Root dhakelti hai = root pressure (R se R)."

2 TRANSPIRATION ★★★ MOST ASKED

DEFINITION

- **Transpiration:** the loss of water in the form of **vapour** from the aerial parts of a plant, mainly through the **stomata** of leaves

How transpiration pulls water up ★★★

- Water evaporates from leaf cells and leaves through the stomata
- Leaf cells lose water, so they draw water from the **xylem of the leaf**
- This creates a **suction (pull)** on the unbroken water column in the xylem
- Water and minerals rise from roots to leaves: the **transpiration pull**

Two functions of transpiration ★★★

Function	How
Absorption and upward movement of water and minerals	the suction pulls the water column up the xylem
Temperature regulation	evaporation of water cools the leaf (like sweating)

BUJHO *Paani toh neeche girta hai, phir 50 metre ke ped mein upar kaise chadhta hai?*

PAHELI *Patte stomata se paani ka vapour chhodte hain, isse xylem ke paani ke column par **khichav (pull)** lagta hai. Isi **transpiration pull** se paani upar aata hai.*







MEME BREAK

*Transpiration pull sabko baantta hai: har patte tak **paani aur minerals**, xylem ke through!*

REMEMBER

Root pressure alone cannot lift water to the top of tall trees. The main force in the daytime is the **transpiration pull**.

3 TRANSLOCATION IN PHLOEM ★★★

DEFINITION

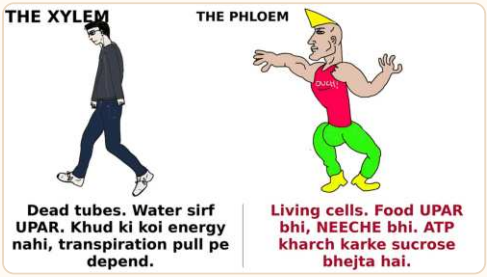
- **Translocation:** transport of soluble products of photosynthesis (and amino acids) through the **phloem**
- Food goes to storage organs (roots, fruits, seeds) and to growing parts; it can move **upward and downward**

How translocation works: uses energy ★★★

- **Sucrose** is loaded into the sieve tubes using energy from **ATP**
- This raises the **osmotic pressure** of the phloem tissue
- **Water enters** the sieve tubes, so the pressure rises further
- The pressure moves the food to tissues with **lower pressure**, where it is used or stored

EXAMPLE

In **spring**, sugar stored in the root or stem is sent **up** to the buds, which need energy to grow. Phloem moves food as the plant needs.



MEME BREAK *Xylem bas transpiration ke bharose; phloem ATP laga ke dono taraf maal bhejta hai!*

Xylem vs phloem ★★★

Point	Xylem	Phloem
Carries	water and minerals	food (sucrose, amino acids)
Direction	one way: **upward**	**both ways**, up and down
Cells	vessels, tracheids (mostly dead)	sieve tubes, companion cells (living)
Energy	no ATP of its own; transpiration pull, root pressure	**uses ATP**
Process	conduction of water	translocation

MNEMONIC: X FOR H_2O, P FOR PIZZA

"Xylem = paani (X jaise paani ka nal, sirf upar). Phloem = Pizza, yaani khana: upar bhi, neeche bhi."

THINK OF IT THIS WAY

Xylem is a one-way **water pipe** run by suction; phloem is a two-way **delivery service** that pays with ATP.



★★★ *Transpiration ke 2 functions aur transpiration pull, ye har saal aata hai, Nitesh Bhaiya ki guarantee!*

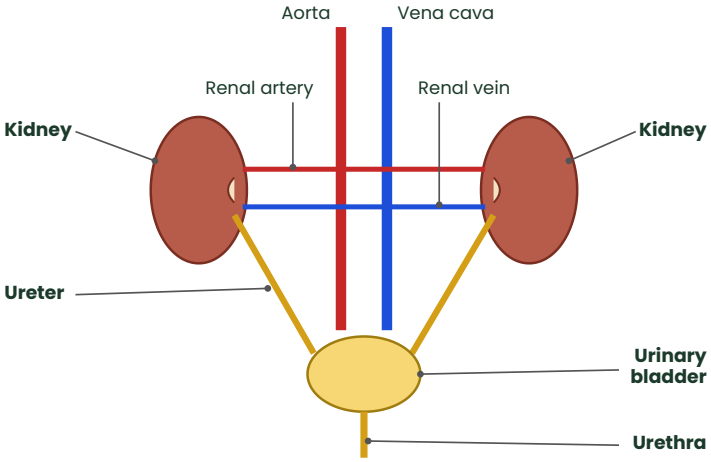
4 EXCRETION IN HUMAN BEINGS ★★★ MOST ASKED

DEFINITION

- **Excretion:** the biological process of removing **harmful nitrogenous metabolic wastes** (like urea, uric acid) from the body
- Unicellular organisms remove wastes by simple **diffusion** from the body surface into the surrounding water

Human excretory system: labelled diagram ★★★

A pair of **kidneys**, a pair of **ureters**, a **urinary bladder** and a **urethra**.



Parts and their jobs ★★★

Part	Job
Kidneys (in the abdomen, one on each side of the backbone)	filter blood and make urine
Ureters	carry urine from the kidneys to the bladder
Urinary bladder	muscular bag that stores urine; it is under nervous control
Urethra	passes urine out of the body

MNEMONIC: URINE KA ROUTE

"Kabhi Ulta Bolna Uchit nahi" = Kidney → Ureter → Bladder → Urethra.



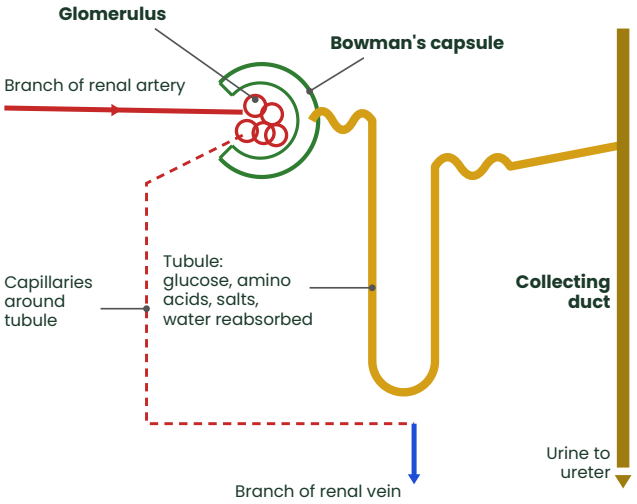
★★★ *Excretory system ka labelled diagram aur urine ka route, board ka favourite hai, Nitesh Bhaiya ki guarantee!*

× WRONG	✓ RIGHT
Phloem carries water up	Xylem carries water; phloem carries food
Urea is made in the kidney	Urea is made in the liver, filtered by the kidney
Urine is stored in the kidney	Stored in the urinary bladder
Dialysis reabsorbs glucose	No reabsorption in an artificial kidney

5 NEPHRON: THE FILTRATION UNIT ★★★ MOST ASKED

DEFINITION

- **Nephron:** the basic **filtration unit** of the kidney. Each kidney has a very large number of nephrons packed close together
- Parts: **Bowman's capsule** (a cup) with a **glomerulus** (a cluster of thin-walled capillaries) inside, a long **tubule** and a **collecting duct**



How urine is formed ★★★

1. **Filtration:** the renal artery brings blood to the glomerulus; water, glucose, amino acids, salts and urea are filtered into **Bowman's capsule**.
2. **Selective reabsorption:** in the tubule, useful substances (glucose, amino acids, salts) and a major part of the water go **back into the blood** of the capillaries.
3. The amount of water reabsorbed depends on how much **extra water** the body has and how much waste must be removed.
4. **Collecting duct:** the remaining fluid is **urine**; it goes to the ureter and is stored in the bladder.



MEME BREAK

*Roz ~180 L filtrate, par bahar sirf **1 se 2 L urine**. Baaki sab tubule mein **reabsorb**: kidney ki smart investment!*

BUJHO *Glomerulus mein toh glucose bhi filter ho jaata hai, phir urine mein glucose kyun nahi aata?*



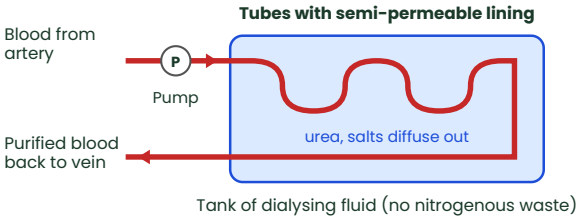
PAHELI *Kyunki **tubule** glucose, amino acids aur zyada paani ko wapas blood mein **reabsorb** kar leta hai. Healthy insaan ke urine mein glucose nahi hota.*



6 ARTIFICIAL KIDNEY (HAEMODIALYSIS) ★★☆

When the kidneys fail

- Infection, injury or less blood flow can make the kidneys stop working; poisonous wastes then **build up** in the body
- An **artificial kidney** removes nitrogenous wastes from the blood by **dialysis**



How it works ★★☆

1. Long tubes with a **semi-permeable lining** sit in a tank of **dialysing fluid**.
2. The fluid has the **same osmotic pressure as blood** but has **no nitrogenous wastes**.
3. The patient's blood is passed through the tubes; wastes **diffuse** out into the fluid.
4. The purified blood is pumped back into the patient.

KEY DIFFERENCE

A natural kidney **reabsorbs** useful substances; an artificial kidney has **no reabsorption**.

MNEMONIC: URINE FORMATION

"Filter, Reabsorb, Urine" = FRU: pehle chhaano, phir kaam ka wapas lo, bacha urine.

7 EXCRETION IN PLANTS ★★★

How plants get rid of wastes ★★★

- ✔ **Oxygen** (from photosynthesis) and **CO_2** (from respiration) leave by diffusion through stomata
- ✔ **Excess water** is lost by transpiration
- ✔ Many wastes are stored in **cell vacuoles** and in **leaves that fall off**
- ✔ Some wastes are stored as **resins and gums**, mostly in **old xylem**
- ✔ Some wastes are released into the **surrounding soil**



MEME BREAK

*Purane patte waste lekar gir jaate hain: plant ka **easy excretion** trick!*

8 PRACTICE MCQS

1. Water and minerals in plants move through: ★★★
(a) phloem (b) xylem
(c) cortex (d) epidermis

2. The main force that lifts water in tall trees during the day is: ★★★
(a) root pressure (b) transpiration pull
(c) osmosis in leaves (d) gravity

3. Translocation of food in plants needs: ★★★
(a) no energy (b) energy from ATP
(c) sunlight only (d) root pressure

4. The basic filtration unit of the kidney is: ★★★
(a) neuron (b) nephron
(c) ureter (d) alveolus

5. The capillary cluster inside Bowman's capsule is the: ★★★
(a) glomerulus (b) tubule
(c) collecting duct (d) villus

6. Which is NOT normally present in the urine of a healthy person? ★★☆
(a) water (b) urea
(c) salts (d) glucose

7. Urine is stored in the: ★★☆
(a) kidney (b) ureter
(c) urinary bladder (d) urethra

8. Plants store some waste products as resins and gums in: ★★☆
(a) old xylem (b) young leaves
(c) root hairs (d) phloem sap

9. **Assertion (A):** Transpiration helps in the upward movement of water in plants.
Reason (R): Loss of water vapour from leaves creates a suction that pulls water up the xylem. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** An artificial kidney reabsorbs glucose and amino acids into the blood.
Reason (R): The dialysing fluid has the same osmotic pressure as blood but no nitrogenous wastes. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How are water and minerals transported in plants? Give two functions of transpiration. **(3 marks)**
2. Draw a labelled diagram of the human excretory system. Which organ stores urine? **(3 marks)**
3. Draw the structure of a nephron and label Bowman's capsule, glomerulus, tubule and collecting duct. How is urine formed? **(5 marks)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Xylem conducts water and minerals.
2	(b)	Transpiration pull in the daytime.
3	(b)	Sucrose is loaded using ATP.
4	(b)	Nephron = filtration unit.
5	(a)	Glomerulus sits in the capsule.
6	(d)	Glucose is fully reabsorbed.
7	(c)	The bladder stores urine.
8	(a)	Resins and gums in old xylem.
9	(a)	R explains A.
10	(d)	A false: no reabsorption. R true.

Model answer points

- **Q1:** root ions draw water in; xylem column pulled by transpiration; functions: water and mineral uptake, temperature control
- **Q2:** kidneys, ureters, urinary bladder, urethra (plus aorta, vena cava, renal artery and vein); urine is stored in the urinary bladder
- **Q3:** filtration in glomerulus into Bowman's capsule; reabsorption in tubule; urine in collecting duct

Day 45 checklist: tick before you sleep

- ☐ I can explain root pressure and transpiration pull
- ☐ I can draw and compare xylem and phloem transport
- ☐ I can draw the human excretory system and a nephron
- ☐ I can explain urine formation and haemodialysis
- ☐ I solved all 10 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.



LEVEL 45 COMPLETE +140 XP earned. Next: Level 46



BUJHO *Samajh gaya! Xylem paani upar, phloem khana dono taraf, aur nephron chhaan ke wapas leta hai.*

PAHELI *Shabaash! Kal Level 46: Chapter 9 ka* ***Boss Level.***



NITESH SIR SAYS
Nephron ka diagram 5 baar bana lo, labels ke saath. Exam mein ***diagram = pakke marks.*** *Kal milte hai* ***Day 46 pe!***

DAY 46
LEVEL 46

BIOLOGY | CHAPTER 9 | BOSS LEVEL

CHAPTER 9 BOSS LEVEL

Goal: test all of Life Processes in 90 minutes.



by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Case & Long | 05 Answer Key | +200 XP Boss Level | Level 46 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 9 REVISION MAP ★★★ MOST ASKED

Chapter 9 mind map: revise in 5 minutes

NUTRITION
$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (sunlight, chlorophyll); holozoic, saprotrophic, parasitic

RESPIRATION ★★★
Glucose → pyruvate (cytoplasm); aerobic in mitochondria: $CO_2 + H_2O$ + ATP; yeast: ethanol; muscle: lactic acid

LIFE PROCESSES

TRANSPORT IN HUMANS
4-chambered heart, double circulation, arteries, veins, capillaries, lymph; BP 120/80

TRANSPORT IN PLANTS
Xylem: water up (transpiration pull); phloem: food both ways (uses ATP)

EXCRETION ★★★
Kidney → ureter → bladder → urethra; nephron: filtration in glomerulus, reabsorption in tubule; dialysis; plants: leaves, resins, gums, soil

Must-know: digestive juices ★★★

Juice / enzyme	From	Acts on
Salivary amylase	salivary glands	starch → sugar
Pepsin (with HCl)	gastric glands	proteins
Bile (no enzyme)	liver	emulsifies fats
Trypsin	pancreas	proteins
Lipase	pancreas	fats → fatty acids + glycerol

MNEMONIC: 4 LIFE PROCESSES
"Life ki kahani N-R-T-E se suno" = Nutrition, Respiration, Transportation, Excretion.



BOSS INTRO
*Notes ek baar padh ke so gaye? Aaj nahi! Aaj **diagrams label** karke Boss harana hai.*

2 SECTION A: MCQS ★★★ MOST ASKED

1. The site of photosynthesis in a plant cell is: ★★★
(a) mitochondria (b) chloroplast
(c) nucleus (d) ribosome

2. Opening and closing of stomata is controlled by: ★★★
(a) guard cells (b) xylem
(c) root hairs (d) phloem

3. The enzyme that digests proteins in the stomach is: ★★★
(a) amylase (b) lipase
(c) pepsin (d) trypsin

4. Bile is secreted by the liver and stored in: ★★☆
(a) pancreas (b) gall bladder
(c) stomach (d) small intestine

5. Anaerobic respiration in yeast gives: ★★★
(a) lactic acid + energy (b) $CO_2 + H_2O$ + energy
(c) ethanol + CO_2 + energy (d) pyruvate only

6. Glucose is broken down into pyruvate in: ★★★
(a) cytoplasm (b) mitochondria
(c) chloroplast (d) nucleus

7. Oxygenated blood from the lungs reaches the heart through: ★★★
(a) pulmonary artery (b) pulmonary vein
(c) aorta (d) vena cava

8. The heart of a fish has: ★★☆
(a) two chambers (b) three chambers
(c) four chambers (d) one chamber

9. In human lungs, the exchange of gases takes place in: ★★★
(a) bronchi (b) trachea
(c) alveoli (d) larynx

How to attack the Boss: 90 minutes

- Section A (MCQs): 15 min; read every A/R option slowly
- Sections B and C: 35 min; draw diagrams in pencil
- Sections D and E: 30 min; label every part you draw
- Last 10 min: check with the key, then score yourself



★★★ *A/R aur diagram-based MCQs Chapter 9 se har paper mein aate hain, Nitesh Bhaiya ki guarantee!*

10. Excess water in plants is removed mainly by: ★★☆

(a) translocation (b) transpiration
(c) photosynthesis (d) root pressure

11. Lymph carries digested and absorbed fat from the: ★★☆

(a) stomach (b) small intestine
(c) liver (d) kidney

12. The correct path of urine is: ★★★

(a) kidney → urethra → bladder → ureter (b) kidney → ureter → bladder → urethra
(c) kidney → bladder → ureter → urethra (d) ureter → kidney → bladder → urethra

13. **Assertion (A):** The ventricles of the human heart have thicker muscular walls than the atria.
Reason (R): Ventricles have to pump blood into various organs of the body. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

14. **Assertion (A):** Aquatic animals breathe much faster than terrestrial animals.
Reason (R): Water holds much less dissolved oxygen than air. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

15. **Assertion (A):** Villi increase the surface area for the absorption of digested food.
Reason (R): Villi are found in the inner lining of the large intestine. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MNEMONIC: BLOOD PATH IN THE HEART

"Right = Rukha (O_2 kam), Left = Laal (O_2 zyada)."
Body → vena cava → RA → RV → lungs → LA → LV → aorta → body.

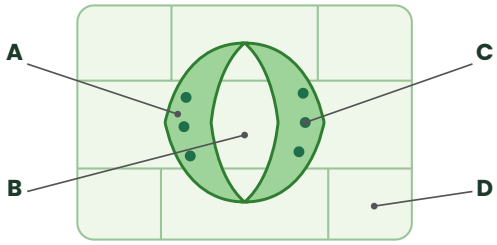
3 SECTIONS B AND C ★★★ MOST ASKED

Section B: Short Answers (2 marks each)

1. Why is diffusion not enough to meet the oxygen needs of multicellular organisms like humans?
2. Name the products of glucose breakdown in (i) yeast, (ii) muscles during heavy exercise. Where does glycolysis occur?
3. State two functions of hydrochloric acid in the stomach.
4. Identify A, B, C and D in the diagram of a stoma below. What makes the pore open?

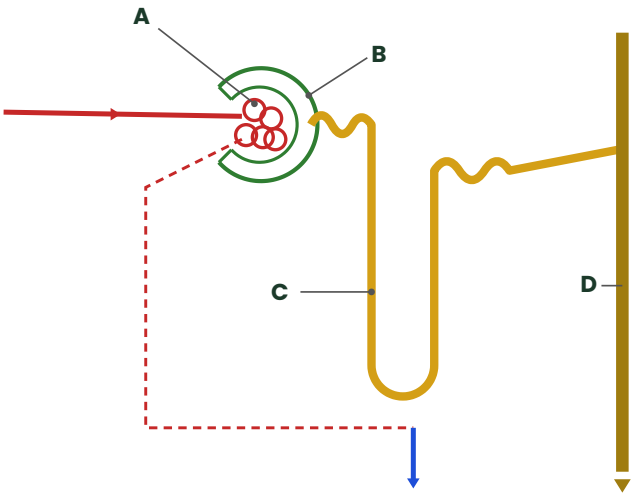
TIP

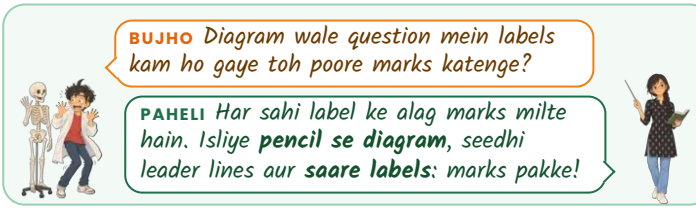
2 marks = **two clear points**; 3 marks = three points or a diagram.



Section C: Short Answers (3 marks each)

1. What is double circulation? Why must oxygenated and deoxygenated blood be kept apart in mammals and birds?
2. What is translocation? How does it take place in plants, and why does it need energy?
3. Write the role of (a) salivary amylase, (b) bile, (c) villi in digestion.
4. Identify parts A, B, C and D of the nephron below and write the function of C.





Bonus rapid-fire round: quick answers ★★☆

Question	Answer
Energy currency of the cell	ATP
Normal blood pressure	120/80 mm Hg
Valves in the heart stop	backflow of blood
Chambers: fish / frog	2 / 3
Lacteals in villi absorb	fat
Filtrate made by kidneys per day	about 180 L
Protein-digesting enzyme of pancreas	trypsin

MCQ TIP

In A/R questions, first decide if **A** is true, then **R**, and only then ask: does R **explain** A?

4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: the variegated leaf (4 marks)

Riya kept a potted plant with variegated (green and white) leaves in the dark for three days. Then she kept it in sunlight for six hours, plucked a leaf, boiled it in alcohol on a water bath, washed it and added iodine solution.

1. Why was the plant kept in the dark for three days? **(1)**
2. Why was the leaf boiled in alcohol? **(1)**
3. Which part of the leaf turned blue-black, and what does the experiment prove? **(2)**

Case 2: the blood pressure check (4 marks)

During a health camp, Mr Rao's blood pressure was 150/100 mm Hg. The doctor said the normal value is about 120/80 mm Hg and asked him to cut salt and walk daily.

1. Name the instrument used to measure blood pressure. **(1)**
2. What do 120 and 80 stand for? **(1)**
3. Name Mr Rao's condition. What causes it, and why is it harmful? **(2)**

Case 3: the limewater test (4 marks)

Aman took two test tubes of fresh limewater. Into tube X he pumped ordinary air with a syringe; into tube Y he blew his exhaled breath through a straw.

1. Which tube turns milky first, and why? **(1)**
2. Name the gas that turns limewater milky. **(1)**
3. Give two differences between aerobic and anaerobic respiration. **(2)**



MEME BREAK

*Might-ochondria? Nahi bhai, **mitochondria** hi powerhouse hai: aerobic respiration ka ATP wahin banta hai!*

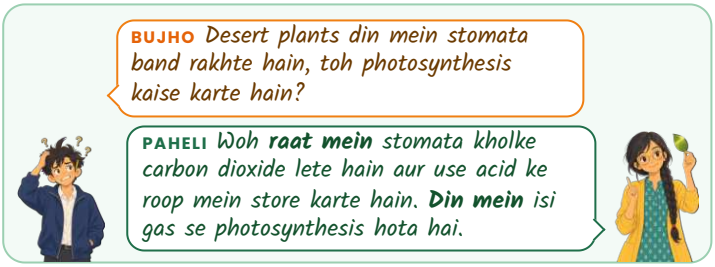
× WRONG	✓ RIGHT
Bile is an enzyme	Bile has no enzyme; it emulsifies fats
Pulmonary vein carries deoxygenated blood	It carries oxygenated blood to the heart
Lactic acid is made in yeast	Yeast makes ethanol; muscles make lactic acid
Phloem moves food only downward	Phloem moves food both ways

Section E: Long Answers (5 marks each)

1. Draw a labelled diagram of the human alimentary canal. Describe the digestion of food in the small intestine.
   OR
   Draw a labelled diagram of the human respiratory system. Explain how air is breathed in and out, and why the lungs always hold some air.
2. Draw a sectional view of the human heart and label: right atrium, left ventricle, aorta, pulmonary artery, vena cava. Trace the path of blood through it.
   OR
   Draw the structure of a nephron. Explain how urine is formed. How does an artificial kidney work?
3. (a) What are the raw materials for photosynthesis? Write the three events of photosynthesis. (b) How do desert plants take up CO_2 without losing much water?

Common mistakes that cost marks

- ✔ Writing "oxygen is a raw material" for photosynthesis; it is a **product**
- ✔ Drawing arrows the wrong way in the heart: blood enters **atria**, leaves from **ventricles**
- ✔ Forgetting **"selective"** in selective reabsorption
- ✔ Leaving labels off a diagram: each label carries marks





★★★ ***Heart, nephron aur digestive system ke diagrams: ek na ek Boss paper mein pakka, Nitesh Bhaiya ki guarantee!***

Most asked in Chapter 9: Eduspark PYQ analysis

Topic	Rating
Human digestive system (diagram), enzymes	★★★
Heart, double circulation (diagram)	★★★
Nephron and urine formation (diagram)	★★★
Breakdown of glucose pathways	★★★
Photosynthesis, stomata	★★☆
Transpiration, translocation	★★☆
Artificial kidney, plant excretion	★☆☆

Diagram drill: draw these today

- ✔ Human alimentary canal with glands
- ✔ Human respiratory system with alveoli
- ✔ Sectional view of the heart and double circulation
- ✔ Human excretory system and a nephron

5 ANSWER KEY ★★★ MOST ASKED

Answer key: Section A (MCQs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(b)	2	(a)	3	(c)	4	(b)	5	(c)
6	(a)	7	(b)	8	(a)	9	(c)	10	(b)
11	(b)	12	(b)	13	(a)	14	(a)	15	(c)

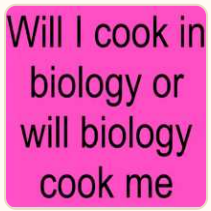
Why: Q10 transpiration removes extra water; Q11 lymph carries fat from the intestine; Q13–14 both true, R explains A; Q15 A true, R false (villi are in the **small** intestine).

Model answers: Sections B and C

- **B1:** large body volume; diffusion is too slow to reach cells deep inside; need a transport system
- **B2:** (i) ethanol + CO_2; (ii) lactic acid; glycolysis in the cytoplasm
- **B3:** makes the medium acidic for pepsin; kills germs in food
- **B4:** A guard cell, B stomatal pore, C chloroplast, D epidermal cell; water entering guard cells makes them swell and open the pore
- **C1:** blood passes twice through the heart in one cycle; separation gives efficient O_2 supply for high energy needs (warm-blooded)
- **C2:** transport of food in phloem; sucrose loaded with ATP, osmotic pressure rises, water enters, food moves to low-pressure areas
- **C3:** (a) starch → sugar; (b) emulsifies fats, makes medium alkaline; (c) increase surface area for absorption
- **C4:** A glomerulus, B Bowman's capsule, C tubule, D collecting duct; C reabsorbs glucose, amino acids, salts, water

Model answers: Sections D and E

- **Case 1:** to remove (destarch) stored starch; alcohol removes chlorophyll; only the **green** part turns blue-black, so chlorophyll is needed for photosynthesis
- **Case 3:** tube Y, exhaled air has more CO_2; carbon dioxide; aerobic uses O_2, gives CO_2 + H_2O and more energy, in mitochondria; anaerobic without O_2, less energy
- **Case 2:** sphygmomanometer; 120 systolic (contraction), 80 diastolic (relaxation); hypertension, caused by constriction of arterioles; may rupture an artery, internal bleeding
- **E1:** small intestine gets bile (emulsifies fats) and pancreatic juice (trypsin, lipase, amylase); intestinal juice completes digestion; villi absorb. OR: diaphragm and rib muscles enlarge the chest, air rushes in; residual volume left for gas exchange
- **E2:** body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta → body. OR: filtration, selective reabsorption, urine; dialysis by diffusion, no reabsorption
- **E3:** CO_2, water, minerals, sunlight, chlorophyll; light absorbed, water split into hydrogen and oxygen, CO_2 reduced to carbohydrate; desert plants take CO_2 at night



MEME BREAK
*Will I cook in biology? Agar aaj ka Boss **50+ / 62** hai, toh haan bhai, tum **cook** kar rahe ho!*

Score card: how did you fight the boss?

	A	B	C	D	E	Total
Marks	15	8	12	12	15	**62**
Mine	__	__	__	__	__	____

Day 46 checklist: tick before you sleep

- [ ] I revised the whole Chapter 9 mind map
- [ ] I solved all 15 MCQs and checked them with the key
- [ ] I drew and labelled the stoma and nephron diagrams
- [ ] I wrote all three case-based answers in full
- [ ] I scored myself and marked my weak topics

My score: ____ / 5

Total 62 marks. 50+ = Boss defeated! Kam aaye? Day 41–45 dobara.



Boss Level tracker: write your percentage

Boss Level	Day	My %
Chapter 9: Life Processes	46	_____
Chapter 10: Control and Coordination	50	_____
Chapter 11: Reproduction	54	_____
Chapter 12: Heredity	57	_____
Chapter 13: Our Environment	60	_____

LEVEL 46 COMPLETE +200 XP earned. Next: Level 47



BUJHO *Boss khatam! Nutrition se excretion tak, poora Chapter 9 clear.*

PAHELI *Shabaash! Kal Level 47: **Control and Coordination**.*

Coming up: Day 47

Control and Coordination begins: **neurons, synapse, reflex arc** and the human brain. Keep your nephron and heart diagrams handy for weekly revision.



NITESH SIR SAYS
*Life Processes board mein sabse zyada marks wala chapter hai. Diagrams roz ek baar banao. **Kal milte hai Day 47 pe!***

BIOLOGY | CHAPTER 9 COMPLETE | LEVEL 46 ASTRA SERIES

CHAPTER 9 SWAHAA!

Life Processes | Days 41–46



BUJHO *Photosynthesis, digestion, respiration, heart aur nephron! Chapter 9 ke saare boss khatam, Swahaa!*

PAHELI *Shabash Bujho! Ab body kaise control hoti hai, woh dekhenge. Brain ready rakhna!*

CHAPTER XP EARNED
+900 XP

LEVELS CLEARED
6 levels (Days 41–46)

BOOK PROGRESS
Level 46 / 60 5900 XP total

WHAT YOU HAVE MASTERED

- ✔ Photosynthesis and autotrophic nutrition
- ✔ Aerobic and anaerobic respiration
- ✔ Xylem, phloem and transpiration
- ✔ Human digestive system and enzymes
- ✔ Heart, blood and double circulation
- ✔ Nephron and excretion

NEXT CHAPTER Day 47: Control and Coordination

Chapter 9 cleared!

DAY 47
LEVEL 47

BIOLOGY | CHAPTER 10 | PART 1 ASTRA SERIES

NERVOUS SYSTEM & THE HUMAN BRAIN

Goal: neurons, reflexes and the brain.



by Nitesh Prakash

01 Receptors | 02 Neuron | 03 Reflex Arc | 04 Human Brain | 05 Limitations | +125 XP today | Level 47 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHY DO WE NEED COORDINATION? ★★★ MOST ASKED

DEFINITION

- **Stimulus:** any change in the surroundings (light, heat, touch, sound, smell) to which an organism responds
- **Coordination:** the working together of different organs in a controlled, systematic way so that the body gives a **proper response** to a stimulus
- Movement in response to a stimulus is controlled; animals use two systems: the **nervous system** and the **endocrine (hormonal) system**

Receptors: the sensors of the body ★★☆

Receptor	Where	Detects
Gustatory	tongue	taste
Olfactory	nose	smell
Photoreceptors	eye	light
Phonoreceptors	inner ear	sound, balance
Thermoreceptors	skin	heat and cold
Touch receptors	skin	touch, pressure, pain

RECEPTOR KYA HAI?

Receptors are **specialised tips of nerve cells** in our sense organs. They detect information from the environment and start the nerve impulse.



BUJHO *Nervous system aur coordination, ek hi cheez hai kya?*

PAHELI *Nahi Bujho! Coordination **kaam** hai, aur nervous system uska ek **tool**. Animals mein doosra tool hai **hormones**, plants mein sirf hormones.*



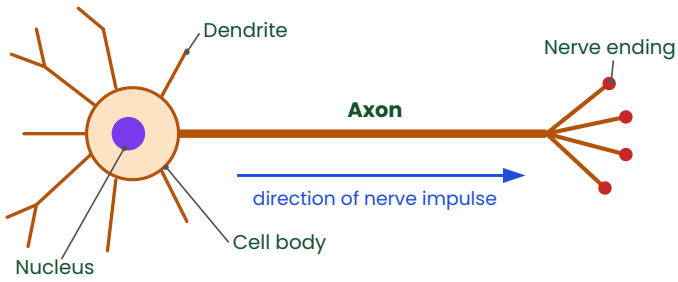


MEME BREAK

*Ek neuron ne signal pakda aur poori chain **activate**! Dendrite → cell body → axon → nerve ending, yahi order.*

2 NEURON: THE UNIT OF THE NERVOUS SYSTEM ★★★

A **neuron (nerve cell)** is the structural and functional unit of the nervous system. It is the **longest cell** of the human body and carries messages as **electrical impulses**.

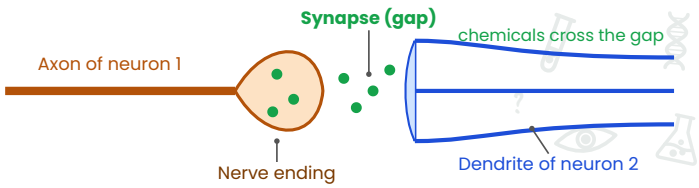


Parts and their jobs ★★★

Part	Job
Dendrite	picks up information from receptors or another neuron
Cell body	has the nucleus; the impulse passes through it
Axon	long fibre that carries the impulse away from the cell body
Nerve ending	releases chemicals that pass the message on

How a message travels (NCERT) ★★★

1. Information is picked up at the **dendrite tip**, which sets off a **chemical reaction** creating an **electrical impulse**.
2. The impulse travels: **dendrite → cell body → axon → nerve ending**.
3. At the nerve ending the electrical impulse makes the neuron release **chemicals**; they cross the gap (**synapse**) and start a new impulse in the dendrite of the next neuron.
4. A similar synapse passes the impulse from a neuron to a **muscle cell or gland**.



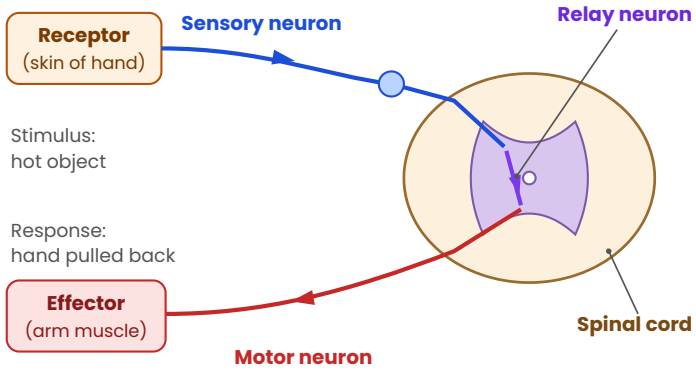
3 REFLEX ACTION AND REFLEX ARC

★★★ MOST ASKED

DEFINITION

- **Reflex action:** a sudden, **automatic** and involuntary response to a stimulus, done **without thinking** (without the conscious control of the brain)
- **Reflex arc:** the pathway along which the nerve impulse travels during a reflex action

Examples: pulling the hand back from a hot object, blinking when an object comes near the eye, sneezing, coughing, the knee-jerk, the pupil shrinking in bright light.



Why reflex arcs are in the spinal cord ★★★

- ✔ Thinking in the brain is **too slow** for an emergency; the spinal cord gives a **quick** response.
- ✔ Input nerves meet output nerves in a bundle in the spinal cord, so the reply is sent straight back.
- ✔ The information **also reaches the brain**, which becomes aware of the event a moment later.
- ✔ Reflex arcs evolved early; they work well even in animals with little or no thinking power.

MNEMONIC: REFLEX ARC IN ORDER

"Ram Se Raavan Mara Ekdum"
Receptor → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector



MEME BREAK
*Reflex ka seedha rasta: receptor se spinal cord, wahan se muscle. Dimaag ko **news baad mein** milti hai.*

★★★ Neuron ka diagram aur synapse ka kaam: 2–3 marks pakke, Nitesh Bhaiya ki guarantee!

Voluntary vs involuntary vs reflex ★★☆

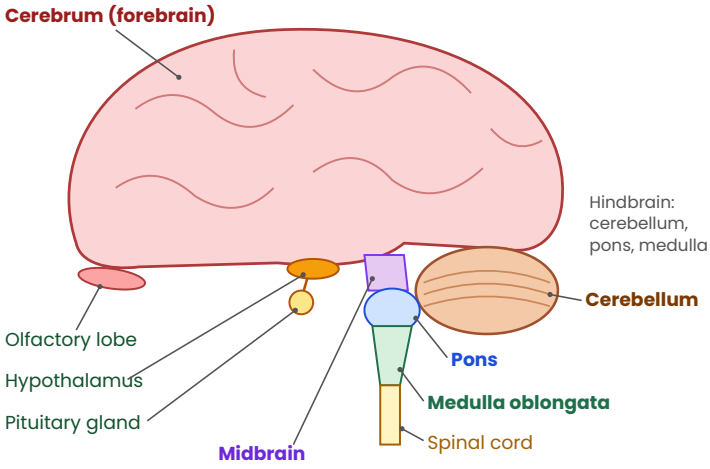
Action	Control	Examples
Voluntary	by our will; forebrain (cerebrum)	writing, talking, walking
Involuntary	automatic; midbrain and hindbrain (medulla)	heartbeat, breathing, salivation
Reflex	sudden and very fast; mostly the spinal cord	blinking, hand off a hot plate



MEME BREAK
*Haath garam tawe se hata, dimaag ko pata baad mein chala. Ye kaam **spinal cord** ka hai, aliens ka nahi!*

4 HUMAN BRAIN ★★★ MOST ASKED

The **brain** and the **spinal cord** form the **central nervous system (CNS)**. The brain is the main coordinating centre; it has three regions: **forebrain, midbrain** and **hindbrain**.



★★★ Brain ka labelled diagram aur parts ke kaam, ye Nitesh Bhaiya ki guarantee wala question hai!

Brain + spinal cord + nerves

- **CNS** = brain + spinal cord: they take decisions and send orders.
- **PNS (peripheral nervous system)** = **cranial nerves** (from the brain) + **spinal nerves** (from the spinal cord); they link the CNS with all body parts.
- **Spinal cord:** carries messages to and from the brain and handles most reflex actions.

★★★ Reflex arc ka labelled diagram har saal pucha jata hai, Nitesh Bhaiya ki guarantee!

Parts of the brain and their functions ★★★

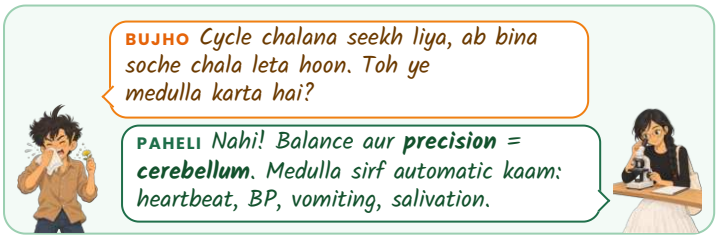
Region	Part	Main functions
Forebrain	Cerebrum	thinking, memory, intelligence; voluntary actions; areas for **hearing, smell, sight**
	Olfactory lobe	sense of smell
	Hypothalamus	hunger, thirst, body temperature, sleep; controls the **pituitary gland**
Midbrain		reflex movements of the head, neck and eyes
Hindbrain	Cerebellum	precision of voluntary actions; **posture and balance** (walking straight, riding a cycle, picking up a pencil)
	Pons	regulates breathing
	Medulla	involuntary actions: **blood pressure, heartbeat, salivation, vomiting**, swallowing

MNEMONIC: HINDBRAIN TEAM

Cycle, Pranayam, Machine
Cerebellum = cycle balance, Pons = breathing,
Medulla = automatic machine (BP, heartbeat, vomiting)

How the brain and spinal cord are protected ★★☆

- ✔ The brain sits inside a bony box, the **cranium (skull)**.
- ✔ Inside the box it floats in a fluid-filled balloon, the **cerebrospinal fluid (CSF)**, which acts as a **shock absorber**.
- ✔ The spinal cord is protected by the **vertebral column** (backbone).



EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Many axons are wrapped in a fatty **myelin sheath** that makes the impulse travel faster. A common chemical released at a synapse is **acetylcholine**.
Humans have **12 pairs** of cranial nerves and **31 pairs** of spinal nerves.

✖ WRONG	✔ RIGHT
Reflexes are controlled by the brain	Mostly by the spinal cord; the brain is informed later
Cerebellum controls heartbeat	Medulla controls heartbeat and BP; cerebellum gives balance
Impulse flows axon → dendrite in a neuron	Dendrite → cell body → axon → nerve ending

5 HOW NERVOUS TISSUE CAUSES ACTION ★★☆

1. The nerve impulse reaches a **muscle** through a nerve ending (synapse).
2. Muscle cells have special **proteins** that change their **shape and arrangement** in response to the impulse.
3. This makes the muscle cell **shorter (contract)**, and the body part moves.

Voluntary muscles (arms, legs) move by our will; **involuntary muscles** (heart, gut) work on their own.

6 LIMITATIONS OF NERVE IMPULSES ★★☆

- ✔ They reach only those cells that are **connected by nervous tissue**, not every cell of the body.
- ✔ After making an impulse, a cell needs time to **reset**; it cannot send impulses continuously.
- ✔ So the body also uses **chemical communication (hormones)**, which reach every cell through the blood; plants, with no nervous system, depend only on chemicals.



MEME BREAK

*Exam hall mein wo ek **braincell** bhi impulse ke baad **reset** leta hai. Isliye body ko hormones bhi chahiye hote!*

Exam tips

- ✔ In the reflex arc, draw **arrows**: towards the spinal cord on the sensory neuron, away from it on the motor neuron.
- ✔ Write "**relay neuron**" inside the spinal cord; examiners look for it.
- ✔ Brain diagram: label all three regions plus cerebellum, pons, medulla and hypothalamus.

7 PRACTICE MCQS

1. Which part of a neuron picks up information from the environment? ★★★

(a) axon (b) dendrite
(c) nerve ending (d) myelin

2. The gap between the nerve ending of one neuron and the dendrite of the next is called: ★★★

(a) axon (b) synapse
(c) impulse (d) receptor

3. Which part of the brain maintains posture and balance of the body? ★★★

(a) cerebrum (b) medulla
(c) cerebellum (d) pons

4. Involuntary actions like blood pressure, salivation and vomiting are controlled by the: ★★★

(a) medulla in the hindbrain (b) cerebrum
(c) hypothalamus (d) midbrain

5. In a reflex arc, which comes immediately after the sensory neuron? ★★★

(a) receptor
(b) relay neuron in the spinal cord
(c) motor neuron
(d) effector

6. Olfactory receptors detect: ★★☆

(a) taste
(b) light
(c) smell
(d) sound

7. The brain is protected by a fluid-filled balloon that acts as a shock absorber. This fluid is: ★★☆

(a) lymph
(b) cerebrospinal fluid
(c) plasma
(d) bile

8. **Assertion (A):** Pulling the hand away from a hot object is a reflex action.
Reason (R): The response is generated by the spinal cord, without waiting for a decision from the thinking part of the brain. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** A nerve impulse can reach every cell of the body.
Reason (R): Nerve impulses can reach only those cells that are connected by nervous tissue. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. What is a synapse? How is a message passed across it? **(2 marks)**
2. Name the parts of the brain that control (a) posture and balance, (b) blood pressure, (c) thinking. **(3 marks)**
3. Draw a neat labelled diagram of a neuron. Trace the path of a nerve impulse through it. **(3 marks)**
4. What is a reflex action? Draw the reflex arc for pulling the hand away from a hot object, and explain why such reflexes are controlled by the spinal cord. **(5 marks)**

Model answers

1 Synapse
- Gap between the nerve ending of one neuron and the dendrite of the next (1)
- Electrical impulse at the nerve ending releases chemicals; they cross the gap and start a new impulse (1)

2 Brain parts
- (a) cerebellum (b) medulla (c) cerebrum, 1 mark each

3 Neuron
- Diagram with dendrite, cell body, nucleus, axon, nerve ending (2)
- Dendrite → cell body → axon → nerve ending (1)

4 Reflex arc
- Definition (1); labelled arc: receptor, sensory neuron, spinal cord (relay), motor neuron, effector (2)
- Quick response, no wait for the brain; brain still informed (2)

Answer key: MCQs

Q	Ans	Why
1	(b)	Dendrite tips pick up information.
2	(b)	Synapse: chemicals carry the signal across.
3	(c)	Cerebellum: posture, balance, precision.
4	(a)	Medulla controls involuntary actions.
5	(b)	R S R M E: relay neuron comes next.
6	(c)	Olfactory = smell; gustatory = taste.
7	(b)	CSF cushions the brain in the cranium.
8	(a)	Both true; R explains the speed of A.
9	(d)	A false; R true (only connected cells).

Day 47 checklist: tick before you sleep

- ☐ I can define stimulus, receptor and coordination
- ☐ I can draw and label a neuron and trace the impulse
- ☐ I can draw the reflex arc and explain it
- ☐ I can label the brain and give each part's job
- ☐ I solved all 9 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 47 COMPLETE +125 XP earned. Next: Level 48



BUJHO *Samajh gaya! Neuron ka order, reflex arc ka RSRME, aur hindbrain ka Cycle, Pranayam, Machine.*

PAHELI *Shabaash! Kal Level 48: plants mein coordination aur hormones.*

Day 47 hit-list: Eduspark PYQ analysis

Topic	Rating
Reflex arc: diagram and explanation	★★★
Neuron: labelled diagram	★★★
Brain parts and functions	★★★
Synapse and limitations of nerve impulses	★★☆



NITESH SIR SAYS
*Diagram hi marks dilata hai: **neuron, reflex arc aur brain** roz ek baar bana ke dekho. **Kal milte hai Day 48 pe!***

DAY 48
LEVEL 48

BIOLOGY | CHAPTER 10 | PART 2 ASTRA SERIES

COORDINATION IN PLANTS

Goal: plant movements, tropisms and plant hormones.



by Nitesh Prakash

01 Mimosa | 02 Tendrils | 03 Tropisms | 04 Auxin | 05 Hormones

+125 XP today | Level 48 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

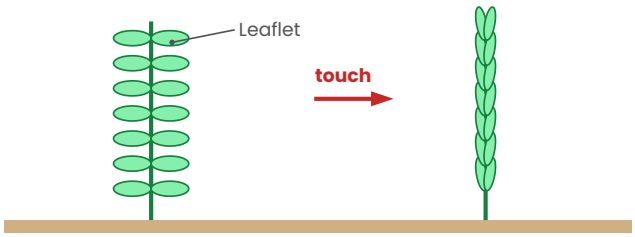
1 COORDINATION IN PLANTS ★★☆

Plants have **no nervous system and no muscles**, yet they respond to light, gravity, water, touch and chemicals. They do it in two ways:

Two kinds of plant movement ★★★

Type	Growth involved?	Example
Immediate response to a stimulus	no, independent of growth	leaves of the touch-me-not (Mimosa) fold
Movement due to growth	yes, slow and directional	tendrils coil; shoots bend to light

2 TOUCH-ME-NOT: QUICK RESPONSE ★★★ MOST ASKED



Before touch: leaflets open — After touch: leaflets fold

How Mimosa folds its leaves ★★★

1. Touch is sensed at one point, but the movement happens **at a different point**, so information must travel.
2. Plant cells pass this information using **electrical-chemical means**, from cell to cell (no special nerve tissue).
3. Cells at the leaf base **change shape** by changing the amount of **water** in them: they swell or shrink.
4. So the leaflets fold quickly. **No growth** is involved.

KITNI SPEED?

Mimosa folds in **seconds** (no growth). Tropisms take **hours to days** because they need growth.



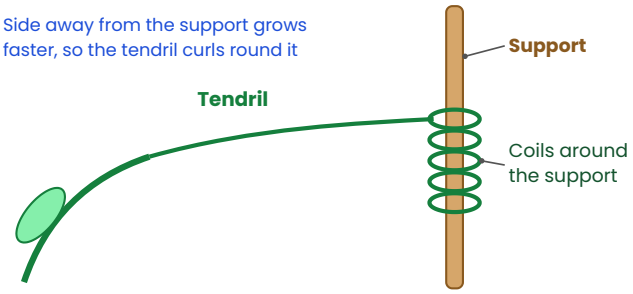
★★★ *Mimosa (touch-me-not) vs tendril ka difference har saal, Nitesh Bhaiya ki guarantee!*

BUJHO *Plant ke paas muscles hi nahi, toh Mimosa ki pattiyan hilti kaise hain?*

PAHELI *Paani ka khel hai Bujho! Cells mein **paani kam-zyada** hota hai, cells ka **shape badalta** hai, aur patti band. Muscle protein wala system animals ka hai.*

3 MOVEMENT DUE TO GROWTH ★★☆

Some plants (pea, sweet pea, grapevine) climb with **tendrils**. Tendrils are sensitive to **touch**; when they touch a support they **grow around it**.



WHY THE TENDRIL CURLS

The part of the tendril **in contact** with the support grows **slowly**; the part **away** from it grows **faster**. This uneven growth makes the tendril coil round the support.

DEFINITION

- **Tropism (tropic movement):** a slow, **directional** growth movement of a plant part in response to a stimulus
- Growth **towards** the stimulus = **positive** tropism; **away** from it = **negative** tropism



MEME BREAK

*Tendril ka scene: jab tak support **touch** nahi hota, bas wait. Touch hua, toh **growth** se lipat gaya.*

4 TROPISMS: DIRECTIONAL GROWTH

★★★ MOST ASKED

The four tropisms of the syllabus ★★★

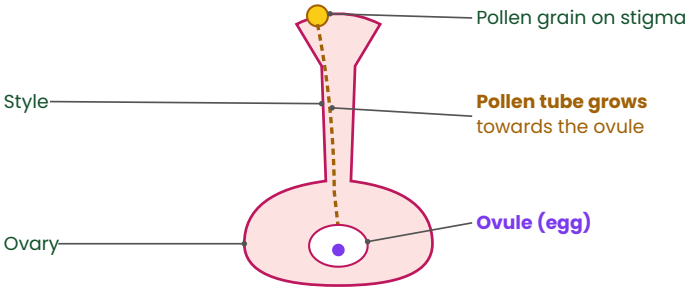
Tropism	Stimulus	Shoot	Root
Phototropism	light	+ bends towards light	− bends away
Geotropism	gravity	− grows upward	+ grows downward
Hydrotropism	water		+ grows towards water
Chemotropism	chemicals	pollen tube grows towards the ovule	

MNEMONIC: FOUR TROPISMS

"Pappu Ghar Hamesha Chalta"
Photo = light, Geo = gravity, Hydro = water, Chemo = chemical

YAAD RAKHNE KI TRICK: SHOOT VS ROOT

Shoot suraj ki taraf, zameen se door.
Root zameen aur paani ki taraf, roshni se door.



Chemotropism: after pollination, the **pollen tube** grows down the style towards the **ovule**, attracted by chemicals; it carries the male gamete to the egg.

Exam tips

- ✔ Always say whether a tropism is **positive or negative**, and for which part (shoot or root).
- ✔ Mimosa is **not** a tropism: it is an immediate response with no growth.
- ✔ Write "auxin is made at the **shoot tip**" in every phototropism answer.



★★★ *Phototropism aur geotropism ke examples har paper mein, Nitesh Bhaiya ki guarantee!*



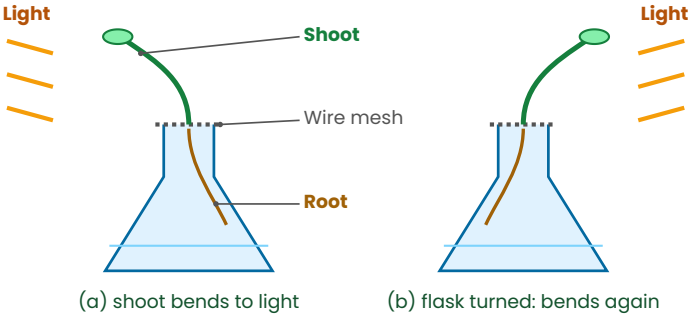
BUJHO *Plant ki toh aankh hi nahi, use pata kaise chalta hai roshni kis taraf hai?*

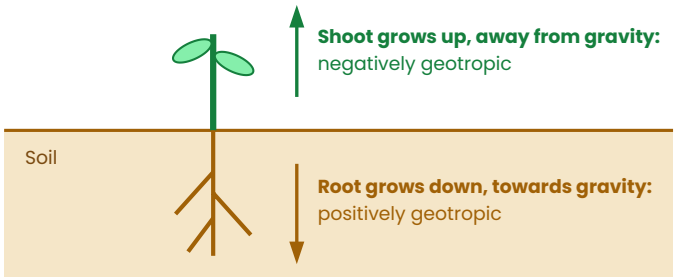


PAHELI *Shoot ki **tip** light sense karti hai. Wahan bana **auxin** chhaon wali side jaata hai, wo side zyada badhti hai, aur shoot roshni ki taraf jhukta hai.*

NCERT Activity: plant in a flask ★★☆

1. Fill a conical flask with water, cover its mouth with a **wire mesh** and place two or three freshly germinated bean seeds on it.
2. Keep it in a cardboard box **open on one side**, facing light from a window.
3. After 2–3 days: **shoots bend towards the light**, **roots bend away**.
4. Turn the flask so the shoots face away from the light: in a few days the old parts stay put, but the **new parts bend towards light again**.





Shoot when it finally finds the sunlight from the window



MEME BREAK

*Shoot ko roshni mil gayi, ab bas usi taraf jhukna hai **(positively phototropic)**. Root ko roshni pasand hi nahi.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Growth in response to touch (as in tendrils) is called **thigmotropism**; the folding of Mimosa leaves is a **nastic** (non-directional) movement. Ethylene is a gaseous hormone that helps fruits ripen.

SOCHO ZARA

A pot kept on its side for a few days: the shoot **turns up** (negative geotropism) and the root **turns down** (positive geotropism). Both are slow growth movements.

5 PLANT HORMONES ★★★ MOST ASKED

DEFINITION

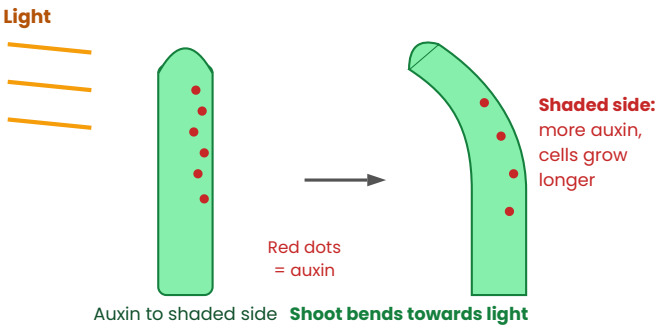
> **Plant hormones (phytohormones)** are chemical substances that coordinate **growth, development** and **responses** to the environment; they are made at one place and **diffuse** to the area of action

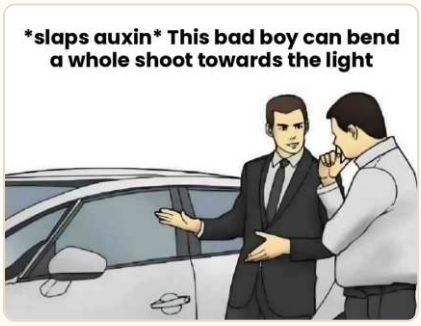
Animals vs plants: passing information

	Animals	Plants
Special tissue	nervous tissue and muscles	none
Signals	electrical impulses + hormones	electrical-chemical signals + hormones
Movement	muscle proteins change shape	cells change water content, or growth

Auxin: how a shoot bends to light ★★★

1. **Auxin** is made at the **tip of the shoot**.
2. When light comes from one side, auxin **diffuses towards the shaded side** of the shoot.
3. Auxin makes cells **grow longer**, so cells on the shaded side elongate more.
4. The shaded side grows faster than the lit side, so the shoot **bends towards light**.





MEME BREAK

*Auxin: shoot tip pe bana, **shady side** pe gaya, cells lambe kiye, aur shoot roshni ki taraf mud gaya.*

AUXIN KE AUR KAAM

Auxin also helps **tendrils** coil (the side away from the support grows faster) and promotes growth at the shoot tip.



★★★ *Auxin se phototropism wala question har saal, Nitesh Bhaiya ki guarantee!*

Plant hormones and their jobs ★★★

Hormone	Main job	Example
Auxin	cell elongation (growth)	shoot bends to light; tendril coiling
Gibberellins	growth of the **stem**	stem elongation
Cytokinins	promote **cell division**	high in fruits and seeds
Abscisic acid	**inhibits** growth	wilting of leaves; stress hormone

MNEMONIC: 3 PROMOTERS, 1 STOPPER

Auxin = Aage badho (lambe cells)
Gibberellin = Girafe jaisa tall stem
Cytokinin = Cell division
Abscisic acid = Absolutely ruko



MEME BREAK

*Teen hormones growth badhate, ek **(abscisic acid)** rokta. Perfectly balanced, jaise plant ko chahiye.*

BUJHO *Hydrotropism aur hydro... matlab root paani dekh ke bhagti hai?*

PAHELI *Bhaagti nahi, **badhti** hai! Tropism = growth movement. Root dheere dheere paani ki taraf **grow** karti hai, isliye ye slow hota hai.*

6 PRACTICE MCQS

1. The folding of leaves of the touch-me-not plant on touching is due to: ★★★

(a) growth of cells
(b) change in the water content of cells
(c) action of muscles
(d) auxin

2. The growth of a pollen tube towards the ovule is an example of: ★★★

(a) hydrotropism
(b) chemotropism
(c) geotropism
(d) phototropism

3. Roots growing downwards into the soil show: ★★★

(a) positive geotropism
(b) negative geotropism
(c) positive phototropism
(d) negative hydrotropism

4. Which plant hormone promotes cell division? ★★★

(a) auxin
(b) gibberellin
(c) cytokinin
(d) abscisic acid

5. Which hormone inhibits growth and causes wilting of leaves? ★★☆

(a) auxin
(b) abscisic acid
(c) cytokinin
(d) gibberellin

6. When light falls on a shoot from one side, auxin: ★★★

(a) moves to the lit side (b) is destroyed
(c) moves to the shaded side (d) moves to the root

7. A tendril coils around a support because: ★★☆

(a) the side touching the support grows faster (b) the side away from the support grows faster
(c) cells lose water (d) of gravity

8. **Assertion (A):** The shoot of a plant bends towards light.
Reason (R): Auxin collects on the shaded side and makes cells there grow longer. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** Movement of Mimosa leaves is a growth movement.
Reason (R): Mimosa leaflets fold because their cells change shape by changing water content. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. How do the leaves of a sensitive plant move on touch? How is this different from the movement of a shoot towards light? **(3 marks)**
2. Name the plant hormone that (a) promotes cell division, (b) inhibits growth, (c) helps the stem grow. **(3 marks)**
3. Describe an activity to show that the shoot is positively phototropic and the root negatively phototropic. Draw a labelled diagram. **(5 marks)**

Model answers

1 Mimosa vs shoot

- Mimosa: no growth; cells change shape by losing/gaining water; quick (1.5)
- Shoot: growth movement; auxin on shaded side; slow and directional (1.5)

2 Hormones

- (a) cytokinin (b) abscisic acid (c) gibberellin, 1 mark each

3 Flask activity

- Steps: flask, wire mesh, germinated seeds, box open on one side (2)
- Result: shoot to light, root away; diagram (3)

× WRONG	✓ RIGHT
Mimosa leaves move by growth	Immediate response; cells change water content, no growth
Auxin collects on the lit side	On the shaded side, which grows faster
Abscisic acid promotes growth	It inhibits growth; the other three promote it

Answer key: MCQs

Q	Ans	Why
1	(b)	Water moves out, cells change shape.
2	(b)	Chemicals attract the pollen tube.
3	(a)	Root goes towards gravity.
4	(c)	Cytokinins: cell division.
5	(b)	ABA: growth inhibitor, wilting.
6	(c)	Auxin diffuses to the shade.
7	(b)	Away side grows faster: coiling.
8	(a)	Both true; R explains A.
9	(d)	A false (no growth); R true.

Day 48 checklist: tick before you sleep

- ☐ I can explain how Mimosa leaves fold without growth
- ☐ I can explain tendril coiling and define tropism
- ☐ I can name four tropisms with examples
- ☐ I can explain auxin and phototropism with a diagram
- ☐ I solved all 9 MCQs and checked them with the key

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–2 phir se padho.



LEVEL 48 COMPLETE +125 XP earned. Next: Level 49



BUJHO Samajh gaya! Pappu Ghar Hamesha Chalta, aur 3 promoters, 1 stopper.

PAHELI Shabaash! Kal Level 49: animals mein hormones.

Day 48 hit-list: Eduspark PYQ analysis

Topic	Rating
Auxin and phototropism (diagram)	★★★
Tropisms with examples	★★★
Plant hormones and functions	★★★
Mimosa vs tendril movement	★★☆



NITESH SIR SAYS

*Plants chup chaap bhi kitna kuch karte hain! **Tropism ka diagram** aur hormones ki table aaj hi yaad karo. **Kal milte hai Day 49 pe!***

DAY 49 LEVEL 49

BIOLOGY | CHAPTER 10 | PART 3 ASTRA SERIES

HORMONES IN ANIMALS & FEEDBACK

Goal: name each endocrine gland, its hormone and its job.



by Nitesh Prakash

01 Adrenaline | 02 Endocrine Glands | 03 Thyroid, GH | 04 Insulin | 05 Feedback | +125 XP today | Level 49 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 HORMONES: THE CHEMICAL MESSENGERS ★★★

DEFINITION

- **Hormone:** a chemical messenger secreted by an **endocrine gland** directly into the **blood**, which carries it to target cells far away
- **Endocrine glands** are **ductless**: they pour their secretion straight into the blood

Why animals need hormones too ★★★

- Nerve impulses reach only cells **connected by nerves**, and a nerve cell needs time to reset before the next impulse
- Many changes must happen in **all cells at once** and **last long**: growth, puberty, sugar balance

Exocrine vs endocrine glands ★★☆

Point	Exocrine gland	Endocrine gland
Duct	has a duct	**ductless**
Secretion goes to	a surface or cavity	**blood** directly
Examples	salivary, sweat, gastric glands	thyroid, pituitary, adrenal

NCERT Q: How does chemical coordination take place in animals? ★★★

- Endocrine glands release hormones into the **blood** in response to a stimulus
- Blood carries them all over the body; only **target cells** with matching receptors respond

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

The adrenal gland has two parts: an outer **cortex** (hormones like cortisol and aldosterone) and an inner **medulla**, which makes adrenaline and noradrenaline.

BUJHO *Hormone bhi nerve jaisa wire se jaata hai na, seedha target tak?*



PAHELI *Nahi Bujho! Hormone ka koi wire nahi. Gland use seedha **blood** mein chhod deta hai, blood poore body mein le jaata hai, aur sirf target cells react karte hain.*



2 ADRENALINE: FIGHT OR FLIGHT ★★★ MOST ASKED

Emergency hormone of the adrenal gland ★★★

Scary moment (a dog chasing you, a sudden exam surprise): the **adrenal glands** release **adrenaline** straight into the blood. It reaches the heart, muscles and other organs and gets the body ready to **fight or run**.

Organ	What adrenaline does	Why it helps
Heart	beats **faster**	more oxygen to muscles
Breathing	rate **rises** (diaphragm, rib muscles contract)	more oxygen in
Digestive system, skin	blood supply **cut down** (small arteries constrict)	blood diverted to skeletal muscles

Flow: from fear to action

- **Stimulus:** sudden danger is sensed by the senses
- **Adrenal glands** pour adrenaline into the blood
- **Response:** faster heartbeat and breathing, more blood to skeletal muscles



MEME BREAK

*Adrenal = **ad** (near) + **renal** (kidney). Gland kidney ke upar topi jaisa baitha hai, aur emergency mein poori body ko alert kar deta hai!*

MNEMONIC: ADRENALINE KE 3 TEZ

"Dil tez, saans tez, khoon muscles ki taraf tez." Pet aur skin ka khoon kam, kyunki bhaagte waqt khaana digest karna zaroori nahi!

COMMON MISTAKE

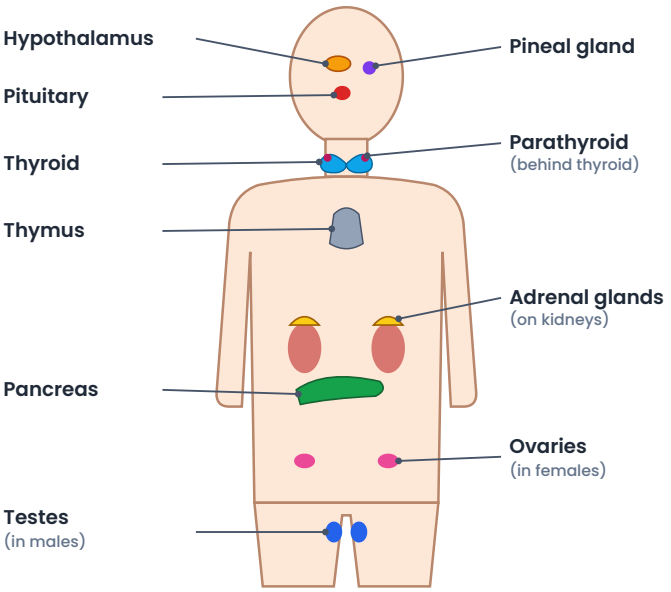
Adrenaline does **not** travel along nerves to the heart. It is carried by the **blood**, and that is why it can act on many organs at the same time.



★★★ *Adrenaline ke effects (heart, breathing, blood flow) har saal pooche jaate hain, Nitesh Bhaiya ki guarantee!*

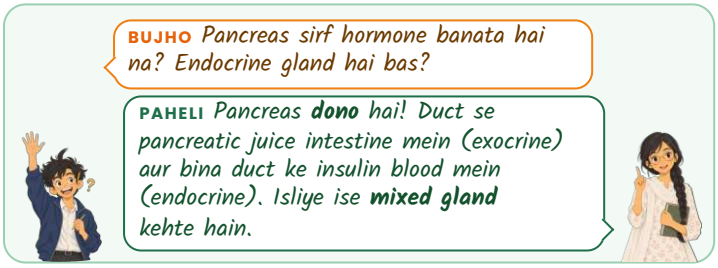
3 ENDOCRINE SYSTEM OF THE HUMAN BODY

★★★ MOST ASKED



MNEMONIC: GLANDS FROM HEAD TO TOE

"Har Pal Pyaar, Thoda Pyaar, Thoda Adjust, Phir Tension Over." Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Thymus, Adrenal, Pancreas, Testes/Ovaries.



Key features of hormones

- ✔ Secreted in **tiny amounts**, yet very powerful
- ✔ Act **slowly** but the effect **lasts longer**
- ✔ Both **too little** and **too much** cause disorders
- ✔ Timing and amount are controlled by **feedback mechanisms**

5 THYROID, IODINE AND GOITRE ★★★ MOST ASKED

Why iodised salt is advised (NCERT) ★★★

- The thyroid gland needs **iodine** to make **thyroxin**
- Thyroxin controls the metabolism of **carbohydrates, proteins and fats** for balanced growth
- Too little iodine → too little thyroxin → **goitre**: the thyroid swells and the neck looks swollen
- Iodised salt supplies enough iodine in daily food

MNEMONIC: KAM HUA TOH KYA HUA

"Iodine kam, gala jam (goitre). GH kam, height kam (dwarfism). Insulin kam, sugar ka gham (diabetes)."

Note: the four small **parathyroid** glands sit on the back of the thyroid; write **thyroxin** (NCERT spelling).

4 GLAND, HORMONE AND FUNCTION

★★★ MOST ASKED

The table you must learn line by line ★★★

Gland	Hormone	Main function
Hypothalamus	releasing hormones	tell the pituitary to release its hormones (e.g. growth hormone releasing factor)
Pituitary	growth hormone	growth and development of the body
Pineal	melatonin	sleep–wake cycle
Thyroid	thyroxin	metabolism of carbohydrates, proteins and fats
Parathyroid	parathormone	calcium level in blood
Thymus	thymosin	development of immunity in children
Adrenal	adrenaline	prepares body for emergency
Pancreas	insulin	lowers blood sugar level
Testes	testosterone	male puberty changes
Ovaries	oestrogen	female puberty changes, menstrual cycle



MEME BREAK

*Yahi lab scientist wala feeling aata hai jab table ki 10 lines ek baar mein yaad ho jaati hain. Gland, hormone, kaam: **teen column, poore marks!***

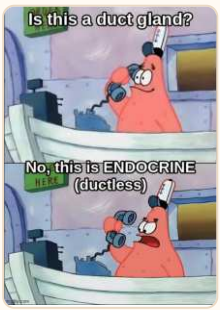


★★★ *Gland–hormone matching ka MCQ ya table har paper mein milta hai, Nitesh Bhaiya ki guarantee!*

6 GROWTH HORMONE: HEIGHT CONTROL ★★☆

Growth hormone from the pituitary ★★☆

Condition	In childhood	Result
Normal	right amount	normal height
Dwarfism	**too little** growth hormone	very short stature
Gigantism	**too much** growth hormone	very tall stature



MEME BREAK

*Endocrine gland ka koi duct nahi hota, phone seedha **blood** pe lagta hai. Duct wala gland exocrine hai, yaad rakho!*

7 HORMONES OF PUBERTY ★★☆

Testosterone and oestrogen ★★☆

Point	Testosterone	Oestrogen
Made by	**testes** (males)	**ovaries** (females)
Starts at	puberty	puberty
Main role	development of male sex organs and male features	development of female sex organs and the menstrual cycle
Visible changes	deeper voice, facial hair, more muscle	breast development, broader hips

MNEMONIC: T-T AND O-O

"T se Testes, T se Testosterone; O se Ovary, O se Oestrogen."
Gland aur hormone ka pehla akshar same!

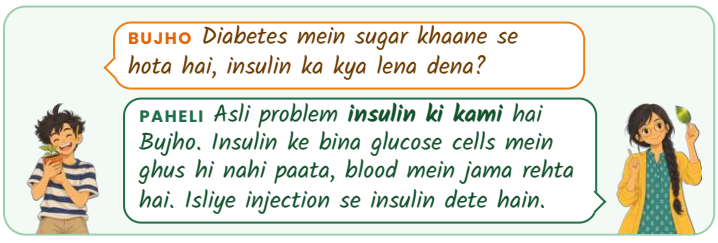
8 INSULIN AND DIABETES ★★★ MOST ASKED

Insulin keeps blood sugar in check ★★★

- **Insulin** is secreted by the **pancreas** (by special groups of cells in it)
- It helps cells take up glucose and **lowers** the blood sugar level
- Too little insulin → sugar level stays **high** → **diabetes**, which harms many organs

NCERT Q: Why are some diabetes patients given insulin injections? ★★★

- Their pancreas does not secrete **enough insulin**, so blood sugar remains high
- Injected insulin does the missing job and brings blood sugar **down to normal**
- It is injected because insulin is a protein and would be **digested** if taken by mouth (extra point)



EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

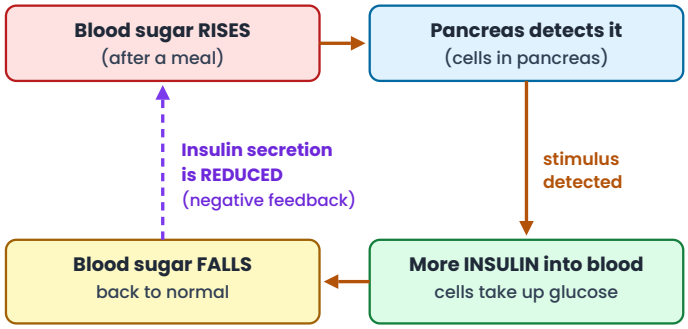
The pancreas also makes **glucagon**, which **raises** blood sugar when it falls too low. Insulin and glucagon work as a pair to keep sugar steady.

Common mistakes that cost marks

- ✔ Writing that **thyroxin** controls blood sugar (it is **insulin**)
- ✔ Calling the pituitary hormone "growth enzyme": it is **growth hormone**
- ✔ Saying hormones travel through **nerves** (they travel in **blood**)
- ✔ Mixing up goitre (iodine) with diabetes (insulin)

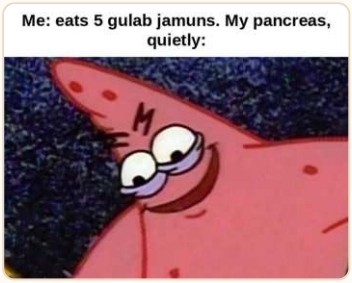
Remember: puberty changes start slowly because they are run by hormones carried in the blood, not by nerves.

9 FEEDBACK MECHANISM: BLOOD SUGAR ★★★ MOST ASKED



How feedback controls the amount ★★★

- Hormones must be released in the **right amount** at the **right time**
- Sugar rises → pancreas detects it → **more insulin** → sugar falls
- Sugar falls → insulin secretion is **reduced**: the result itself switches the gland down
- This self-checking loop is called a **feedback mechanism**



MEME BREAK

*Gulab jamun gaya, sugar badhi, aur pancreas ne chupke se **insulin** chhod diya. Feedback system kabhi chhutti nahi leta!*

10 NERVOUS VS HORMONAL COORDINATION ★★★ MOST ASKED

Compare: NCERT exercise favourite ★★★

Point	Nervous	Hormonal
Message	electrical impulse (chemical at synapse)	chemical (hormone)
Carried by	nerve fibres	blood
Speed	very fast	slow
Effect	short-lived	long-lasting
Reaches	only cells linked by nerves	all cells; target cells respond
Example	reflex action	growth, puberty

MNEMONIC: NERVE VS HORMONE

"Nerve = WhatsApp call (fast, ek bande tak). Hormone = newspaper (slow, sabke ghar, der tak asar)."

Exam tips

- ✔ In a comparison answer, write **at least four points** in a two-column table
- ✔ Draw the feedback loop with arrows: easy marks

11 PRACTICE MCQS

1. Which gland secretes growth hormone? (NCERT) ★★★
(a) adrenal (b) thyroid
(c) pituitary (d) pancreas

2. Iodine, whose lack causes goitre, is needed to make: ★★★
(a) insulin (b) thyroxin
(c) adrenaline (d) testosterone

3. Which of these is NOT an effect of adrenaline? ★★★
(a) faster heartbeat (b) faster breathing
(c) more blood to the digestive system (d) more blood to skeletal muscles

4. Endocrine glands release their secretions: ★★☆
(a) into ducts (b) directly into the blood
(c) into the stomach (d) on the skin

5. Excess growth hormone in childhood causes: ★★☆
(a) dwarfism (b) goitre
(c) gigantism (d) diabetes

6. The gland that acts as both an exocrine and an endocrine gland is: ★★☆
(a) thyroid (b) pancreas
(c) pituitary (d) adrenal

7. Which hormone regulates the calcium level in blood? ★☆☆
(a) thyroxin (b) parathormone
(c) thymosin (d) melatonin

8. Less insulin is secreted once blood sugar falls. This is an example of: ★★☆
(a) reflex action (b) feedback mechanism
(c) tropism (d) osmosis

9. **Assertion (A):** Hormonal coordination is slower than nervous coordination.
Reason (R): Hormones are carried to target cells by the blood. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Iodised salt is advised in our diet.
Reason (R): Iodine is needed by the pancreas to make insulin. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Day 49 checklist: tick before you sleep

- [ ] I can name all ten endocrine glands and show them on a diagram
- [ ] I can match each gland with its hormone and function
- [ ] I can explain adrenaline, goitre, dwarfism and gigantism
- [ ] I can draw the blood sugar feedback loop
- [ ] I can compare nervous and hormonal coordination

My score: ____ / 5

Short and long answer questions

1. How does our body respond when adrenaline is secreted into the blood? **(2 marks)**
2. Why are some patients of diabetes treated by giving injections of insulin? **(2 marks)**
3. Draw a diagram of the human endocrine system and label the thyroid, adrenal gland, pancreas and pituitary. Write one hormone of each. **(5 marks)**
4. Compare nervous and hormonal mechanisms of control and coordination in animals (any four points). **(3 marks)**

Answer key: MCQs

Q	Ans	Why
1	(c)	Pituitary makes growth hormone.
2	(b)	Thyroid needs iodine for thyroxin.
3	(c)	Blood to digestive system is reduced.
4	(b)	Ductless: straight into blood.
5	(c)	Excess GH in childhood: gigantism.
6	(b)	Juice via duct; insulin into blood.
7	(b)	Parathormone controls calcium.
8	(b)	The result switches the gland down.
9	(a)	Blood transport is slow; R explains A.
10	(c)	A true; R false: thyroid makes thyroxin.

Model answers (marking points)

1 Adrenaline response
- Heart beats faster, so more oxygen reaches muscles (1)
- Breathing rate rises; blood to digestive system and skin falls, more to skeletal muscles (1)

2 Insulin injections
- Pancreas secretes too little insulin, blood sugar stays high (1)
- Injected insulin lowers blood sugar to normal (1)

3 Endocrine diagram
- Correct positions and labels (3); pituitary: growth hormone, thyroid: thyroxin, adrenal: adrenaline, pancreas: insulin (2)

LEVEL 49 COMPLETE +125 XP earned. Next: Level 50



BUJHO *Samajh gaya! Gland seedha blood mein hormone chhodta hai, aur feedback amount control karta hai.*

PAHELI *Shabaash! Kal Level 50: Chapter 10 ka Boss Level.*



NITESH SIR SAYS
Gland, hormone, kaam: ye teen column wali table roz ek baar likho. Diagram mein glands ki position pakki karo. ***Kal Boss Level, taiyaar rehna!***

DAY 50
LEVEL 50

BIOLOGY | CHAPTER 10 | BOSS LEVEL

CONTROL AND COORDINATION BOSS

Goal: finish the chapter test and score 80% or more.



by Nitesh Prakash

01 MCQs + A/R | 02 Short Answers | 03 Diagrams | 04 Case-based | 05 Long Answers | +200 XP Boss Level | Level 50 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 10 REVISION MAP ★★★ MOST ASKED

Chapter 10 mind map: revise in 5 minutes ★★★

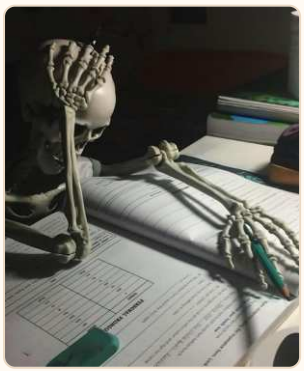
Branch	Key points
Nervous system	receptors → sensory neuron → CNS → motor neuron → effector; neuron: dendrite, cell body, axon, nerve ending; synapse is chemical
Reflex arc	quick, automatic; controlled by **spinal cord**; brain informed later
Brain	forebrain (thinking, senses, hypothalamus), midbrain, hindbrain (cerebellum: balance; pons and medulla: involuntary actions)
Plants	no nerves; touch (Mimosa, water changes), tropisms (photo, geo, hydro, chemo); hormones auxin, gibberellin, cytokinin, abscisic acid
Animal hormones	adrenaline, thyroxin, GH, insulin, testosterone, oestrogen; feedback control

How to attack the Boss: 90 minutes

- Section A (12 MCQs incl. A/R): 15 minutes, no blanks
- Sections B and C: draw neat labelled diagrams, they carry marks
- Case-based: read the passage once, then answer each part
- Keep 10 minutes to check arrows in the reflex arc and labels

MNEMONIC: BRAIN KE TEEN BOSS

"Cerebrum = CEO (soch, yaad, senses). Cerebellum = balance coach. Medulla = autopilot (BP, saans, ulti, laar)."



MEME BREAK

*Boss test se pehle ki raat: skeleton bhi padh raha hai! Tension mat lo, mind map dekho aur **diagrams practise karo**.*



★★★ ***Reflex arc ka diagram aur brain ke parts ke functions: Chapter 10 ke sabse pakke sawaal, Nitesh Bhaiya ki guarantee!***

2 SECTION A: MCQS (1 MARK EACH) ★★★ MOST ASKED

1. The gap between two neurons is called a: ★★★
(a) dendrite (b) synapse
(c) axon (d) impulse

2. Posture and balance of the body are maintained by the: ★★★
(a) cerebrum (b) medulla
(c) cerebellum (d) pons

3. Blood pressure, salivation and vomiting are controlled by the: ★★★
(a) medulla (b) cerebellum
(c) hypothalamus (d) cerebrum

4. The plant hormone that promotes cell division is: ★★★
(a) auxin (b) gibberellin
(c) cytokinin (d) abscisic acid

5. Growth of a pollen tube towards the ovule is an example of: ★★☆
(a) phototropism (b) geotropism
(c) hydrotropism (d) chemotropism

6. Leaves of the touch-me-not plant fold on touch because: ★★☆
(a) of growth at the base (b) cells change shape by losing water
(c) auxin is destroyed (d) of gravity

7. The plant hormone that inhibits growth and causes wilting of leaves is: ★★☆
(a) abscisic acid (b) auxin
(c) cytokinin (d) gibberellin

8. Deficiency of which hormone causes diabetes? ★★★
(a) thyroxin (b) adrenaline
(c) insulin (d) growth hormone

9. Which gland is called the master gland? ★★☆
(a) thyroid (b) pituitary
(c) adrenal (d) pancreas



BUJHO *Reflex mein brain ka koi role nahi?*

PAHELI *Hota hai, par baad mein! Turant action **spinal cord** leti hai, dard ka pata brain ko baad mein chalta hai.*



10. In a reflex arc, the correct sequence is: ★★★

(a) receptor → motor neuron → spinal cord → sensory neuron → muscle
(b) receptor → sensory neuron → spinal cord → motor neuron → muscle
(c) muscle → motor neuron → brain → receptor
(d) receptor → brain → sensory neuron → muscle

11. **Assertion (A):** A plant shoot kept near a window bends towards the light.
Reason (R): Auxin diffuses to the shaded side of the shoot and makes cells there grow longer. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

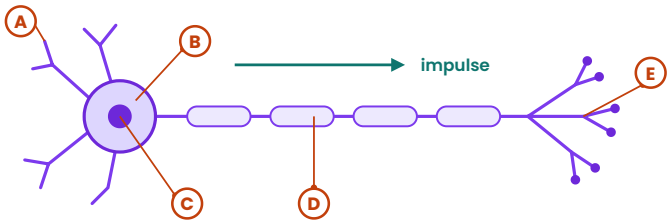
12. **Assertion (A):** The brain is protected by a fluid-filled balloon inside the cranium.
Reason (R): The fluid (cerebrospinal fluid) absorbs shocks. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

3 SECTION B: SHORT ANSWERS (2 MARKS EACH)

★★★ MOST ASKED

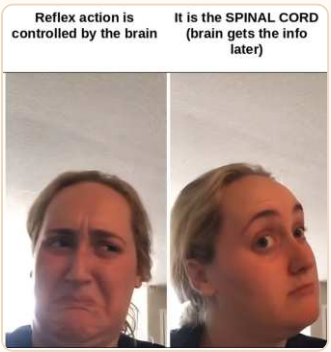
B1. Diagram-based: the neuron ★★★



1. Identify the parts marked A to E.
2. In which direction does the impulse travel: from A to E or from E to A?

B2 to B4 ★★☆

1. What is a synapse? How does an impulse cross it? **(2)**
2. How does the tendril of a pea plant coil around a support? **(2)**
3. Why is the use of iodised salt advised? **(2)**



MEME BREAK
*Reflex action ka control **spinal cord** mein hota hai. Brain ko khabar baad mein milti hai, isliye dard thoda late lagta hai!*

4 SECTION C: SHORT ANSWERS (3 MARKS EACH)

★★★ MOST ASKED

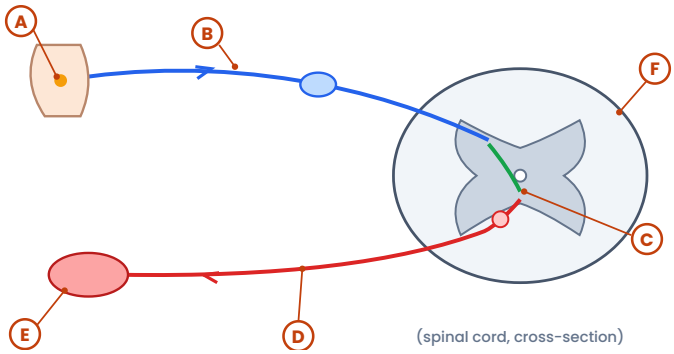
C1 to C3 ★★★

1. What are plant hormones? Name one plant hormone each that (a) promotes cell division, (b) inhibits growth, (c) promotes stem elongation. **(3)**
2. Distinguish between the movement in touch-me-not leaves and the movement of a shoot towards light (any three points). **(3)**
3. How does our body respond when adrenaline is secreted into the blood? Why is it called an emergency hormone? **(3)**

5 SECTION D: CASE-BASED (4 MARKS EACH)

★★★ MOST ASKED

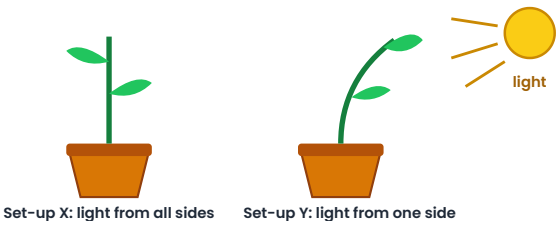
Case 1: the hot pan ★★★



Rohan touched a hot pan and pulled his hand back before he even felt the pain.

1. Identify the parts A, C and E. **(1)**
2. Name the type of neuron marked B and D. **(1)**
3. Where is this action controlled, and why is the brain not involved first? **(2)**

Case 2: the window plant ★★★



Two potted plants were kept for a week: X in a room with light from all sides, Y near a window.

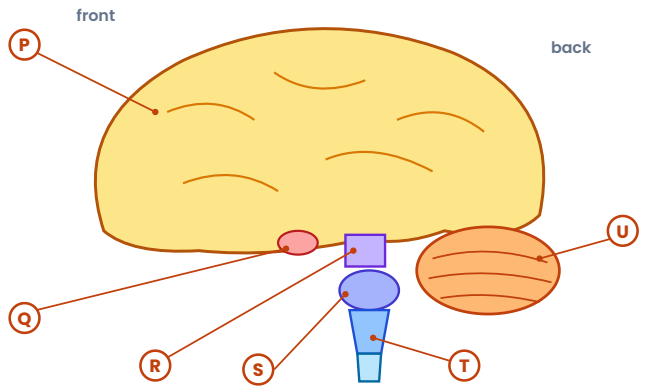
1. Name the response shown by the shoot of Y. **(1)**
2. Which hormone causes it, and where is it made? **(1)**
3. Explain how this hormone makes the shoot bend. How would the roots respond? **(2)**

DIAGRAM MARKS
Leader lines must touch the part; always put **arrows** on a reflex arc.

6 SECTION E: LONG ANSWERS (5 MARKS EACH)

★★★ MOST ASKED

E1. Diagram-based: the human brain ★★★



1. Identify the parts marked P to U. **(3)**
2. Write one function each of P, U and S. **(2)**

E2. Hormones in animals ★★★

1. Name the endocrine glands that secrete (a) thyroxin, (b) insulin, (c) growth hormone, (d) adrenaline. **(2)**
2. What is goitre? How can it be prevented? **(1)**
3. Explain the feedback mechanism with the example of blood sugar. **(2)**



MEME BREAK

*Arrows ki direction sahi = poore marks. Receptor se muscle tak, beech mein **spinal cord**. Gigachad energy!*

Model answers: Section B

B1 Neuron

- A dendrite, B cell body, C nucleus, D axon, E nerve ending (½ each, max 1½)
- Impulse travels from A (dendrite) to E (nerve ending) (½)

B2 Synapse

- Gap between the nerve ending of one neuron and the dendrite of the next (1)
- Electrical impulse releases chemicals at the nerve ending; they cross the gap and start a new impulse (1)

B3 Tendril

- Cells on the side away from the support grow faster than those in contact (1)
- So the tendril bends and coils around the support; this growth is controlled by auxin (1)

B4 Iodised salt

- Iodine is needed to make thyroxin (1); its lack causes goitre (1)

Common mistakes that cost marks

- ✔ Writing that the brain controls a reflex action (it is the **spinal cord**)
- ✔ Drawing reflex arc arrows from muscle to receptor
- ✔ Calling cerebellum the centre of thinking (that is the **cerebrum**)
- ✔ Saying plants have nerves: they use **hormones** and water movement

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	Junction between neurons.
2	(c)	Cerebellum: posture and balance.
3	(a)	Medulla controls involuntary actions.
4	(c)	Cytokinins promote cell division.
5	(d)	Pollen tube grows towards a chemical.
6	(b)	Cells lose water and change shape.
7	(a)	ABA inhibits growth, wilting of leaves.
8	(c)	Too little insulin: high blood sugar.
9	(b)	Pituitary controls other glands.
10	(b)	Receptor → sensory → cord → motor → muscle.
11	(a)	Auxin on shaded side; R explains A.
12	(a)	CSF cushions the brain; R explains A.



MEME BREAK

*Thyroxin = metabolism, **insulin** = blood sugar. Ye mix kiya toh fan bhi disappoint ho jayega!*

Model answers: Section C

C1 Plant hormones

- Chemicals that coordinate growth and responses in plants (1)
- (a) cytokinin (b) abscisic acid (c) gibberellin or auxin (⅔ each)

C2 Mimosa vs phototropism

- Mimosa: not growth related, quick, cells change shape by water movement
- Phototropism: growth related, slow, auxin makes shaded side grow more (1 each point)

C3 Adrenaline

- Heart beats faster, breathing faster, blood diverted to skeletal muscles (2)
- Prepares the body to fight or run in danger, so emergency hormone (1)



★★★ Neuron, reflex arc aur brain: teen diagrams pakke, toh Chapter 10 mein poore marks, Nitesh Bhaiya ki guarantee!

Model answers: Section D

1 The hot pan

- A receptor in skin, C relay neuron in spinal cord, E effector muscle (1)
- B sensory neuron, D motor neuron (1)
- Controlled by the spinal cord; going to the brain first would take too long, so the arc gives a quick response that protects the body (2)

2 The window plant

- Positive phototropism (1)
- Auxin, made at the shoot tip (1)
- Auxin moves to the shaded side, cells there grow longer, so the shoot bends towards light; roots bend away from light (2)

Score card: how did you fight the boss?

Score (out of 12 in A)	Verdict
11 to 12	Boss defeated, legend level
8 to 10	Strong, revise the slips
below 8	Re-read Days 47 to 49, then retry

Bonus rapid-fire round: answer in one breath

Question	Answer
Longest part of a neuron	axon
Thinking part of the brain	cerebrum (forebrain)
Centre for hunger and body temperature	hypothalamus
Bony box protecting the brain	cranium
Plant hormone for stem elongation	gibberellin
Root growth towards water	hydrotropism
Hormone needing iodine	thyroxin
Gland on top of kidneys	adrenal

Boss Level tracker: write your percentage

Section A ____ / 12 Sections B to E ____ / 50 Total ____ %

Exam strategy for Chapter 10

- ✔ Start long answers with a neat labelled diagram, then write the points
- ✔ Use the words **receptor, sensory neuron, relay neuron, motor neuron, effector** in every reflex answer
- ✔ For plant answers, always name the hormone: **auxin** for bending, **ABA** for wilting

Day 50 checklist: tick before you sleep

- ☐ I attempted all 12 MCQs and checked them with the key
- ☐ I can label the neuron, reflex arc and brain diagrams
- ☐ I can explain phototropism with the role of auxin
- ☐ I can match glands with hormones and explain feedback
- ☐ I marked my weak topics to revise from Days 47 to 49

My score: ____ / 5

Model answers: Section E

E1 Brain

- P cerebrum, Q hypothalamus, R midbrain, S pons, T medulla, U cerebellum (½ each)
- P: thinking, memory, senses; U: balance and posture; S: involuntary actions such as regulating breathing (with medulla) (2)

E2 Hormones

- (a) thyroid (b) pancreas (c) pituitary (d) adrenal (½ each)
- Goitre: swollen thyroid due to iodine deficiency; prevented by iodised salt (1)
- Sugar rises → pancreas releases more insulin → sugar falls → insulin secretion reduced (2)

Most asked in Chapter 10: Eduspark PYQ analysis

Topic	Rating
Reflex arc (diagram and sequence)	★★★
Parts of the brain and their functions	★★★
Neuron diagram	★★★
Phototropism and auxin	★★★
Gland–hormone–function, goitre, diabetes	★★★
Plant hormones table	★★☆
Nervous vs hormonal comparison	★★☆

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Nerve impulses in some human nerves travel at up to about 100 metres per second, which is why a reflex is over before you even think about it.

LEVEL 50 COMPLETE +200 XP earned. Next: Level 51



BUJHO *Boss khatam! Neuron, reflex arc, brain, plant hormones, animal hormones: sab clear.*

PAHELI *Shabaash! Chapter 10 complete. Kal se Chapter 11: reproduction.*



NITESH SIR SAYS

Is chapter mein diagram hi asli hathiyar hai: neuron, reflex arc, brain, aur phototropism ka set-up. Har din ek baar bina dekhe banao. ***Boss defeated, ab agla level!***

Coming up: Day 51

- Chapter 11 starts: **How do Organisms Reproduce?**
- DNA copying, variation, binary and multiple fission, budding, regeneration

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter.

BIOLOGY | CHAPTER 10 COMPLETE | LEVEL 50

CHAPTER 10 SWAHAA!

Control and Coordination | Days 47–50



BUJHO *Neuron, reflex arc, brain, auxin aur insulin, sab clear! Mera braincell aaj full activate hai!*

PAHELI *Bilkul! Ab dekhte hain organisms apne jaise naye organisms kaise banate hain.*

CHAPTER XP EARNED
+575 XP

LEVELS CLEARED
4 levels (Days 47–50)

BOOK PROGRESS
Level 50 / 60 6475 XP total

WHAT YOU HAVE MASTERED

- ✔ Neuron, synapse and reflex arc
- ✔ Tropic movements in plants
- ✔ Endocrine glands and hormones
- ✔ Human brain and its parts
- ✔ Plant hormones
- ✔ Feedback mechanism

NEXT CHAPTER Day 51: How do Organisms Reproduce?

Chapter 10 cleared!

DAY 51
LEVEL 51

BIOLOGY | CHAPTER 11 | PART 1

ASEXUAL REPRODUCTION & VARIATION

Goal: see how one parent makes many, and why copies vary.



by Nitesh Prakash

01 DNA Copying | 02 Variation | 03 Fission | 04 Regeneration | 05 Budding

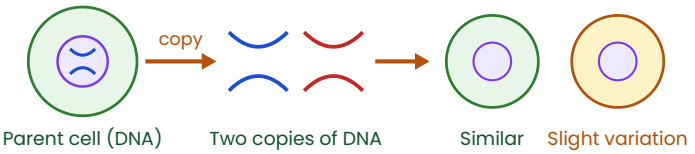
+125 XP today | Level 51 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DO ORGANISMS CREATE EXACT COPIES? ★★★

DEFINITION

- **Reproduction:** the process by which living organisms produce new individuals of **their own kind**; it is not needed for an individual to stay alive, but it is needed for the **species** to continue
- **DNA** (deoxyribonucleic acid) in the chromosomes of the nucleus carries the **information** to make proteins; different proteins give different body designs
- The basic event in reproduction is the **creation of a DNA copy**, along with extra cellular apparatus, so that one cell becomes two



Why copies are never 100% exact ★★☆

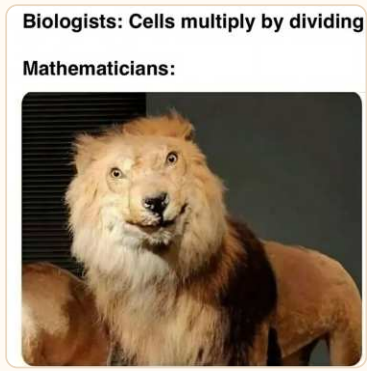
- Cells copy DNA using **biochemical reactions**, which are never absolutely accurate
- So each DNA copy is **similar** to the original but may have small changes: this is **variation**
- Too big a change: the new cell may not survive; small changes are the raw material of variation
- Sexual reproduction adds even more variation (Day 52)

Importance of variation ★★★

- **Consistency of DNA copying** keeps the body design suited to the organism's **niche** (its place in the ecosystem)
- **Variation** helps a species survive when the niche changes suddenly
- Example: if water temperature rises, most bacteria die; a few **heat-resistant variants** survive and keep the species alive
- So variation is useful for the **survival of the species** over time, not for the individual



★★★ *"Why is variation useful for the species but not always for the individual?"* **Nitesh Bhaiya ki guarantee: ye 3 marks ka sawaal pakka aata hai!**



MEME BREAK

Single cell ka family planning: ***divide karo aur multiply ho jao.*** *Binary fission mein 1 se 2, phir 4, phir 8!*



BUJHO *Toh Amoeba ki beti bilkul photocopy hoti hai na? Xerox machine jaisi?*

PAHELI *Almost! DNA copying ke reactions* ***100% accurate nahi*** *hote, isliye har copy mein thoda sa* ***variation*** *aa jaata hai. Wahi species ko bachata hai.*

2 MODES OF REPRODUCTION ★★☆

Asexual vs sexual: first look

Feature	Asexual	Sexual
Parents	**one**	usually **two**
Gametes	not formed	formed and fuse
Offspring	nearly identical copies	show more variation
Speed	fast, many offspring	slower

MNEMONIC

"Fatafat Fission, Fir Fragment, Re-bano, Bud nikalo, Veg bhi, Spore bhi"

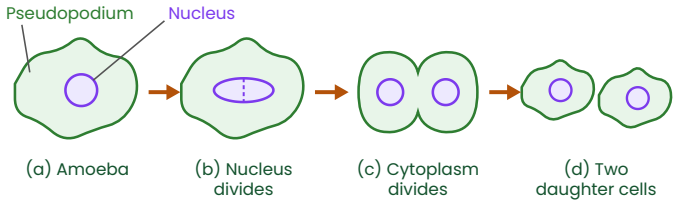
Six asexual modes in order: **F**ission, **F**ragmentation, **R**egeneration, **B**udding, **V**egetative propagation, **S**pore formation (last two on Day 52).

3 FISSION IN UNICELLULAR ORGANISMS ★★★

DEFINITION

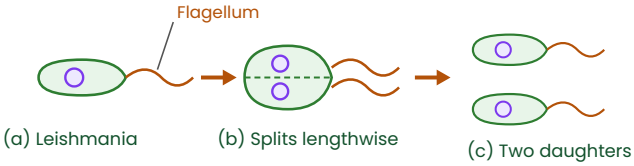
- **Fission:** a single-celled organism splits into two or more **new cells**; the parent cell disappears into its daughters
- **Binary fission:** division into **two** equal halves, e.g. **Amoeba**, bacteria, **Leishmania**, Paramecium

Binary fission in Amoeba ★★★



1. The **nucleus divides** first (DNA copied and shared)
2. The **cytoplasm** then constricts in the middle
3. Two **daughter Amoebae** are formed; Amoeba can split in **any plane**

Binary fission in Leishmania ★★☆



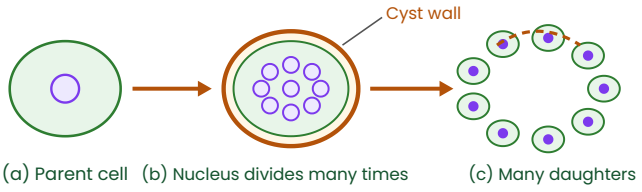
- Leishmania causes **kala-azar**; it has a whip-like **flagellum** at one end
- It splits in a **definite orientation** (lengthwise) with respect to this structure, unlike Amoeba



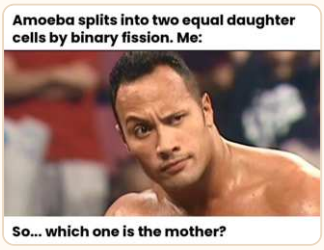
BUJHO *Amoeba aur Leishmania dono binary fission karte hain, toh fark kya hai?*

PAHELI *Amoeba **kisi bhi plane** mein divide ho sakta hai. Leishmania ek **definite plane** (lengthwise, flagellum ke respect mein) mein hi divide hota hai.*

Multiple fission in Plasmodium ★★★



- **Multiple fission:** one cell divides into **many** daughter cells at the same time
- Plasmodium (the **malaria** parasite) forms many daughter cells; the nucleus divides many times first
- In many organisms a **cyst** (protective wall) forms in unfavourable conditions; it bursts later to release the daughters



MEME BREAK *Binary fission mein parent hi gayab ho jaata hai, toh "mother" kaun? **Dono daughters barabar** hain, bhai.*

YAAD RAKHNE KI TRICK

"AL Binary, PM Multiple"

Amoeba and **L**eishmania do **binary** fission; **P**lasmodium does **M**ultiple fission. Like AL and PM on a clock!

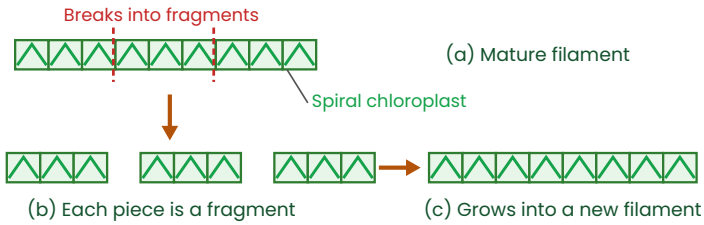
EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

Under ideal conditions some bacteria (e.g. *E. coli*) can divide by binary fission about every **20 minutes**, so one cell can become millions in a few hours. This is why food spoils fast in warm weather.

4 FRAGMENTATION ★★☆

DEFINITION

- **Fragmentation:** a simple multicellular organism breaks into smaller pieces (**fragments**) on maturing; each fragment grows into a new individual, e.g. **Spirogyra**



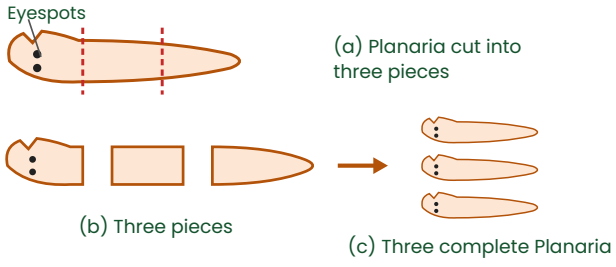
WHY NOT ALL MULTICELLULAR ORGANISMS?

Complex organisms have **specialised** tissues and organs; a random piece cannot rebuild the whole body. They need **specific reproductive cells**, made in reproductive organs.

5 REGENERATION ★★★

DEFINITION

- **Regeneration:** the ability of a fully differentiated organism to give rise to new individuals from its **body parts**, e.g. **Planaria**, **Hydra**



How regeneration happens ★★☆

- It is carried out by **specialised cells**
- These cells **proliferate** (make large numbers of cells)
- The mass of cells then **differentiates** into different cell types and tissues, in an organised sequence (development)

REGENERATION IS NOT REPRODUCTION

Most organisms do not normally **depend on being cut** into pieces to reproduce, so regeneration is not the same as reproduction.

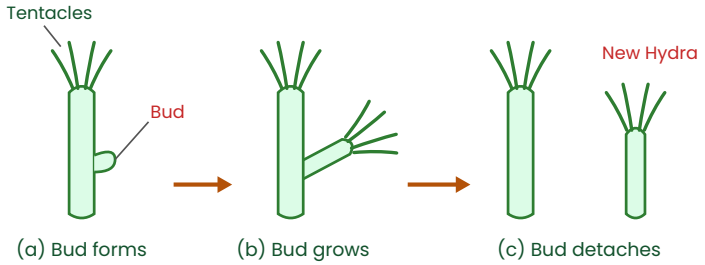


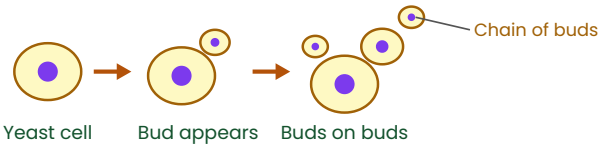
MEME BREAK *Planaria ko kaatoge toh **teen Planaria** mil jayenge. Gussa karna bekaar hai, ye toh regeneration ka master hai!*

6 BUDDING ★★★

DEFINITION

> **Budding:** a **bud** (outgrowth) develops on the parent due to repeated cell division at **one specific site**; it grows into a tiny individual and **detaches** on maturing, e.g. **Hydra**, **yeast**





Budding: key points ★★☆

- Hydra uses **regenerative cells** for budding
- In yeast the buds may stay attached and form a **chain**
- The bud gets a full DNA copy, so it is like the parent



BUJHO *Budding aur regeneration mein confusion hai, dono mein naya Hydra banta hai!*

PAHELI *Budding mein **bud apne aap** ek jagah ugta hai. Regeneration mein **kate hue tukde** se poora organism banta hai.*

MNEMONIC

"HY = Hi Bud! aur PH = Phir se ban jao"

Hydra and **Y**east say **Hi** to their **Bud**; **P**lanaria and **H**ydra **regenerate** (phir se ban jaate). Hydra appears in both!

7 ASEXUAL MODES AT A GLANCE ★★★

Mode	How it happens	Example
Binary fission	one cell splits into two	Amoeba, Leishmania
Multiple fission	one cell splits into many	Plasmodium
Fragmentation	body breaks into pieces	Spirogyra
Regeneration	body part forms a whole organism	Planaria, Hydra
Budding	outgrowth detaches	Hydra, yeast

NCERT in-text questions: model answers

1. **Importance of DNA copying?** It passes the body design to the next generation, and small copying errors give variation.
2. **Binary vs multiple fission?** Binary: one cell gives two (Amoeba). Multiple: one cell gives many at once (Plasmodium).
3. **Why can complex organisms not regenerate into new individuals?** Their bodies are highly specialised; most cells cannot divide and rebuild every organ.
4. **Is Hydra reproducing when it buds?** Yes: budding is asexual reproduction in Hydra.

× WRONG	✔ RIGHT
Amoeba divides in a fixed plane	Amoeba can divide in any plane; Leishmania has a definite plane
Regeneration is the same as reproduction	Organisms do not depend on being cut to reproduce
Hydra reproduces only by regeneration	Hydra shows both budding and regeneration
Variation helps every individual	Variation helps the species survive changes

Exam tips

- ✔ In diagram answers of binary fission, show **all stages** with the nucleus dividing first.
- ✔ Always name an **example** with each mode; examples carry marks.
- ✔ Write "Plasmodium: multiple fission", not binary fission.

8 PRACTICE MCQS

1. The basic event in reproduction is the: ★★★

(a) formation of gametes (b) creation of a DNA copy
(c) division of the cytoplasm (d) formation of a bud

2. Leishmania reproduces by: ★★★

(a) multiple fission (b) budding
(c) binary fission (d) fragmentation

3. Multiple fission is seen in: ★★★

(a) Amoeba (b) Plasmodium
(c) Hydra (d) yeast

4. Spirogyra reproduces asexually by: ★★☆
(a) budding (b) regeneration
(c) fragmentation (d) spore formation

5. Which organism shows both budding and regeneration? ★★☆
(a) Planaria (b) Hydra
(c) Spirogyra (d) Plasmodium

6. Variation is useful for: ★★☆
(a) the individual only (b) the survival of the species
(c) faster cell division (d) making exact copies

7. In yeast, buds that stay attached form a: ★☆☆
(a) cyst (b) filament
(c) chain (d) sporangium

8. Regeneration is carried out by: ★★☆
(a) gametes (b) specialised cells
(c) the cyst wall (d) the flagellum

9. **Assertion (A):** Fragmentation is not seen in complex multicellular organisms.
Reason (R): Complex organisms are made of specialised cells, tissues and organs. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Variation in a population helps the species survive.
Reason (R): DNA copying in reproduction is never absolutely accurate. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. **(2 marks)** Distinguish between binary fission in Amoeba and in Leishmania.
2. **(3 marks)** Why is variation beneficial to the species but not necessarily to the individual?
3. **(3 marks, diagram)** Draw labelled diagrams of budding in Hydra. Can Hydra also regenerate?

Answer key

MCQs: 1 (b), 2 (c), 3 (b), 4 (c), 5 (b), 6 (b), 7 (c), 8 (b), 9 (a), 10 (b)

1 Amoeba vs Leishmania
> Amoeba: splits in **any plane**; Leishmania: **definite** lengthwise plane, with respect to its flagellum (1 + 1)

2 Variation
> Copying errors give variation; when the niche changes (heat), some variants survive and the **species** continues
> An individual with a big change may not survive (1 + 1 + 1)

3 Hydra
> Stages: bud forms, grows tentacles, detaches (2); yes, Hydra regenerates from its body parts (1)



NITESH SIR SAYS
*Har mode ke saath **ek example aur ek diagram** yaad karo, number pakke. **Kal milte hai Day 52 pe!***

Today's checklist: tick before you sleep

- ☐ I can explain why DNA copying leads to variation
- ☐ I can draw binary fission in Amoeba and Leishmania
- ☐ I know multiple fission in Plasmodium
- ☐ I can explain fragmentation and regeneration
- ☐ I can draw budding in Hydra and yeast

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 1–3 phir se padho.



LEVEL 51 COMPLETE +125 XP earned. Next: Level 52



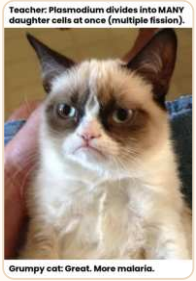
BUJHO *Samajh gaya! Fission, fragmentation, regeneration, budding: sab ek parent se.*

PAHELI *Shabaash! Kal Level 52: plants mein reproduction aur flower ka magic.*

Day 51 hit-list: Eduspark PYQ analysis

Topic	Rating
Binary fission in Amoeba (diagram)	★★★
Budding in Hydra (diagram)	★★★
Importance of variation	★★★
Regeneration in Planaria	★★☆
Multiple fission, fragmentation	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.



MEME BREAK
*Plasmodium ek cell se **bahut saare** bana deta hai. Isliye mosquito net lagao, grumpy cat ki tarah gussa mat karo!*

DAY 52 LEVEL 52

BIOLOGY | CHAPTER 11 | PART 2 ASTRA SERIES

PLANT REPRODUCTION & THE FLOWER

Goal: learn how plants multiply and how a flower makes seeds.



by Nitesh Prakash

01 Vegetative | 02 Tissue Culture | 03 Spores | 04 Flower LS | 05 Fertilisation | +125 XP today | Level 52 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

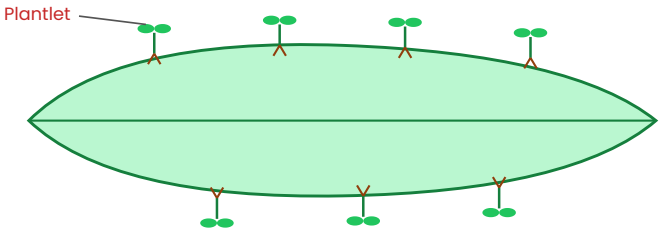
1 VEGETATIVE PROPAGATION ★★★

DEFINITION

- **Vegetative propagation:** new plants develop from **vegetative parts** of the parent (root, stem, leaf) under suitable conditions, without seeds
- It is a mode of **asexual** reproduction; the new plants are genetically like the parent

Natural methods (NCERT examples) ★★★

Plant	Part used	How
Bryophyllum	leaf	buds in the leaf-margin notches fall on soil and grow
Potato	stem (tuber)	"eyes" are buds that sprout into shoots
Sugarcane	stem	stem pieces with nodes are planted
Rose	stem	stem cuttings strike roots
Banana	stem (suckers)	no seeds; grown only vegetatively



Buds in the notches of the leaf margin grow into new plants

Artificial methods ★★☆

- **Cutting:** a piece of stem with buds is planted in moist soil (rose, sugarcane)
- **Layering:** a branch is bent and covered with soil while still on the parent; it roots, then is cut off (jasmine)
- **Grafting:** a cut stem (scion) of one plant is joined to the rooted stem (stock) of another so they grow as one (mango)
- Gardeners use these to copy a plant with **good qualities** exactly

NCERT ACTIVITY

Place a **Bryophyllum leaf** on wet cotton and a **potato** piece with an "eye" in moist soil; in a few days plantlets and shoots appear.

Advantages of vegetative propagation ★★★

1. Plants bear **flowers and fruits earlier** than those grown from seeds
2. Plants that have **lost the capacity to make seeds** (banana, orange, rose, jasmine) can be grown
3. All new plants are **genetically similar** to the parent, so good characters are kept



★★★ *"Write two advantages of vegetative propagation." Nitesh Bhaiya ki guarantee: ye 2 marks har saal ghoom ke aata hai!*

MNEMONIC

"Patti Bryo ki, Tana Aloo ka"

Bryophyllum grows from the **leaf** (patti); potato grows from the **stem** (tana, the tuber). Aloo is a stem, not a root!



MEME BREAK *Aloo **stem** hai, root nahi. Uski "aankhein" actually **buds** hain. Hello there, confusion!*

2 TISSUE CULTURE ★★☆

Steps

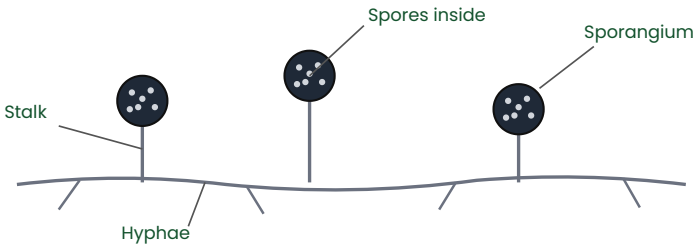
1. Cells are taken from the **growing tip** of a plant
2. They are placed in an **artificial nutrient medium** and divide to form a mass of cells: the **callus**
3. The callus is moved to a medium with **hormones** for growth and differentiation
4. **Plantlets** form; they are shifted to soil to grow into mature plants

Uses: many plants from **one parent**, disease-free, fast; common for **ornamental plants** such as orchids.

3 SPORE FORMATION ★★☆

DEFINITION

- **Spore formation:** many simple multicellular organisms make **spores** in blob-like structures called **sporangia**, e.g. **Rhizopus** (bread mould)



Rhizopus: key points ★★☆

- Thread-like **hyphae** spread on the bread; vertical stalks carry round **sporangia**
- Spores are tiny **asexual reproductive bodies** made in large numbers
- Spores have **thick walls** that protect them till they meet a **moist surface**, then they grow
- NCERT activity: keep a moist slice of bread in a cool, moist, dark place; a white-to-black cottony mould appears

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

A single sporangium can release thousands of spores, which float in air. This is why bread, fruit and even leather goods get mouldy in the rainy season.

4 WHY SEXUAL REPRODUCTION? ★★★

Two big reasons ★★★

- **Variation:** DNA copying errors pile up slowly; combining DNA from **two individuals** creates far more new combinations, which helps the species survive
- **Half chromosomes in gametes:** germ cells carry **half** the chromosomes and half the DNA, so when two fuse, the zygote gets the **normal number** back and DNA does not double every generation
- Germ cells are specialised: one is **large with stored food** (female), one is **small and motile** (male)



BUJHO *Agar gametes mein poore chromosomes hote toh kya problem thi?*

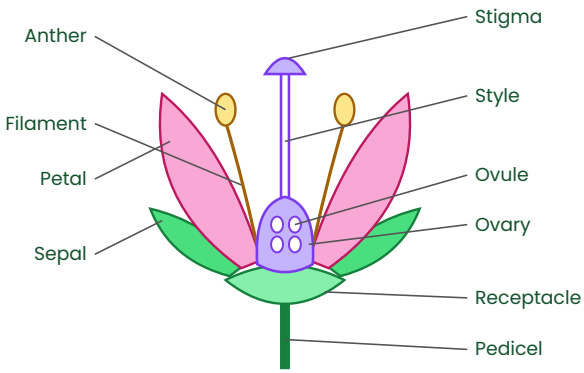
PAHELI *Har generation mein DNA **double** hota jaata! Isliye gametes mein **aadhe** chromosomes hote hain, zygote mein phir se **poore**.*

5 SEXUAL REPRODUCTION IN FLOWERING PLANTS ★★★

DEFINITION

- The **flower** is the reproductive organ; its reproductive parts are the **stamens** (male) and the **carpel or pistil** (female)

Longitudinal section (LS) of a flower ★★★



Stamen = anther + filament (male part, makes pollen) | **Pistil (carpel)** = stigma + style + ovary (female part, has ovules)

Unisexual vs bisexual flowers ★★☆

Type	Has	Examples
Unisexual	either stamens or pistil	papaya, watermelon
Bisexual	both stamens and pistil	Hibiscus, mustard



BUJHO *Hibiscus aur papaya dono phool hain, toh unisexual aur bisexual ka chakkar kya hai?*

PAHELI *Hibiscus **bisexual** hai: ek hi phool mein stamens aur pistil dono. Papaya **unisexual** hai: ek phool mein sirf ek.*



MEME BREAK *Papaya ka phool **unisexual** hota hai, sirf stamens ya sirf pistil. Bisexual flower ke saamne bhi wo kam nahi hai!*

Pollination ★★★

- **Pollination:** transfer of **pollen grains** from the anther to the **stigma**
- **Self-pollination:** pollen reaches the stigma of the **same flower**
- **Cross-pollination:** pollen reaches the stigma of **another flower** of the same kind, carried by wind, water or animals (insects, birds)

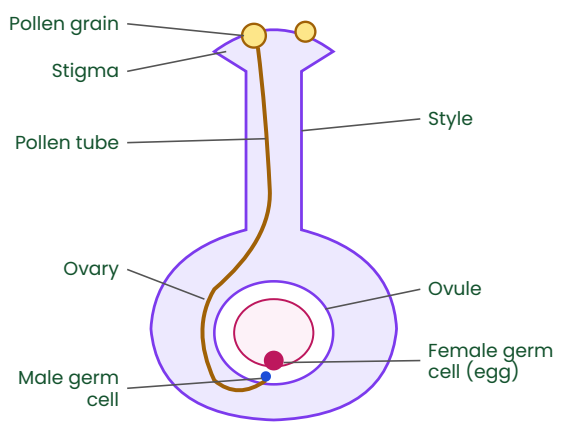
YAAD RAKHNE KI TRICK

"Self = apne ghar ka phool, Cross = padosi ka phool"

Same flower: self-pollination. Another flower of the same kind: cross-pollination. Agents: **Wind, Water, Animals** (WWA).

6 FERTILISATION AND AFTER ★★★

Germination of pollen on stigma ★★★



1. The pollen grain lands on the **stigma** and grows a **pollen tube**
2. The tube grows down through the **style** and reaches the **ovule** in the ovary
3. The **male germ cell** travels down the tube and fuses with the **female germ cell** (egg): this is **fertilisation**, giving a **zygote**

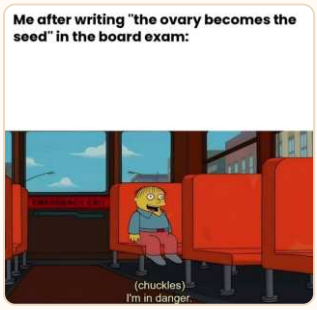
Changes after fertilisation ★★★

This part	becomes
Zygote	**embryo** (divides many times inside the ovule)
Ovule	**seed** (develops a tough coat)
Ovary	**fruit** (grows rapidly and ripens)
Petals, sepals, stamens, style, stigma	usually **shrivel and fall off**

MNEMONIC

"Chhota ovULE se chhota SEED, bada ovARY se bada FRUIT"

The small ovule turns into the seed; the bigger ovary turns into the fruit that holds the seeds. Zygote se embryo, simple!



MEME BREAK

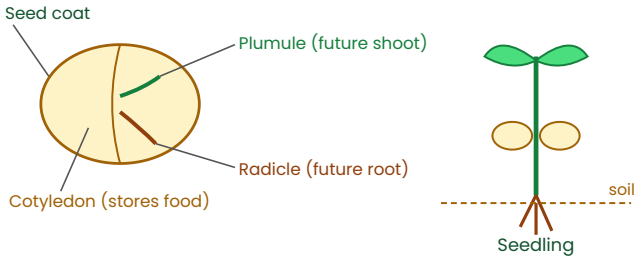
*Board exam mein "ovary becomes seed" likh diya? **Ovule** → **seed** aur **ovary** → **fruit**. Warna danger pakka!*

BUJHO *Pollination aur fertilisation ek hi cheez hai na?*



PAHELI *Nahi! **Pollination** = pollen ka anther se stigma tak jaana. **Fertilisation** = male aur female germ cell ka **fusion**. Pehle pollination, phir fertilisation.*

Seed germination ★★☆



- The seed holds the future plant, the **embryo**: **plumule** (future shoot), **radicle** (future root) and **cotyledons** (stored food)
- With water, air and a suitable temperature the embryo grows into a **seedling**: this is **germination**



MEME BREAK

*Microscope ke neeche pollen tube ka wait karte karte main bhi skeleton ban gaya. **Patience = fertilisation!***

× WRONG	✔ RIGHT
Potato is a modified root	Potato is a modified stem (tuber)
Ovary becomes the seed	Ovule becomes the seed; ovary becomes the fruit
Pollination and fertilisation are the same	Pollination is transfer of pollen; fertilisation is fusion of gametes
Gametes have the full number of chromosomes	Gametes have half; the zygote restores the full number

Exam tips

- ✔ In the flower diagram, label **all six**: sepal, petal, stamen (anther, filament), stigma, style, ovary with ovule.
- ✔ For pollen germination, draw the **pollen tube** reaching the **ovule** and label male and female germ cells.
- ✔ Give one example with every method of vegetative propagation.

7 PRACTICE MCQS

1. In Bryophyllum, new plants arise from buds on the: ★★★

(a) roots (b) stem
(c) leaf margin (d) flowers

2. Which of these is NOT a reason to use vegetative propagation? ★★☆

(a) earlier flowers and fruits (b) growing seedless plants like banana
(c) producing more variation (d) keeping desired characters

3. Rhizopus reproduces asexually by: ★★☆

(a) budding (b) spore formation
(c) fragmentation (d) binary fission

4. The female reproductive part of a flower is the: ★★★

(a) stamen (b) petal
(c) pistil (d) sepal

5. An example of a unisexual flower is: ★★☆

(a) Hibiscus (b) mustard
(c) papaya (d) rose

6. After fertilisation, the ovary develops into the: ★★★

(a) seed (b) fruit
(c) embryo (d) seed coat

7. The pollen tube carries the male germ cell to the: ★★★

(a) anther (b) ovule
(c) sepal (d) filament

8. Gametes have half the number of chromosomes so that: ★★☆

(a) they divide faster (b) the zygote gets the normal number
(c) they can swim (d) variation stops

9. **Assertion (A):** Tissue culture can give many plants from one parent in a short time.
Reason (R): In tissue culture, cells from the growing tip form a callus that is made to form plantlets. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** Cross-pollination needs an agent such as wind, water or animals.
Reason (R): In cross-pollination, pollen of a flower reaches the stigma of the same flower. ★★☆

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Short and long answer questions

1. **(2 marks)** List two advantages of vegetative propagation with one example each.
2. **(3 marks, diagram)** Draw a labelled diagram showing germination of pollen on the stigma.
3. **(5 marks)** Draw the LS of a flower. Explain pollination, fertilisation and the changes that follow in the flower.



NITESH SIR SAYS
Flower ka LS aur pollen tube wala diagram ***daily ek baar*** *banao. Board mein diagram = pakke number.* ***Kal milte hai Day 53 pe!***

Answer key

MCQs: 1 (c), 2 (c), 3 (b), 4 (c), 5 (c), 6 (b), 7 (b), 8 (b), 9 (a), 10 (c)

1 Vegetative propagation
- Earlier flowers and fruits (rose); seedless plants grown (banana); 1 + 1

2 Pollen germination
- Pollen grain on stigma, pollen tube through style, ovule with egg; 3 labelled parts = 3 marks

3 Flower LS
- Diagram with six labels (2); pollination (1); fertilisation (1); ovule to seed, ovary to fruit, others fall off (1)

Today's checklist: tick before you sleep

- [ ] I can explain vegetative propagation with examples
- [ ] I know the steps of tissue culture
- [ ] I can draw Rhizopus and explain spores
- [ ] I can draw and label the LS of a flower
- [ ] I can explain fertilisation and ovule to seed, ovary to fruit

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Page 2–3 phir se padho.



LEVEL 52 COMPLETE +125 XP earned. Next: Level 53



BUJHO *Samajh gaya! Ovule se seed, ovary se fruit, aur pollen tube ovule tak.*

PAHELI *Shabaash! Kal Level 53: reproduction in human beings.*

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

In flowering plants one male gamete fuses with the egg (zygote) and the other fuses with two polar nuclei to form food tissue (endosperm). This is called double fertilisation.

Day 52 hit-list: Eduspark PYQ analysis

Topic	Rating
LS of flower (diagram)	★★★
Germination of pollen on stigma (diagram)	★★★
Ovule to seed, ovary to fruit	★★★
Advantages of vegetative propagation	★★★
Spore formation in Rhizopus	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 53 | LEVEL 53

BIOLOGY | CHAPTER 11 | PART 3 | ASTRA SERIES

REPRODUCTION IN HUMAN BEINGS

Goal: master human reproduction and reproductive health.



by Nitesh Prakash

01 Puberty | 02 Male System | 03 Female System | 04 Placenta | 05 Reproductive Health | +125 XP today | Level 53 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 PUBERTY: THE BODY GETS READY ★★☆

DEFINITION

- **Puberty:** the stage of adolescence when the reproductive tissues mature and the body becomes capable of sexual reproduction
- In boys it usually begins around **13–14 years**, in girls a little earlier, around **10–12 years** (varies person to person)
- Body growth slows down only after the reproductive organs mature, because making germ cells needs a **mature body**

Changes at puberty ★★☆

Both	Boys	Girls
Thick hair in armpits and genital area	Facial hair: beard, moustache	Breast size increases
Thinner hair on arms, legs, face	Voice begins to crack and deepen	Skin of nipples darkens
Skin becomes oily, pimples may appear	Testes start making sperms; penis occasionally erect	Menstruation (periods) begins
Growth spurt; awareness of own body	Driven by **testosterone**	Driven by **oestrogen**



BUJHO *Mere dost ki moustache aa gayi, meri nahi. Kya mujhme kuch gadbad hai?*

PAHELI *Bilkul nahi Bujho! Puberty ka time har insaan mein alag hota hai. Kuch log 11 mein, kuch 15 mein. Sab normal hai.*



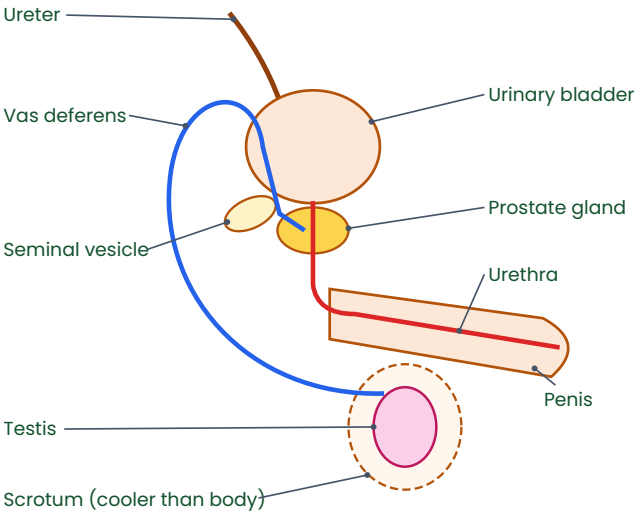
WHY SO MANY CHANGES?

Sexual reproduction needs **germ cells** (sperms and eggs) and, in a woman, a body that can carry a baby. Puberty prepares both. These changes are slow and gradual, and are signals that the body is **sexually maturing**, not that a person is ready to be a parent.



★★★ *Boys, girls aur common changes ki teen list alag likho, 2 marks seedhe. Nitesh Bhaiya ki guarantee!*

2 MALE REPRODUCTIVE SYSTEM ★★★ MOST ASKED



Parts and their jobs ★★★

Part	Function
Testes (2)	Make **sperms** (male germ cells) and the hormone **testosterone**
Scrotum	Pouch that holds the testes **outside** the abdomen, about 2–3 °C cooler
Vas deferens	Carries sperms from the testis; joins the tube from the bladder
Seminal vesicle, prostate	Add fluid: sperms move easily and get **nutrition**
Urethra	Common passage for **sperms and urine**
Penis	Transfers sperms into the female body



★★★ *Testes bahar kyun? Sperm banane ke liye body se kam temperature chahiye. Har saal poochte hai, Nitesh Bhaiya ki guarantee!*

Path of sperms: in order ★★★

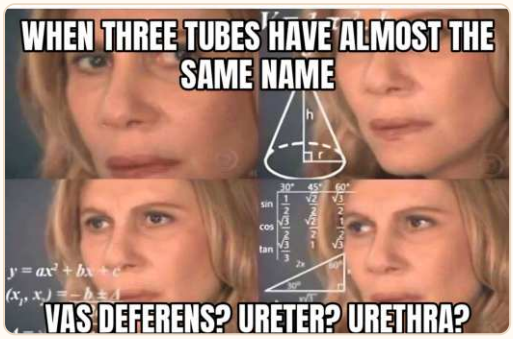
Testis → Vas deferens → (joined by seminal vesicle and prostate fluid) → Urethra → Penis → outside

MNEMONIC: YAAD RAKHNE KI TRICK

"T-V SUP": **T**estis, **V**as deferens, **S**eminal vesicle, **U**rethra, **P**enis. Sperm ka safar ek T-V SUP jaisa: seedha forward!

Hormone and germ cell

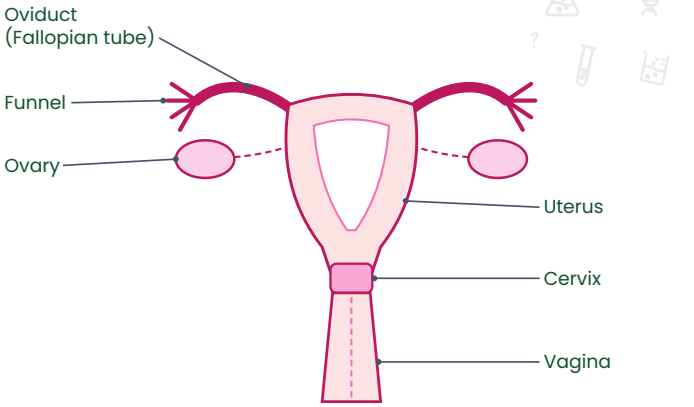
- **Testosterone** (from testes): regulates sperm formation and brings about the puberty changes in boys
- **Sperm:** tiny cell, mostly genetic material, with a **long tail** that helps it swim towards the female germ cell
- Sperms travel in a fluid (semen) made by the seminal vesicles and prostate gland



MEME BREAK

Vas deferens = sperm, ureter = urine, urethra = dono ka exit!

3 FEMALE REPRODUCTIVE SYSTEM ★★★



Parts and their jobs ★★★

Part	Function
Ovaries (2)	Make **eggs** (ova) and the hormones **oestrogen** and progesterone
Oviduct (fallopian tube)	Carries the egg from the ovary to the uterus; **site of fertilisation**
Uterus (womb)	Elastic bag-like organ where the baby develops
Cervix	Narrow neck by which the uterus opens into the vagina
Vagina	Receives sperms; also the birth canal

- ✔ A girl is born with **thousands of immature eggs** in her ovaries; at puberty some start maturing
- ✔ **One egg** is released every month by one of the two ovaries
- ✔ The two oviducts unite into the uterus



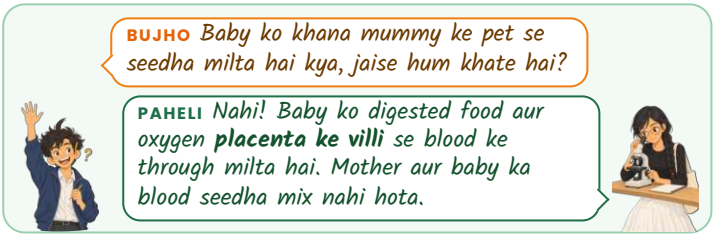
★★★ Oviduct = fertilisation, uterus = baby ka ghar, cervix = gate. Diagram mein ye teen label Nitesh Bhaiya ki guarantee!

4 FERTILISATION TO BIRTH ★★★ MOST ASKED

1. **Fertilisation:** sperms enter through the vagina, travel up and meet the egg in the **oviduct**; their fusion gives a **zygote**
2. **Division:** the zygote divides repeatedly and forms a ball of cells, the early embryo
3. **Implantation:** the embryo gets embedded in the thick, **blood-rich lining** of the uterus
4. **Placenta** forms and nourishes the embryo (see box); the embryo grows into a foetus
5. **Gestation** is about **9 months** (about 40 weeks); birth happens by rhythmic contractions of the uterus muscles

Placenta: the life-line ★★★

- A special **disc-like tissue** embedded in the uterine wall, joining the embryo to the mother
- On the embryo's side it has finger-like **villi**; on the mother's side are **blood spaces** that surround the villi
- Villi give a **large surface area**: glucose and oxygen pass from mother to embryo
- Wastes made by the embryo pass into the mother's blood through the placenta for removal



5 MENSTRUATION ★★☆

1. Every month the uterus **prepares** for a possible pregnancy: its inner lining becomes **thick and spongy** with a rich blood supply
2. The released egg lives for only about **one day**; if it is not fertilised, the thick lining is not needed
3. The lining **breaks down** and comes out through the vagina as blood and mucus: **menstruation**
4. This cycle repeats roughly every **28 days** and each period usually lasts about **2 to 8 days**

EXTRA KNOWLEDGE (HIGH LEVEL) not for assessment

The first menstruation at puberty is called **menarche**; the stopping of the cycle at around 45–50 years is called **menopause**.

6 REPRODUCTIVE HEALTH ★★★ MOST ASKED

DEFINITION

- **Reproductive health:** physical, mental, social and emotional well-being in all matters of reproduction, including the freedom to decide when, and how many, children to have

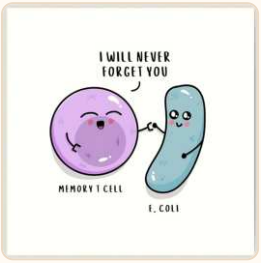
EXAM TIP

Write the **site** with each event: fertilisation in the **oviduct**, implantation in the **uterus** wall, exchange at the **placenta**. Sites alone carry marks in MCQs.

Sexually transmitted diseases (STDs) ★★★

Caused by	Diseases
Bacteria	Gonorrhoea, Syphilis
Viruses	Warts, HIV-AIDS

They spread through sexual contact; HIV can also spread through infected blood and needles. A **condom** (barrier) reduces the risk to a large extent.



MEME BREAK
Memory T cell kabhi germ ko nahi bhoolta. Par HIV inhi immune cells ko target karta hai, isliye AIDS mein body ki defence ki memory bhi kamzor ho jaati!

Contraception: preventing pregnancy ★★★

Method	How it works
Barrier (condom)	Stops sperms reaching the egg; also protects against STDs
Hormonal (oral pills)	Change hormone balance so **eggs are not released**; may have side effects
IUCD (loop, Copper-T)	Placed in the uterus to prevent pregnancy; may cause irritation of the uterus
Surgical	**Vasectomy:** vas deferens blocked (male). **Tubectomy:** oviduct blocked (female)

MNEMONIC: YAAD RAKHNE KI TRICK

"B-H-I-S" = **Bus Hai Itna Simple**: **B**arrier, **H**ormonal, **I**UCD, **S**urgical. Surgical mein: **V**asectomy = **V**as deferens (male), **T**ubectomy = fallopian **T**ube (female).



MEME BREAK
*Copper-T, loop = **IUCD**. Option mein dikhe toh pehchaan lena, ye surgical nahi hai, device hai!*

Female foeticide and sex ratio ★★☆

- Surgery can be misused to remove a pregnancy for wrong reasons, e.g. killing a female foetus after illegal sex determination: **female foeticide**
- Prenatal sex determination is **prohibited by law** in India
- It has lowered the **child sex ratio** sharply in some places; a healthy society needs a balanced ratio of girls and boys



MEME BREAK
*Copper-T ek **device** hai, surgery nahi!*

× WRONG	✓ RIGHT
Fertilisation happens in the uterus	It happens in the **oviduct** (fallopian tube)
Ureter carries sperms	**Vas deferens** carries sperms; ureter carries urine
Oral pills protect against STDs	Only the **condom** (barrier) gives protection from STDs
Placenta mixes mother's and baby's blood	Exchange happens across the **villi**; the bloods do not mix



NITESH SIR SAYS
*Is chapter ko sharmaa ke mat chhodo. Diagrams aur functions **board mein pakke 3–5 marks** laate hai. Labelled diagram roz ek baar banao!*

7 PRACTICE MCQS

1. Sperms are formed in the: ★★★
(a) vas deferens (b) testes
(c) prostate gland (d) seminal vesicle

2. The testes lie in the scrotum outside the abdomen because sperm formation needs: ★★★
(a) a higher temperature (b) a lower temperature
(c) more oxygen (d) more light

3. Fertilisation of the egg in human females normally takes place in the: ★★★
(a) ovary (b) uterus
(c) oviduct (d) vagina

4. The common passage for urine and sperms in human males is the: ★★☆
(a) ureter (b) vas deferens
(c) urethra (d) seminal vesicle

5. Which of these is a viral STD? ★★☆
(a) Gonorrhoea (b) Syphilis
(c) AIDS (d) Tuberculosis

6. Copper-T is an example of: ★★★
(a) barrier method (b) IUCD
(c) surgical method (d) oral pill

7. Which method blocks the vas deferens? ★★☆
(a) tubectomy (b) vasectomy
(c) Copper-T (d) condom

8. In the human embryo, the villi of the placenta mainly: ★★☆

(a) produce eggs
(b) increase surface area for exchange
(c) make testosterone
(d) store urine

9. **Assertion (A):** The lining of the uterus breaks down every month if the egg is not fertilised.
Reason (R): The thick, spongy lining is needed only to nourish an implanted embryo. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

10. **Assertion (A):** Oral contraceptive pills protect a person from sexually transmitted diseases.
Reason (R): Oral pills change the hormonal balance so that eggs are not released. ★★☆

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Short and long answer questions

1. Why are the testes located outside the abdominal cavity? Name the hormone they secrete and give one function of it. **(2 marks)**
2. What is placenta? Describe its function in the human body. **(3 marks)**
3. Draw a labelled diagram of the human female reproductive system. What happens to the egg if it is not fertilised? **(5 marks)**

Case-based question (4 marks)

Reena's biology teacher shows a chart of the female reproductive system. An egg is released from the right ovary on day 14 of the cycle and is fertilised a day later.

1. Name the part where fertilisation took place. **(1)**
2. Where will the embryo get implanted? **(1)**
3. Name the tissue that supplies the embryo with food and oxygen, and state how its structure suits this job. **(2)**

Answer key: MCQs

Q	Ans	Why
1	(b)	Testes make sperms and testosterone.
2	(b)	About 2–3 °C below body temperature.
3	(c)	Sperm meets egg in the oviduct.
4	(c)	Urethra carries both.
5	(c)	HIV is a virus; the other two are bacterial.
6	(b)	Copper-T is an intra-uterine device.
7	(b)	Vasectomy = vas deferens blocked.
8	(b)	More area for food, O_2 and wastes.
9	(a)	R explains why the lining is shed.
10	(d)	Pills give no STD protection; R is true.

Model answers

1 Testes outside

- Sperm formation needs a temperature lower than body temperature (1 mark)
- Testosterone: regulates sperm formation and puberty changes in boys (1 mark)

2 Placenta

- Disc-like tissue in the uterine wall joining embryo and mother
- Villi (embryo side) bathed in maternal blood spaces: large surface area
- Glucose and O_2 pass to the embryo; its wastes pass to the mother's blood

3 Female system

- Diagram: ovaries, oviducts with funnels, uterus, cervix, vagina (any five labels, 3 marks)
- Unfertilised egg lives about one day; the uterine lining breaks down and is shed as menstruation (2 marks)

Case-based answers

- (i) Oviduct (fallopian tube)
- (ii) In the thick, blood-rich lining of the uterus
- (iii) Placenta: its villi give a large surface area, bathed in the mother's blood

Day 53 checklist: tick before you sleep

- ☐ I can list the changes at puberty in boys and girls
- ☐ I can draw and label the male reproductive system
- ☐ I can draw and label the female reproductive system
- ☐ I can explain fertilisation, implantation and the placenta
- ☐ I know the STDs and all four contraceptive methods

My score: ____ / 5

LEVEL 53 COMPLETE +125 XP earned. Next: Level 54



BUJHO *Samajh gaya! Oviduct mein fertilisation, uterus mein baby, aur placenta se khana.*

PAHELI *Shabaash! Kal Level 54: Chapter 11 ka* ***Boss Level.***

Day 53 hit-list: Eduspark PYQ analysis

Topic	Rating
Male / female system diagram and labels	★★★
Placenta: structure and function	★★★
Contraceptive methods; STDs	★★★
Why testes lie in the scrotum	★★☆

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 54
LEVEL 54

BIOLOGY | CHAPTER 11 | BOSS LEVEL

CHAPTER 11 BOSS LEVEL

Goal: test all of reproduction in 90 minutes.



by Nitesh Prakash

01 Revision Map | 02 MCQs | 03 Short Answers | 04 Case & Long | 05 Answer Key | +200 XP Boss Level | Level 54 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 11 REVISION MAP ★★★ MOST ASKED

Chapter 11 mind map: revise in 5 minutes

ASEXUAL
Binary and multiple fission, fragmentation, regeneration, budding, spores

VEGETATIVE ★★★
Bryophyllum leaf, potato, layering, grafting; tissue culture

HOW DO ORGANISMS REPRODUCE?

FLOWERING PLANTS
Stamen + carpel, pollination, pollen tube, ovule to seed, ovary to fruit

HUMANS ★★★
Testes, ovaries, oviduct, placenta; STDs and contraception

Must-know pairs: write them daily ★★★

Organism / part	Remember
Amoeba, Leishmania	Binary fission
Plasmodium	Multiple fission
Spirogyra	Fragmentation
Planaria, Hydra	Regeneration
Hydra, yeast	Budding
Rhizopus	Spore formation (sporangia)
Bryophyllum	Buds on leaf margins
Anther / ovary	Pollen grains / ovules



BUJHO *Ovary aur ovule same cheez hai na? Dono seed ban jaate hai?*

PAHELI *Nahi Bujho! **Ovule** seed banta hai aur **ovary** fruit. Ek ovary ke andar kai ovules ho sakte hai, isliye ek fruit mein kai seeds.*





MEME BREAK
*Kal Biology test hai aur aaj Boss Level? Perfect! Ye page solve karo, kal ka test **easy level** lagega.*

How to attack the Boss: 90 minutes

1. **Section A** (12 marks): 15 minutes. Mark doubtful ones and come back
2. **Sections B and C** (15 marks): 30 minutes. Draw neat, labelled diagrams in pencil
3. **Sections D and E** (18 marks): 40 minutes. Read every case twice
4. Last 5 minutes: check labels, arrows and units of every diagram
5. Start with the section you are most sure of; never leave an MCQ blank

2 SECTION A: MCQS (1 MARK EACH) ★★★

1. Amoeba reproduces by: ★★★
(a) budding (b) binary fission
(c) fragmentation (d) spore formation

2. Multiple fission is seen in: ★★★
(a) Plasmodium (b) Hydra
(c) yeast (d) Spirogyra

3. Which organism reproduces by budding? ★★☆
(a) Planaria (b) Leishmania
(c) Hydra (d) Rhizopus

4. Spirogyra breaks into pieces and each piece grows into a new individual. This is: ★★☆
(a) regeneration (b) fragmentation
(c) budding (d) fission

5. Buds produced in the notches of the leaf margin grow into new plants in: ★★★
(a) rose (b) potato
(c) Bryophyllum (d) banana

6. Rhizopus reproduces asexually by: ★★☆
(a) binary fission (b) budding
(c) spore formation (d) fragmentation

7. Pollen grains are produced in the: ★★☆
(a) ovary (b) anther
(c) stigma (d) style

8. After fertilisation, the ovule develops into the: ★★★
(a) fruit (b) seed
(c) embryo sac (d) flower

9. Which of these is NOT a part of the carpel? ★★☆

(a) stigma (b) style
(c) ovary (d) anther

10. Which contraceptive method also protects against STDs? ★★★

(a) oral pills (b) Copper-T
(c) condom (d) vasectomy

11. **Assertion (A):** DNA copying during reproduction is not perfectly accurate.
Reason (R): Variations produced during reproduction help a species survive in a changing environment. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

12. **Assertion (A):** Human testes are located outside the abdominal cavity in the scrotum.
Reason (R): Sperm formation requires a temperature lower than the normal body temperature. ★★★

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

3 SECTIONS B AND C ★★★ MOST ASKED

Section B: Short Answers (2 marks each)

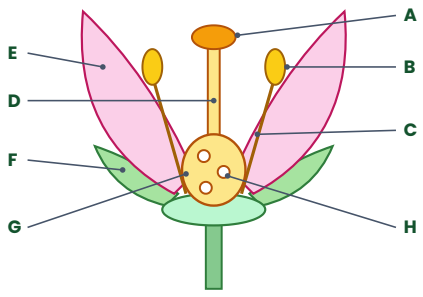
B1. What is DNA copying? Why is it essential for reproduction?
B2. Differentiate between self-pollination and cross-pollination.
B3. List two advantages of vegetative propagation. Name one plant grown this way.



MEME BREAK
*Text ratta maar liya, diagram skip? Boss Level mein **flower LS** zaroor aata hai!*

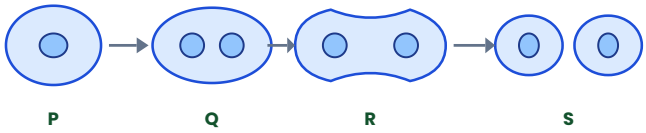
Section C: Short Answers (3 marks each)

C1. The figure shows the longitudinal section of a flower. Name the parts A to H. Which parts form the male and the female reproductive organs?



Section C (continued)

C2. The figure shows stages P to S of a process. Name the process and one organism showing it. How is it different from multiple fission?



C3. Why do the testes lie in the scrotum? Name the hormone they make. What will happen if the vas deferens is blocked?

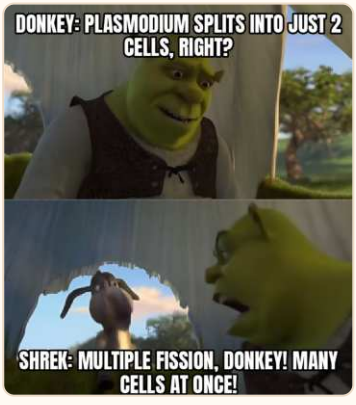
4 SECTIONS D AND E ★★★ MOST ASKED

Section D, Case 1: Meena's nursery (4 marks)

Meena grows roses and sugarcane from stem cuttings, bananas from suckers and Bryophyllum from leaves. She also grafts a good mango variety onto a strong rootstock.
(i) Name this group of methods. **(1)**
(ii) How does a Bryophyllum leaf give new plants? **(1)**
(iii) Give two advantages of these methods over growing from seeds. OR Why is tissue culture useful for nurseries? **(2)**

Case 2: Planning a healthy family (4 marks)

A health worker explains contraceptive methods to a young couple: condoms, oral pills, Copper-T, and surgery for those who want no more children.
(i) Which method also protects against STDs? Name one bacterial STD. **(1)**
(ii) How do oral pills prevent pregnancy? **(1)**
(iii) What is Copper-T? Name the surgical methods for males and females. **(2)**



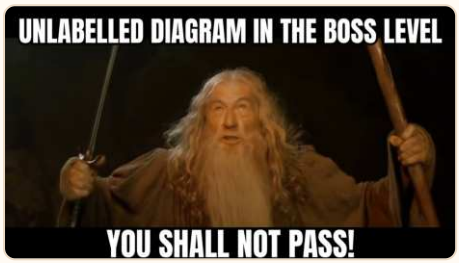
MEME BREAK
*Binary fission = **2** cells. Multiple fission (Plasmodium) = **bahut saare** cells ek saath. Donkey mat bano!*

Section E: Long Answers (5 marks each)

E1. Draw a labelled diagram showing the germination of pollen on the stigma. Describe fertilisation in a flowering plant and state what the zygote, ovule and ovary develop into.
E2. (a) Draw a labelled diagram of the human female reproductive system. (b) How does the developing embryo get nourishment? (c) What happens if the egg is not fertilised?
OR (a) Draw a labelled diagram of the human male reproductive system. (b) Give the functions of the testes, seminal vesicle and prostate gland.

Diagram drill: labels the examiner looks for ★★★

- **Binary fission (Amoeba):** nucleus dividing, cytoplasm dividing, two daughter cells
- **Budding (Hydra):** parent, bud, tentacles of the bud
- **Flower LS:** sepal, petal, stamen (anther, filament), carpel (stigma, style, ovary, ovule)
- **Pollen germination:** pollen grain, pollen tube, style, ovule, female germ cell
- **Human systems:** testis, vas deferens, urethra, prostate; ovary, oviduct, uterus, cervix, vagina



MEME BREAK

*Bina label ka diagram? **Marks nahi milenge!** Har diagram mein 4–6 labels, arrows ruler se.*

× WRONG	✓ RIGHT
Plasmodium divides by binary fission	**Multiple** fission: many daughter cells at once
Ovary becomes the seed	**Ovule** becomes the seed; ovary becomes the fruit
Pollen tube grows through the ovary wall first	It grows down the **style** to reach the ovule
Hydra only regenerates	Hydra shows both **budding** and regeneration

BOSS TIP

Har answer mein **keyword** likho: “multiple fission”, “micropyle”, “villi”, “implantation”. Examiner keywords dhoondhta hai, lambi kahani nahi.

5 ANSWER KEY AND MARKING POINTS ★★★

Answer key: Section A (MCQs)

Q	Ans	Why
1	(b)	Amoeba splits into two.
2	(a)	Plasmodium: many cells at once.
3	(c)	Hydra forms buds.
4	(b)	Pieces regrow: fragmentation.
5	(c)	Bryophyllum leaf-margin buds.
6	(c)	Sporangia release spores.
7	(b)	Anther makes pollen.
8	(b)	Ovule becomes the seed.
9	(d)	Anther belongs to the stamen.
10	(c)	Only a barrier blocks STDs.
11	(b)	Both true; R gives the benefit, not the cause.
12	(a)	R explains A.

Model answers: Sections B and C

B1 DNA copying

- Making a copy of the DNA of a cell before it divides; two copies separate into two cells (1)
- Passes the body design and information to the next generation; small errors give variation (1)

B2 Pollination

- Self: pollen reaches the stigma of the same flower or plant (1)
- Cross: pollen goes to a flower of another plant by wind, water or insects (1)

B3 Vegetative propagation

- Faster flowering and fruiting; all plants are genetically like the parent (1)
- Plants without seeds (banana, orange, rose) can be grown; e.g. potato (1)

C1 Flower LS

- A stigma, B anther, C filament, D style, E petal, F sepal, G ovary, H ovule (2)
- Male: stamen (B + C). Female: carpel or pistil (A + D + G) (1)

C2 Binary fission

- Binary fission in Amoeba; P to S: nucleus divides, then the cytoplasm (2)
- Multiple fission (Plasmodium) gives many daughter cells at once (1)

C3 Testes

- Sperm formation needs a lower temperature (1); testosterone (1)
- Sperms cannot be carried out, so no fertilisation: the idea behind vasectomy (1)

Model answers: Sections D and E

D1 Meena’s nursery

- (i) Vegetative propagation (1)
- (ii) Buds in the notches of the leaf margin fall on soil and grow into plants (1)
- (iii) Faster, identical to parent; seedless plants grown. OR Tissue culture gives many disease-free plants quickly (2)

D2 Healthy family

- (i) Condom; gonorrhoea or syphilis (1)
- (ii) They change the hormone balance so that eggs are not released (1)
- (iii) An IUCD placed in the uterus (1); vasectomy and tubectomy (1)

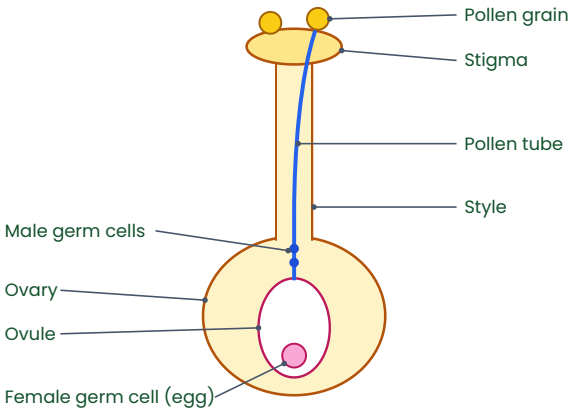
E1 Pollen germination

- Diagram: stigma, pollen grain, pollen tube, style, ovary, ovule, female germ cell (2)
- Pollen tube carries the male germ cell to the egg in the ovule; they fuse to form the zygote (1)
- Zygote → embryo, ovule → seed, ovary → fruit; petals, sepals, stamens wither (2)

E2 Female system

- Diagram: ovaries, oviducts with funnels, uterus, cervix, vagina (2)
- Placenta: villi bathed in mother’s blood supply glucose and oxygen; wastes go back (2)
- Lining breaks and is shed as menstruation (1). OR male: testes make sperms and testosterone; vesicles and prostate add nourishing fluid

Model diagram for E1: pollen on the stigma



★★★ *Pollen tube ka diagram har saal ka favourite hai. Nitesh Bhaiya ki guarantee!*

Bonus rapid-fire round: quick answers

Question	Answer
Process in which DNA makes a copy of itself	DNA replication (copying)
Organism that regenerates from body pieces	Planaria
Sac that holds spores in Rhizopus	Sporangium
Method to grow many plants from tissue	Tissue culture
Male reproductive part of a flower	Stamen
Where the pollen tube enters the ovule	Through the micropyle
Hormone made by the testes	Testosterone
Tissue that nourishes the human embryo	Placenta
Viral STDs	Warts, HIV-AIDS
Approximate human gestation	About 9 months

Score card: how did you fight the boss?

Marks out of 45	Your level
40 and above	Boss defeated: ready for boards
30 to 39	Strong: fix the diagrams once more
Below 30	Re-read Days 51–53, then retry

Day 54 checklist: tick before you sleep

- [ ] I attempted the full Boss test in 90 minutes
- [ ] I can draw the flower LS and pollen germination diagrams
- [ ] I can name the asexual method for each organism
- [ ] I can draw both human reproductive systems with labels
- [ ] I checked every answer with the marking points

My score: ____ / 5

LEVEL 54 COMPLETE

+200 XP earned. Next: Level 55



BUJHO *Boss khatam! Fission se placenta tak, sab yaad ho gaya.*

PAHELI *Shabaash! Kal Level 55: **Heredity** aur Mendel ke experiments.*



NITESH SIR SAYS

*Chapter 11 se board mein lagbhag **har saal ek diagram** aata hai: flower LS, pollen tube ya human system. Roz ek diagram banao, **Kal milte hai Day 55 pe!***

NCERT exercise: quick answers

- **Variation:** helps the species survive a changed environment, though it may not help the individual
- **Spore formation:** spores have thick walls that protect them until they find moist conditions to grow
- **Regeneration** fails in complex animals: their cells are highly specialised and organised into organ systems
- **Vegetative propagation** is used for plants that have lost the capacity to make seeds, e.g. banana, orange, rose, jasmine
- **Fertilised egg:** the zygote divides and forms the embryo, which gets implanted in the uterine wall
- **Contraception** methods: barrier, hormonal, IUCD and surgical (vasectomy, tubectomy)

BUJHO *Fragmentation aur regeneration mein kya farak hai? Dono mein tukde hi toh hote hai!*



PAHELI *Fragmentation mein simple body (Spirogyra) khud tootkar har tukda naya banta hai. Regeneration mein kate hue tukde se **specialised cells** poora organism banate hai (Planaria, Hydra).*



Most asked in Chapter 11: Eduspark PYQ analysis

Topic	Rating
Pollen germination and fertilisation in plants	★★★
Human reproductive systems (diagrams)	★★★
Modes of asexual reproduction	★★★
Vegetative propagation and its advantages	★★☆
Contraception and STDs	★★☆

Coming up: Day 55

Heredity begins: variation, Mendel's pea plants, monohybrid cross and the 3:1 ratio. DNA copying leads straight into genes.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

BIOLOGY | CHAPTER 11 COMPLETE | LEVEL 54 ASTRA SERIES

CHAPTER 11 SWAHAA!

How do Organisms Reproduce? | Days 51–54



BUJHO *Fission, budding, flower ka LS aur pollen tube bhi! Chapter 11 Swahaa!*

PAHELI *Great going! Next stop Heredity: Mendel, pea plants aur Punnett square.*

CHAPTER XP EARNED
+575 XP

LEVELS CLEARED
4 levels (Days 51–54)

BOOK PROGRESS
Level 54 / 60 7050 XP total

WHAT YOU HAVE MASTERED

- ✔ Fission, budding, fragmentation, regeneration
- ✔ Flower parts and pollination
- ✔ Human reproductive systems
- ✔ Vegetative propagation and spores
- ✔ Fertilisation in flowering plants
- ✔ Reproductive health

NEXT CHAPTER Day 55: Heredity

Chapter 11 cleared!

DAY 55 | LEVEL 55

BIOLOGY | CHAPTER 12 | PART 1 | ASTRA SERIES

MENDEL & THE MONOHYBRID CROSS

Goal: learn how traits pass on and predict ratios.



by Nitesh Prakash

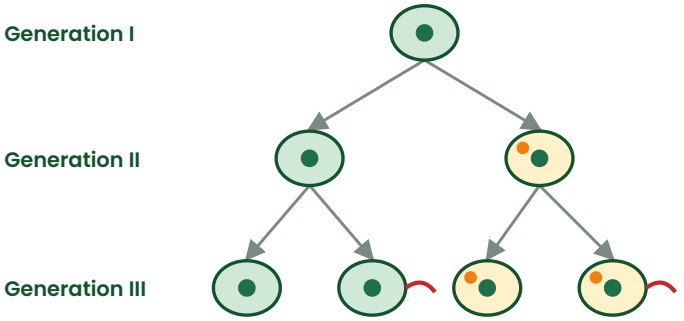
01 Variation | 02 Heredity | 03 Mendel's Pea | 04 Monohybrid | 05 Two Laws | +125 XP today | Level 55 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 VARIATION DURING REPRODUCTION ★★☆

DEFINITION

- **Variation:** the differences in characters (traits) among individuals of the same species, including between parents and their offspring
- It arises because DNA copying during reproduction is **never 100% accurate**, and in sexual reproduction DNA from **two parents** combines



Orange spot and red tail = new variations from DNA copying

Asexual vs sexual: how much variation? ★★☆

Asexual reproduction	Sexual reproduction
one parent, copies of DNA	two parents, DNA of both mixes
very little variation, from small copying errors only	**much greater** variation (new combinations every generation)
offspring almost exactly like the parent	offspring similar, not identical

EXAM POINT

Not every variation is useful; some are harmful and some neutral. The useful ones help a few individuals survive when conditions change, so sexual reproduction (more variation) gives a species a better chance to last.



BUJHO *Variation matlab galti? Toh ye toh bura hua na?*

PAHELI *Nahi Bujho! Garmi badhi toh jin bacteria mein **heat-resistant** variation hai, wahi bachenge. Variation = species ka **survival insurance**.*



Why variation is useful

- ✔ The **environment selects** the variants that suit it; others may die
- ✔ Example: if water gets much hotter, most bacteria die, but a few **heat-resistant** variants survive and multiply
- ✔ So variation helps the **species survive**, even if many individuals die

NCERT in-text questions

1 Trait A in 10%, trait B in 60% of an asexual population: which arose earlier?

> **Trait B.** Variations pile up over generations, so the trait found in more individuals has had more time to spread.

2 How does variation promote survival?

> Some variants suit a changed environment (heat, drought, disease) and survive, so the species does not die out.

2 HEREDITY AND INHERITED TRAITS ★★★

DEFINITION

- **Heredity:** the transmission of characters (traits) from parents to offspring, generation after generation
- **Inherited traits:** features a child gets from the parents through genes, e.g. earlobe type, eye colour, hair type

REMEMBER

A child shows features of **both** parents because both contribute genetic material, so children never look exactly like either parent.

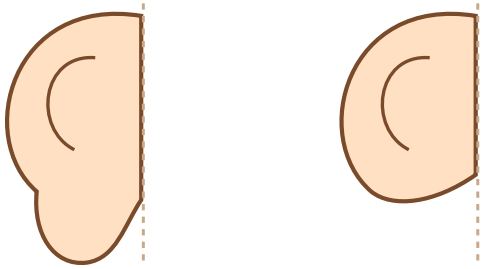


MEME BREAK

*Hair, earlobe, height: sab ka source ek hi, **DNA**. Mom-dad ka DNA mix hua toh tum bane!*

NCERT activity: earlobe survey ★★☆

1. Look at the lower part of the ear (the earlobe) of everyone in your class.
2. It is either **free** (hangs below the point where it joins the head) or **attached** (joins the side of the head closely).
3. Count both types and find the percentage of each. Both are common, so earlobe type is an **inherited trait** that varies.



Free earlobe
lobe hangs free below the joint

Attached earlobe
lobe joins the side of the head

3 MENDEL AND HIS PEA PLANTS ★★★

Gregor Johann Mendel (1822–1884)

- An Austrian monk who did breeding experiments on the **garden pea** (*Pisum sativum*) and worked out the first rules of heredity
- He studied **one or two traits at a time**, counted the offspring and used maths to find patterns
- Called the **Father of Genetics**; his work was recognised only after his death

Why did Mendel choose the pea plant? ★★★

- ✔ Many clear **contrasting traits** (tall/dwarf, round/wrinkled seeds)
- ✔ Flowers are **bisexual** and normally **self-pollinate**, so pure lines are easy to get
- ✔ Cross-pollination can be done **by hand** when needed
- ✔ **Short life cycle** and many seeds per plant, so many generations and large numbers to count
- ✔ Easy to grow in a garden

Seven pairs of contrasting traits ★★☆

Character	Dominant	Recessive
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green
Flower colour	Violet	White
Pod shape	Inflated (full)	Constricted
Pod colour	Green	Yellow
Flower position	Axial	Terminal
Stem height	Tall	Dwarf



★★★ *Monohybrid Punnett square aur ratios 3:1, 1:2:1 har saal aate hain, Nitesh Bhaiya ki guarantee!*

MNEMONIC: SEVEN DOMINANT TRAITS

"Rohit, Yadav, Virat, Iyer, Gill, Axar, Tilak"

Team India ki **dominant** XI: **R**ound, **Y**ellow, **V**iolet, **I**nflated, **G**reen pod, **A**xial, **T**all. Recessive = inke ulte.

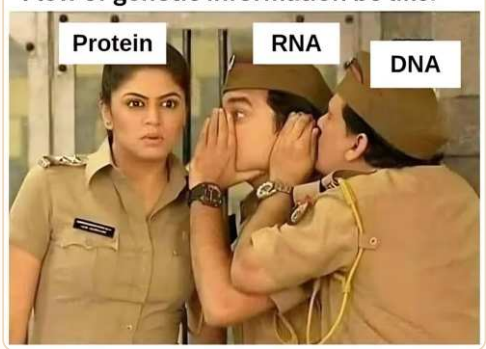


MEME BREAK
*F1 ka tall plant (Tt) apne **dwarf parent** ko yaad kar raha hai. Tension mat lo, F2 mein dwarf **wapas aata hai**!*

4 KEY TERMS OF GENETICS ★★★

Know these words exactly

Term	Meaning
Gene	a section of DNA that carries the information for one trait (one protein)
Alleles	two alternative forms of the same gene, e.g. T (tall) and t (dwarf)
Dominant	the trait that shows up even when only one copy is present (T)
Recessive	the trait that shows up only when both copies are recessive (tt)
Genotype	the gene makeup of an organism, e.g. TT, Tt, tt
Phenotype	the visible character, e.g. tall or dwarf
Homozygous	both alleles same: TT or tt (pure line)
Heterozygous	both alleles different: Tt (hybrid)



MEME BREAK
*Gene = DNA ka chhota hissa jo **protein** ka recipe deta hai. Wahi protein trait banata hai (detail Day 56 mein).*



BUJHO *Tt aur TT dono tall dikhte hain, toh dono same hue na?*

PAHELI ***Phenotype*** *same (dono tall), par* ***genotype*** *alag! TT pure tall hai, Tt hybrid hai jisme dwarf ka* ***t*** *chhupa hai.*



5 MONOHYBRID CROSS ★★★ MOST ASKED

DEFINITION

- **Monohybrid cross:** a cross between two plants that differ in **one pair** of contrasting traits, e.g. tall × dwarf

Parents (P): Tall **TT** × Dwarf **tt**
Gametes: T (from tall) and t (from dwarf)
F1: all **Tt** → all plants **tall** (no medium height)
Selfing F1: Tt × Tt → gametes T, t from each parent
The dwarf trait is hidden in F1 but not lost

F2 Punnett square (Tt × Tt) ★★★

	T	t
T	**TT** Tall	**Tt** Tall
t	**Tt** Tall	**tt** Dwarf

Ratio	F2 result
Phenotypic	Tall : Dwarf = **3 : 1**
Genotypic	TT : Tt : tt = **1 : 2 : 1**

MNEMONIC: MONOHYBRID RATIOS

"Photo mein 3:1, Gene mein 1:2:1"

Phenotype = jo photo mein dikhe (tall/dwarf) = 3:1. **Genotype** = andar ka gene hisaab (TT, Tt, tt) = 1:2:1.

6 MENDEL'S LAWS ★★★

Law of Dominance

- In a hybrid (Tt), of two contrasting alleles only one shows its effect (**dominant**); the other stays hidden (**recessive**)
- Proof: every F1 plant of TT × tt is tall; none is dwarf or medium

Law of Segregation (purity of gametes)

- The two alleles of a gene **separate** when gametes form, so each gamete carries **only one** allele
- Proof: the hidden dwarf trait (tt) **reappears** in F2 in one-fourth of the plants



MEME BREAK

*Bhai, **t** kahin gaya nahi. Recessive allele F1 mein chhup ke baitha hai, F2 mein **tt** banke nikalta hai.*

COMMON MISTAKES

- Writing F1 as "medium height": wrong, F1 is **all tall** (no blending)
- Mixing up ratios: 3:1 is **phenotypic**, 1:2:1 is **genotypic**
- Writing a gamete as "Tt": a gamete carries **one** allele only, T or t

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Mendel's actual F2 count for stem height was **787 tall : 277 dwarf**, about 2.84 : 1, close to the 3 : 1 he predicted.

Practice MCQs: Eduspark PYQ analysis

1. Which of these is **not** a reason why Mendel chose the pea plant? ★★☆
(a) contrasting traits (b) self-pollination
(c) long life cycle (d) many seeds

2. A cross between TT and tt gives F1 plants that are: ★★★
(a) all dwarf (b) all tall
(c) half tall (d) all medium

3. The genotypic ratio of the F2 generation in a monohybrid cross is: ★★★
(a) 3 : 1 (b) 1 : 2 : 1
(c) 9 : 3 : 3 : 1 (d) 1 : 1

4. In 400 F2 plants of a Tt × Tt cross, the expected number of dwarf plants is: ★★☆
(a) 100 (b) 200
(c) 300 (d) 400

5. An organism with genotype Tt is said to be: ★★☆
(a) homozygous (b) heterozygous
(c) recessive (d) pure line

6. Which pair is a pair of alleles? ★☆☆
(a) T and t (b) T and R
(c) TT and tt (d) R and y

7. Which trait in pea is **recessive**? ★★☆
(a) violet flower (b) axial flower
(c) green seed (d) inflated pod

8. Variation is greatest in organisms produced by: ★★☆
(a) binary fission (b) budding
(c) sexual reproduction (d) fragmentation

9. **Assertion (A):** Dwarf plants reappear in the F2 generation of a cross between pure tall and pure dwarf pea plants.
Reason (R): The two alleles of a gene separate during gamete formation, and each gamete gets only one of them. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** All F1 plants of a TT × tt cross are tall.
Reason (R): Variation does not occur in sexual reproduction. ★★☆
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer key: MCQs

Q	Ans	Why
1	(c)	Pea has a short life cycle; long is wrong.
2	(b)	T is dominant, so Tt is tall.
3	(b)	TT : Tt : tt = 1 : 2 : 1.
4	(a)	One-fourth of 400 = 100 dwarf.
5	(b)	Two different alleles = heterozygous.
6	(a)	Alternative forms of the same gene.
7	(c)	Yellow seed is dominant; green is recessive.
8	(c)	Two parents mix DNA: most variation.
9	(a)	Segregation explains the return of tt.
10	(c)	A true (dominance); R false.

Short and long answer questions

1. Define variation. Why is variation in a population useful for the survival of a species? Give one example. **(3 marks)**
2. Distinguish between genotype and phenotype, giving one example of each. **(2 marks)**
3. With the help of a labelled Punnett square, show a cross between pure tall (TT) and pure dwarf (tt) pea plants up to F2. State the phenotypic and genotypic ratios and the law it proves. **(5 marks)**

Model answers

1 Variation

- Differences among individuals of a species, caused by inaccurate DNA copying and mixing of parents' DNA
- Helps some individuals survive changed conditions, e.g. heat-resistant bacteria in hot water

2 Genotype vs phenotype

- Genotype = gene makeup (TT, Tt, tt); phenotype = visible trait (tall, dwarf)

3 Monohybrid cross

- P: TT × tt → F1 all Tt (tall) → F1 selfed → F2 Punnett square as on page 62
- Phenotype 3 tall : 1 dwarf; genotype 1 TT : 2 Tt : 1 tt; laws of dominance and segregation

Case-based question (4 marks)

A student crosses pure violet-flowered pea plants (VV) with pure white-flowered plants (vv) and then self-pollinates the F1 plants.

1. What is the flower colour of F1 plants? **(1)**
2. Which colour is dominant? **(1)**
3. Out of 800 F2 plants, how many are white, and how many are pure violet (VV)? **(2)**

EXAM TIPS

- Gametes ko **single letter** (T, t) mein likho; har box mein genotype aur phenotype likho.
- Number wale sawal: dwarf = total ÷ 4, tall = total × 3/4, Tt = total ÷ 2.

Day 55 checklist: tick before you sleep

- [ ] I can explain how variation arises and why it helps survival
- [ ] I know the earlobe example of an inherited trait
- [ ] I can list the seven contrasting traits of pea
- [ ] I can draw the monohybrid Punnett square and give 3:1 and 1:2:1
- [ ] I can state the laws of dominance and segregation

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Section 5 phir se padho.



Case-based answers

- (i) All violet (Vv)
- (ii) Violet is dominant over white
- (iii) White = 800 ÷ 4 = **200**; pure violet VV = 800 ÷ 4 = **200** (Vv = 400)

LEVEL 55 COMPLETE +125 XP earned. Next: Level 56



BUJHO *Samajh gaya! F1 mein dwarf chhupa, F2 mein 1/4 wapas, aur ratio 3:1.*

PAHELI *Shabaash! Kal Level 56: dihybrid cross aur sex determination.*

Day 55 hit-list: Eduspark PYQ analysis

Topic	Rating
Monohybrid cross with Punnett square	★★★
3:1 and 1:2:1 ratios	★★★
Genotype vs phenotype, dominant vs recessive	★★★
Why Mendel chose pea	★★☆
Variation and survival	★★☆
Earlobe example	★☆☆



NITESH SIR SAYS

*Punnett square ko **chart ki tarah** banao: upar ek parent ke gametes, side mein doosre ke. Har box mein dono mila do, ratio khud nikal aayega. **Kal milte hai Day 56 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 56
LEVEL 56

BIOLOGY | CHAPTER 12 | PART 2 ASTRA SERIES

DIHYBRID CROSS & SEX DETERMINATION

Goal: master the 9:3:3:1 cross and how sex is decided.



by Nitesh Prakash

01 Dihybrid | 02 9:3:3:1 | 03 Gene Action | 04 Germ Cells | 05 XX and XY

+125 XP today | Level 56 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 DIHYBRID CROSS ★★★ MOST ASKED

DEFINITION

> **Dihybrid cross:** a cross between two plants that differ in **two pairs** of contrasting traits, e.g. seed shape and seed colour

P: Round yellow **RRYY** × Wrinkled green **rryy**
Gametes: RY and ry
F1: all **RrYy** → all **round yellow**
F1 selfed: RrYy × RrYy; each parent gives **4 kinds** of gametes: RY, Ry, rY, ry
R = round, r = wrinkled; Y = yellow, y = green (R and Y dominant)

F2 Punnett square (RrYy × RrYy) ★★★

Gametes	RY	Ry	rY	ry
RY	**RRYY** RY	**RRYy** RY	**RrYY** RY	**RrYy** RY
Ry	**RRYy** RY	**RRyy** RG	**RrYy** RY	**Rryy** RG
rY	**RrYY** RY	**RrYy** RY	**rrYY** WY	**rrYy** WY
ry	**RrYy** RY	**Rryy** RG	**rrYy** WY	**rryy** WG

RY = round yellow, RG = round green, WY = wrinkled yellow, WG = wrinkled green

F2 result: 16 boxes, 4 phenotypes ★★★ MOST ASKED

Phenotype	Boxes	Type
Round yellow	**9**	parental
Round green	**3**	new combination
Wrinkled yellow	**3**	new combination
Wrinkled green	**1**	parental

Phenotypic ratio **9 : 3 : 3 : 1**. The two **new combinations** prove that seed shape and seed colour are inherited independently.

COUNT IT YOURSELF

Pure **RRYY** and **rryy** fill only 1 box each; the 9 round yellow boxes hold 4 genotypes (RRYY, RRYy, RrYY, RrYy).

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Mendel's actual F2: **315** round yellow, **108** round green, **101** wrinkled yellow, **32** wrinkled green (556 seeds).

2 LAW OF INDEPENDENT ASSORTMENT ★★★

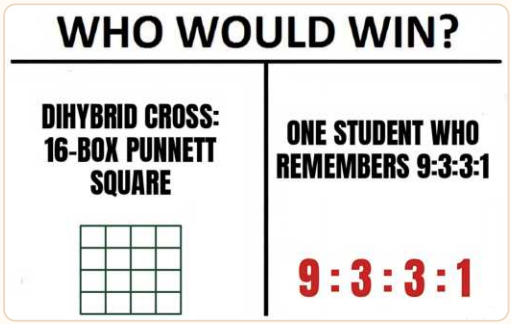
DEFINITION

> **Law of independent assortment:** when two pairs of traits are crossed, the alleles of one pair separate (assort) **independently** of the alleles of the other pair during gamete formation
> So shape and colour do not stay "tied": round can go with green and wrinkled with yellow

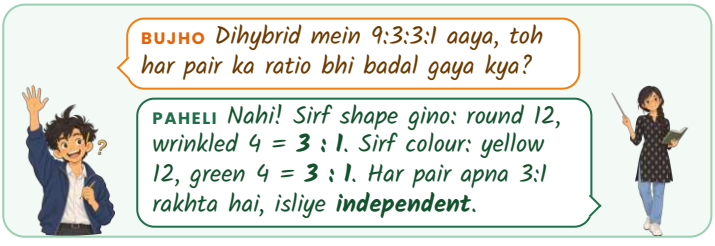
MNEMONIC: 9 : 3 : 3 : 1

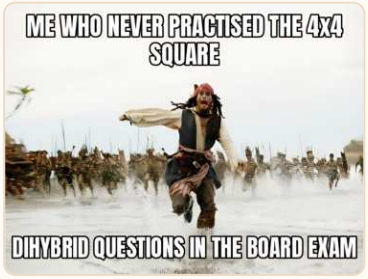
"9 mein dono Dabang, 3-3 mein ek-ek, 1 mein dono Chhupe"

9 = both dominant (round yellow), **3 + 3** = one dominant each (round green, wrinkled yellow), **1** = both recessive (wrinkled green). Total 16.



MEME BREAK
*16 box dekh ke darr gaye? Bas yaad rakho **9 : 3 : 3 : 1**, aur haar Punnett square ki hogi!*





MEME BREAK
*Exam mein bhaagna nahi padega agar aaj ek baar **4 × 4 square** khud bana liya. Practice!*

3 HOW DO TRAITS GET EXPRESSED? ★★★

DEFINITION

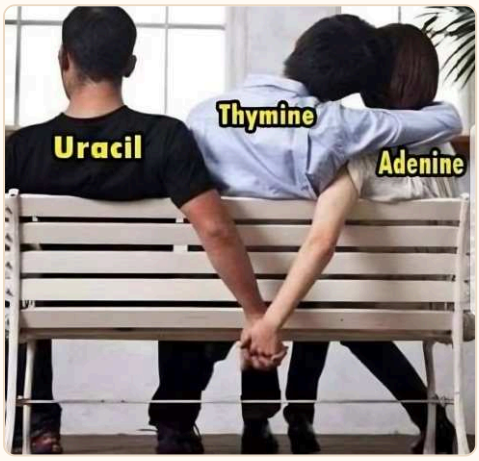
- A **gene** is a section of DNA that carries the information to make **one protein**
- Proteins (often **enzymes**) control the chemical reactions in cells, and these reactions decide the traits

NCERT example: plant height ★★☆

Gene (DNA) → makes a **protein** (enzyme) → enzyme makes a **plant hormone** → hormone controls **height**

Efficient enzyme → more hormone → tall plant; altered gene → less efficient enzyme → less hormone → short plant

- Plant height depends on the amount of a growth **hormone**
- The amount of hormone depends on how **efficient the enzyme** that makes it is
- So the **gene** for that enzyme decides tall or short

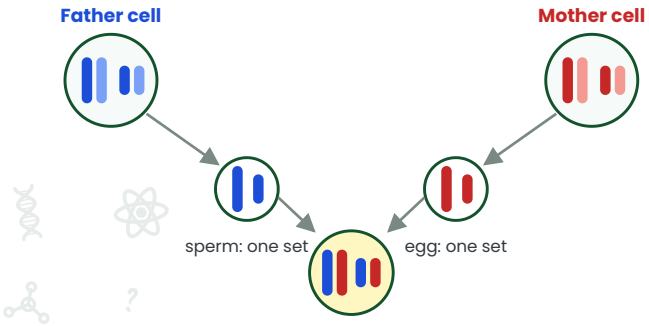


MEME BREAK
*DNA mein **thymine**, RNA mein **uracil**. Gene ka message RNA leke jaata hai protein banane, isliye uracil pe nazar (not for assessment).*

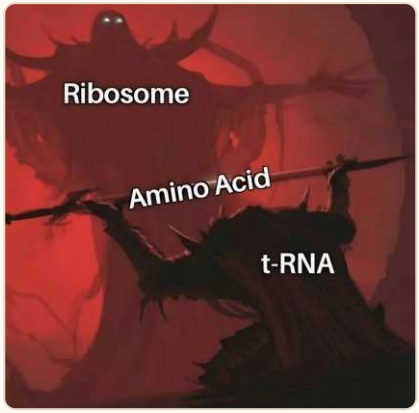
4 BOTH PARENTS CONTRIBUTE EQUALLY ★★☆

How do both parents pass on genes?

- ✔ Genes are carried on **chromosomes**; each body cell has chromosomes in **pairs**, one of each pair from each parent
- ✔ Each **germ cell** (sperm or egg) gets only **one set**: one chromosome from each pair
- ✔ At fertilisation the two sets join, so the **zygote** gets the full number back, half from each parent
- ✔ This keeps the chromosome number **constant** generation after generation and lets both parents contribute equally



Zygote: pairs restored (one of each pair from each parent)



MEME BREAK
*Protein banane ki factory = **ribosome**. Gene ki recipe se amino acids jodkar protein, aur protein se trait!*

5 SEX DETERMINATION ★★★ MOST ASKED

DEFINITION

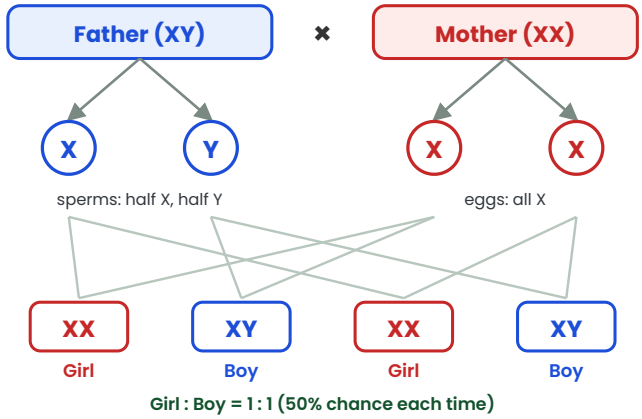
- **Sex determination:** the way the sex (male or female) of a new individual is decided

Different ways in different animals ★★☆

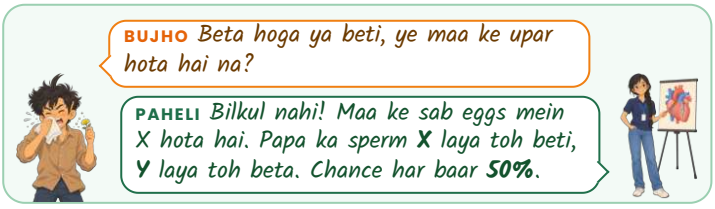
Animal	Sex decided by
Some reptiles, e.g. **turtles**	**temperature** at which the fertilised eggs are kept
Snails	can **change sex**, so sex is not fixed by genes
Humans	**genes**: inherited sex chromosomes

Sex chromosomes in humans ★★★

- ✔ Humans have **23 pairs** of chromosomes: **22 pairs** of autosomes (same in both sexes) and **1 pair** of sex chromosomes
- ✔ Female: **XX** (a perfect pair). Male: **XY** (Y is shorter)
- ✔ All eggs carry **X**; half the sperms carry **X** and half carry **Y**
- ✔ Egg + X sperm → **XX girl**; egg + Y sperm → **XY boy**. So the sex of the child is decided by the **father**



Girl : Boy = 1 : 1 (50% chance each time)





MEME BREAK

*Science clear hai: maa sirf **X** de sakti hai, **Y** sirf papa ke sperm mein. Beti hone pe maa ko blame karna **galat** hai.*

MNEMONIC: SEX DETERMINATION

"Papa ka Y = Ladka, Papa ka X = Ladki"

Maa ka contribution hamesha **X**. Result papa ke sperm pe: 23 pairs = 22 + XX ya 22 + XY.



★★★ *Sex determination ka flow chart (XX × XY) har saal board mein aata hai, Nitesh Bhaiya ki guarantee!*



BUJHO *Toh turtles mein bhi XX aur XY hota hoga?*

PAHELI *Nahi Bujho! Kuch reptiles mein ande kis **temperature** pe rakhe gaye, usse sex decide hota hai. Aur snail toh apna sex **badal** bhi sakta hai.*



NCERT exercise questions

1 Tall violet × short white gives all violet, about half short. Tall parent's genotype?

> **TtWW**: all violet means WW; half short means the tall parent was Tt.

2 How do Mendel's crosses show traits are inherited independently?

> In the dihybrid F2, new combinations (round green, wrinkled yellow) appear.

3 Father A, mother O, daughter O. Can we tell which is dominant?

> **No.** One family is too small, and the father may carry an O allele, so we need many more crosses.

Monohybrid vs dihybrid at a glance ★★☆

Point	Monohybrid	Dihybrid
Trait pairs	one (TT × tt)	two (RRYY × rryy)
F2 boxes	4	16
F2 phenotypes	3 : 1	9 : 3 : 3 : 1
Law shown	dominance, segregation	independent assortment

COMMON MISTAKES

> Writing F1 gametes as only RY and ry: an RrYy plant makes **four** kinds (RY, Ry, rY, ry)

> "Mother decides the sex": wrong, the **father's** sperm (X or Y) decides

> "Humans have 23 chromosomes": no, **23 pairs** = 46 chromosomes

Practice MCQs: Eduspark PYQ analysis

1. The F1 plants of a cross RRYY × rryy are: ★★★
(a) round green (b) wrinkled yellow
(c) round yellow (d) wrinkled green

2. How many kinds of gametes does an RrYy plant form? ★★☆
(a) 1 (b) 2
(c) 3 (d) 4

3. The phenotypic ratio of the F2 generation of a dihybrid cross is: ★★★
(a) 3 : 1 (b) 1 : 2 : 1
(c) 9 : 3 : 3 : 1 (d) 1 : 1 : 1 : 1

4. In a dihybrid F2 of 160 plants, the expected number of wrinkled green plants is: ★★☆
(a) 10 (b) 30
(c) 90 (d) 160

5. A gene controls a trait by making a: ★★☆
(a) carbohydrate (b) protein
(c) fat (d) vitamin

6. The number of autosome pairs in a human cell is: ★★★
(a) 22 (b) 23
(c) 44 (d) 46

7. In some reptiles such as turtles, the sex of the young is decided by: ★★☆
(a) sex chromosomes (b) incubation temperature
(c) the mother (d) diet

8. A child gets an X chromosome from the father. The child will be: ★★★
(a) a boy (b) a girl
(c) either (d) cannot say

9. **Assertion (A):** Round green and wrinkled yellow seeds appear in the F2 of a cross between round yellow and wrinkled green plants.
Reason (R): The alleles for seed shape and seed colour are inherited independently of each other. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

10. **Assertion (A):** The sex of a child is decided by the mother.
Reason (R): The mother has two X chromosomes and all her eggs carry an X chromosome. ★★★
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

EXAM TIPS

> Flow chart mein teen line zaroor: **Parents → Gametes → Offspring**, aur ratio 1 : 1 likho.

> Dihybrid numbers: wrinkled green = total ÷ 16, round yellow = total × 9/16, round green = total × 3/16.

> "23 pairs" aur "46 chromosomes" dono sahi hain; "23 chromosomes" likha toh number katega.

Answer key: MCQs

Q	Ans	Why
1	(c)	R and Y are dominant: RrYy is round yellow.
2	(d)	RY, Ry, rY, ry.
3	(c)	16 boxes give 9 : 3 : 3 : 1.
4	(a)	1/16 of 160 = 10.
5	(b)	Gene → protein (enzyme) → trait.
6	(a)	22 pairs of autosomes + 1 pair sex chromosomes.
7	(b)	Temperature of the fertilised eggs.
8	(b)	X from father + X from mother = XX.
9	(a)	Independent assortment explains new combinations.
10	(d)	A false (father decides); R true.

Short and long answer questions

1. How do genes control the characteristics of an organism? Explain with the example of plant height. **(3 marks)**
2. "The sex of a child is determined by the father." Justify with a labelled flow chart. **(3 marks)**
3. Mendel crossed pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy). Show the F1 and F2 with a Punnett square, give the F2 ratio, and state the law that this cross proves. **(5 marks)**

Model answers

1 Genes and traits

- Gene → protein (enzyme) → hormone → trait; efficient enzyme gives more growth hormone, so a tall plant; altered gene gives less hormone, so a short plant

2 Father decides

- Mother XX gives only X eggs; father XY gives X or Y sperm (flow chart as on page 65)
- X sperm → XX girl; Y sperm → XY boy; so the father's sperm decides

3 Dihybrid cross

- F1 all RrYy (round yellow); F2 from a 4 × 4 square: 9 RY : 3 RG : 3 WY : 1 WG
- Law of independent assortment (new combinations appear)

Case-based question (4 marks)

In a family the first two children are girls. Relatives tell the mother that the third child will surely be a boy, and that she alone is responsible for having girls.

1. How many pairs of chromosomes does a human cell have? **(1)**
2. Which chromosome does the mother always give? **(1)**
3. What is the chance that the third child is a boy? Is the relatives' claim correct? Give a reason. **(2)**

DIAGRAM TIP

Punnett square ki grid ruler se banao, gametes ko circle karo aur ratio neeche likho. Har step ke alag marks milte hain.

Day 56 checklist: tick before you sleep

- [ ] I can draw the 4 × 4 Punnett square of RRYY × rryy
- [ ] I can give 9 : 3 : 3 : 1 and state independent assortment
- [ ] I can explain gene → enzyme → hormone → height
- [ ] I know why both parents contribute equally
- [ ] I can draw the XX × XY flow chart and give the 50% chance

My score: ____ / 5

5/5 = Jalwa hai! 4 se kam? Punnett square phir se banao.



Case-based answers

- (i) 23 pairs (22 pairs autosomes + 1 pair sex chromosomes)
- (ii) An X chromosome
- (iii) 50% every time, whatever happened before; the claim is wrong, since the father's sperm (X or Y) decides

LEVEL 56 COMPLETE +125 XP earned. Next: Level 57



BUJHO *Samajh gaya! 16 box, 9:3:3:1, aur beta ya beti papa ka sperm decide karta hai.*

PAHELI *Shabaash! Kal Level 57: Heredity ka Boss Level.*

Day 56 hit-list: Eduspark PYQ analysis

Topic	Rating
Sex determination in humans (flow chart)	★★★
Dihybrid cross and 9:3:3:1	★★★
Law of independent assortment	★★★
Genes → proteins → traits (plant height)	★★☆
Both parents contribute equally	★★☆
Sex determination in turtles, snails	★☆☆



NITESH SIR SAYS

*Dihybrid square banate waqt **pehle 4 gametes** likho, phir har box mein R-wale saath aur Y-wale saath. Gadbad kabhi nahi hogi. **Kal Boss Level, taiyaar raho!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

DAY 57 · LEVEL 57

BIOLOGY | CHAPTER 12 | BOSS LEVEL

CHAPTER 12 BOSS LEVEL

Goal: test all of Heredity in 90 minutes.



by Nitesh Prakash

01 Revision Map · 02 MCQs · 03 Short Answers · 04 Case & Long · 05 Answer Key · +200 XP Boss Level | Level 57 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 CHAPTER 12 REVISION MAP ★★★ MOST ASKED

Chapter 12 mind map: revise in 5 minutes

VARIATION
Arises in reproduction (DNA copying errors, sexual mixing); most in sexual reproduction; helps survival

MONOHYBRID ★★★
TT × tt → F1 all Tt (tall); F2 3 : 1 phenotype, 1 : 2 : 1 genotype

HEREDITY

DIHYBRID ★★★
RRYY × rryy → F1 RrYy; F2 9 : 3 : 3 : 1 (16 boxes)

LAWS
Dominance, segregation (one allele per gamete), independent assortment

TRAITS AND SEX ★★★
DNA (gene) → protein/enzyme → hormone → trait. Humans: 23 pairs, XX girl, XY boy; father decides

Must-know ratios and terms ★★★

Cross	F2 phenotype	F2 genotype
Monohybrid Tt × Tt	3 : 1	1 : 2 : 1
Dihybrid RrYy × RrYy	9 : 3 : 3 : 1	9 types
Gametes of RrYy	RY, Ry, rY, ry	4 types
XY × XX (sex)	1 boy : 1 girl	XY : XX = 1 : 1

MNEMONIC: 3 MENDEL LAWS
"D-S-I: Dil Se Insaan": Dominance (FI mein ek hi dikhta), Segregation (gamete mein ek hi allele), Independent assortment (do traits alag alag chalte).

Monohybrid F2 at a glance: Tt × Tt ★★★

Gametes	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

3 tall : 1 dwarf (phenotype); 1 TT : 2 Tt : 1 tt (genotype).

MNEMONIC: TWO RATIOS
"Ek gene, teen-ek; do gene, nau-teen-teen-ek": monohybrid 3 : 1, dihybrid 9 : 3 : 3 : 1.

How to attack the Boss: 90 minutes

1. Section A (20 MCQs, 1 mark each): **20 min**
2. Section B (5 × 2 marks): **12 min**
3. Section C (4 × 3 marks): **20 min**
4. Section D (2 cases × 4 marks): **14 min**
5. Section E (2 × 5 marks): **18 min**; check every cross: 6 min

Total 60 marks. In every cross write parents, gametes, Punnett square, then the ratio with labels.



BOSS INTRO
*"I hate biology" bolne wale bhi Heredity mein full marks laate hain: bas **Punnett square** sahi bharna hai. Chalo, Boss se ladein!*

2 SECTION A: MCQS ★★★ MOST ASKED

1. Mendel chose the garden pea because it: ★★☆

(a) has many clear contrasting traits
(b) cannot self-pollinate
(c) has a long life cycle
(d) has no variation

2. A tall plant (TT) is crossed with a dwarf plant (tt). The F1 plants are: NCERT ★★★

(a) all dwarf
(b) all tall (Tt)
(c) 3 tall : 1 dwarf
(d) 1 tall : 1 dwarf

3. The F2 phenotypic ratio of a monohybrid cross is: ★★★

(a) 1 : 2 : 1
(b) 9 : 3 : 3 : 1
(c) 3 : 1
(d) 1 : 1

4. The F2 genotypic ratio of a monohybrid cross is: ★★★

(a) 3 : 1
(b) 1 : 2 : 1
(c) 1 : 1
(d) 9 : 3 : 3 : 1

5. Which pair was NOT among the contrasting traits Mendel studied? ★☆☆

(a) round / wrinkled seed
(b) yellow / green seed
(c) violet / white flower
(d) red / black seed

6. The number of different phenotypes in the F2 of a dihybrid cross is: ★★★

(a) 2
(b) 4
(c) 9
(d) 16

7. In the F2 of RrYy × RrYy, the fraction of round green seeds is: ★★★
(a) 9/16 (b) 3/16
(c) 1/16 (d) 4/16

8. The number of types of gametes formed by a plant of genotype RrYy is: ★★☆
(a) 1 (b) 2
(c) 4 (d) 8

9. The number of pairs of chromosomes in a human body cell is: ★★★
(a) 22 (b) 23
(c) 44 (d) 46

10. A human egg cell normally carries: ★★☆
(a) 22 + X (b) 22 + Y
(c) 44 + XX (d) 22 + XY

11. The sex of a human child is decided by: NCERT ★★★
(a) the mother's egg
(b) the sex chromosome in the father's sperm
(c) the season of birth
(d) the mother's diet

12. In some reptiles such as turtles, the sex of the young is decided by: ★★☆
(a) X and Y chromosomes
(b) the temperature at which the fertilised eggs are kept
(c) the father only
(d) the food of the mother

13. Genes are segments of: ★★☆
(a) protein (b) DNA
(c) the cell membrane (d) starch

14. A pea plant is tall because the gene for tallness makes an enzyme that works efficiently to produce more: ★★☆
(a) plant growth hormone (b) chlorophyll
(c) starch (d) cellulose

15. A pea plant with genotype Tt is: ★★☆
(a) homozygous tall (b) heterozygous tall
(c) homozygous dwarf (d) heterozygous dwarf

3 SECTION B: SHORT ANSWERS (2 MARKS EACH)

★★★ MOST ASKED

Answer in 30–50 words

1. Distinguish between dominant and recessive traits with one example each. **(2)**
2. A man with blood group A marries a woman with blood group O, and their daughter has blood group O. Is this information enough to tell whether blood group A or O is dominant? Why or why not? NCERT **(2)**
3. How do Mendel's experiments show that traits may be dominant or recessive? NCERT **(2)**
4. How is the equal genetic contribution of the male and female parents ensured in the progeny? NCERT **(2)**
5. Why does variation arise during reproduction? Give one way in which it helps a species. **(2)**

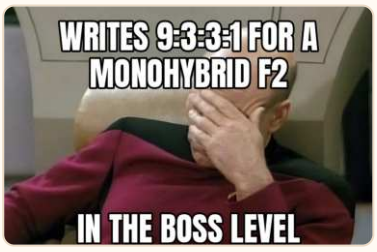
16. In the F2 of the cross TT × tt, the percentage of dwarf plants is: ★★★
(a) 0% (b) 25%
(c) 50% (d) 75%

17. Variation among offspring is greatest in: ★★☆
(a) binary fission (b) budding
(c) sexual reproduction (d) fragmentation

18. **Assertion (A):** In human beings, the sex of the child is determined by the father.
Reason (R): The father produces two types of sperms, carrying X or Y, while the mother produces only X-carrying eggs. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

19. **Assertion (A):** Dwarfness reappears in the F2 generation of Mendel's monohybrid cross.
Reason (R): The recessive trait is lost completely in the F1 generation. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

20. **Assertion (A):** Both parents contribute equally to the inheritance of traits in the child.
Reason (R): Each germ cell carries only one set of chromosomes, that is one copy of each gene. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true



EXAM FACEPALM
Monohybrid = 3 : 1, dihybrid = 9 : 3 : 3 : 1. Ratio galat likha toh Picard ke saath examiner bhi facepalm karega.

Common mistakes that cost marks

× WRONG	✔ RIGHT
F2 genotype of Tt × Tt = 3 : 1	Genotype 1 : 2 : 1; phenotype 3 : 1
Gametes of Tt written as "Tt"	Each gamete gets one allele: T or t
Mother decides the sex of the baby	Father's sperm (X or Y) decides
Recessive trait is lost in F1	It is hidden in F1 and reappears in F2
Human cell has 23 chromosomes	23 pairs = 46; a gamete has 23

4 SECTION C: SHORT ANSWERS (3 MARKS EACH)

★★★ MOST ASKED

Show your working in every cross

1. Pure tall (TT) pea plants are crossed with pure dwarf (tt) plants. Using a Punnett square, show the F1 and F2 generations and write the phenotypic and genotypic ratios of F2. **(3)**
2. How do Mendel's experiments show that different traits are inherited independently? NCERT **(3)**
3. Explain with the example of plant height how genes control the characteristics of an organism. **(3)**
4. Tall plants with violet flowers are crossed with short plants with white flowers. All progeny have violet flowers, but almost half are short. Work out the genotype of the tall parent and explain your answer. NCERT **(3)**



BUJHO *Do betiyaan hui toh dadi ne mummy ko blame kiya. Kya mummy ki galti?*

PAHELI *Bilkul nahi! Mummy ke saare eggs mein X hi hota hai. Baby ka sex **papa ke sperm** (X ya Y) se tay hota hai, aur har baar chance **50 : 50**.*



Allele terms in one line

- **Gene:** a section of DNA that codes for a protein (trait)
- **Alleles:** two forms of one gene, e.g. T (tall) and t (dwarf)
- **Homozygous:** TT or tt; **heterozygous:** Tt
- **Genotype** = gene make-up (Tt); **phenotype** = what you see (tall)



BUJHO *9 : 3 : 3 : 1 mein kaunsa 9 hai aur kaunsa 1? Har baar confuse ho jaata hoon.*

PAHELI *Simple! **9** = dono dominant (round yellow), **3 + 3** = ek dominant ek recessive (round green, wrinkled yellow), **1** = dono recessive (wrinkled green).*



Any cross in 4 steps (full marks)

1. **Parents:** write both genotypes, e.g. TT × tt
2. **Gametes:** one allele of each gene per gamete (T; t)
3. **Punnett square:** gametes on the top and side, fill every box
4. **Result:** phenotype and genotype ratios, with labels

MNEMONIC: WHO DECIDES BOY OR GIRL

"Y aaya toh Boy aaya": Y sirf papa ke paas hai, isliye baby ka sex papa ka sperm tay karta hai.

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

Evolution (deleted this year): helpful variations are passed on more often, so a population slowly changes over many generations. **Not for assessment.**

5 SECTION D: CASE-BASED (4 MARKS EACH)

★★★ MOST ASKED

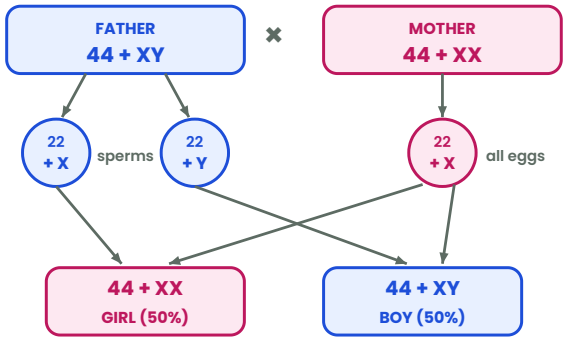
Case 1: Riya's pea garden

Riya crossed pure-breeding pea plants with round yellow seeds (RRYY) with plants having wrinkled green seeds (rryy). She then self-pollinated the F1 plants and collected 1600 F2 seeds.

1. What will be the phenotype and genotype of the F1 plants? **(1)**
2. Write all the types of gametes the F1 plants form. **(1)**
3. Name the four F2 phenotypes with their ratio. How many of the 1600 seeds are expected to be wrinkled and green, and how many round and green? **(2)**

Case 2: Boy or girl?

Study the diagram of sex determination in human beings and answer the questions.



1. Which parent produces two kinds of gametes? Name them. **(1)**
2. What is the sex chromosome make-up of a boy and of a girl? **(1)**
3. A couple already has two daughters. What is the chance that their third child will be a son? Justify it. **(2)**

6 SECTION E: LONG ANSWERS (5 MARKS EACH)

★★☆

Write in points, draw where asked

1. (a) State Mendel's law of dominance and law of segregation. (b) A pure round yellow (RRYY) plant is crossed with a wrinkled green (rryy) plant. Show the F2 with a Punnett square and state the law it proves. **(5)**
2. (a) With a labelled flow diagram, explain how the sex of a child is determined in human beings. (b) How is sex determined in some animals such as turtles and snails? **(5)**



THINK DEEP

*Mummy hamesha X deti hai, toh **X ya Y** papa ka sperm laata hai. Philosoraptor bhi maan gaya!*

Answer key: Section A

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(a)	2	(b)	3	(c)	4	(b)	5	(d)
6	(b)	7	(b)	8	(c)	9	(b)	10	(a)
11	(b)	12	(b)	13	(b)	14	(a)	15	(b)
16	(b)	17	(c)	18	(a)	19	(c)	20	(a)

- **Q7:** round green = 3 of 16 boxes, so 3/16
- **Q19:** R is false: dwarfness is only hidden in F1, not lost

Marking points: Section B

1 Dominant vs recessive
- Dominant shows even in Tt, e.g. tallness (1); recessive shows only in tt, e.g. dwarfness (1)

2 Blood group A or O
- No (1). One family with one child is too small; O could be dominant or recessive here. Many progeny are needed (1)

3 Mendel and dominance
- TT × tt gave all tall F1: tall is dominant (1); dwarf reappeared in 1/4 of F2, so it was hidden, that is recessive (1)

4 Equal contribution
- Germ cells carry one set of chromosomes (1); fusion restores pairs, one copy from each parent (1)

5 Variation
- Errors in DNA copying and mixing of DNA of two parents (1); variants may survive a change, e.g. heat-tolerant bacteria in a heat wave (1)

Marking points: Section C

1 Monohybrid cross
- TT × tt; gametes T and t; F1 all Tt, tall (1)
- Tt × Tt Punnett square drawn correctly (1)
- F2: 3 tall : 1 dwarf; 1 TT : 2 Tt : 1 tt (1)

2 Independent inheritance
- RRYY × rryy gives F1 RrYy, all round yellow (1)
- F2 shows 4 types in 9 : 3 : 3 : 1 (1)
- New combinations (round green, wrinkled yellow) prove shape and colour are inherited independently (1)

3 Genes control traits
- A gene codes for a protein such as an enzyme (1)
- An efficient enzyme makes more growth hormone, so the plant is tall (1)
- An altered gene gives a weak enzyme, less hormone, a short plant (1)

4 Tall violet parent
- Genotype TtWW (1)
- All progeny violet: parent is WW (1)
- Half short: parent is Tt, as Tt × tt gives 1 tall : 1 short (1)



NITESH SIR SAYS

*Heredity mein marks **Punnett square** se aate hain: parents, gametes, square, ratio. Charon steps likho, full marks pakke. **Kal milte hai Day 58 pe!***

Marking points: Section D

1 Riya's pea garden
- (i) All round yellow, RrYy (1)
- (ii) RY, Ry, rY, ry (1)
- (iii) Round yellow 9 : round green 3 : wrinkled yellow 3 : wrinkled green 1 (1); wrinkled green = 1600 ÷ 16 = 100, round green = 300 (1)

2 Boy or girl?
- (i) Father: X-carrying and Y-carrying sperms (1)
- (ii) Boy 44 + XY, girl 44 + XX (1)
- (iii) 50% (1): every fertilisation is a fresh event; an X or a Y sperm is equally likely to fuse with the egg (1)

Marking points: Section E

1 Laws and dihybrid cross
- (a) Dominance: only the dominant trait shows in F1 (1); segregation: the two alleles separate, each gamete gets one (1)
- (b) F1 RrYy; 4 × 4 square below (1.5); 9 : 3 : 3 : 1 (0.5); law of independent assortment (1)

Gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

2 Sex determination
- (a) Flow diagram as in Case 2 (2); mother gives only X eggs, father gives X or Y sperms; X sperm makes a girl, Y sperm a boy (1)
- (b) Turtles: temperature at which the fertilised eggs are kept (1); snails can change sex, so it is not genetically fixed (1)

Day 57 checklist: tick before you sleep

- [ ] I can draw a monohybrid Punnett square and give both ratios
- [ ] I can work a dihybrid cross and explain 9 : 3 : 3 : 1
- [ ] I can state the laws of dominance, segregation and independent assortment
- [ ] I can draw the sex determination flow diagram
- [ ] I scored at least 45 / 60 in this Boss test

My score: ____ / 5

LEVEL 57 COMPLETE +200 XP earned. Next: Level 58



BUJHO *Boss down! 3 : 1, 9 : 3 : 3 : 1 aur XY sab yaad ho gaya.*

PAHELI *Shabaash! Kal Level 58: Our Environment, food chain aur 10% law.*

SCORE CHECK

45+ = Boss defeated. Below 40? Redo Days 55 and 56.

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

BIOLOGY | CHAPTER 12 COMPLETE | LEVEL 57

CHAPTER 12 SWAHAA!

Heredity | Days 55–57



BUJHO 3:1, 1:2:1, 9:3:3:1, sab ratios yaad! Aur XX-XY bhi pakka!

PAHELI Ek last chapter bacha hai, Bujho: Our Environment. Phir poori book complete!

CHAPTER XP EARNED
+450 XP

LEVELS CLEARED
3 levels (Days 55–57)

BOOK PROGRESS
Level 57 / 60 7500 XP total

WHAT YOU HAVE MASTERED

- ✔ Variation and heredity
- ✔ Dihybrid cross (9:3:3:1)
- ✔ How genes control traits
- ✔ Mendel's monohybrid cross (3:1)
- ✔ Dominant and recessive traits
- ✔ Sex determination in humans

NEXT CHAPTER Day 58: Our Environment

Chapter 12 cleared!

DAY 58 LEVEL 58

BIOLOGY | CHAPTER 13 | PART 1 ASTRA SERIES

ECOSYSTEM AND FOOD CHAINS

Goal: ecosystem, food chains, food webs, 10% law.



by Nitesh Prakash

01 Ecosystem | 02 Consumers | 03 Food Chains | 04 Food Web | 05 10% Law | +125 XP today | Level 58 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 WHAT IS AN ECOSYSTEM? ★★★ MOST ASKED

DEFINITION

- **Ecosystem:** all the living organisms of an area together with the non-living components (air, water, soil, light, temperature) with which they interact
- Living things depend on each other and on their physical surroundings for food, shelter and growth; together they form one functional unit

Components of an ecosystem ★★★

Biotic (living)	Abiotic (non-living)
Plants (producers)	Sunlight, temperature
Animals (consumers)	Water, air, rainfall
Bacteria, fungi (decomposers)	Soil, minerals, humidity

Natural vs artificial ecosystems ★★☆

Natural	Artificial (human-made)
Forest, grassland, desert	Garden, park
Pond, lake, river, sea	Aquarium
Run without human help	Crop field, zoo; humans maintain them



BUJHO *Aquarium mein toh sirf machhliyan hain, woh ecosystem kaise hua?*

PAHELI *Machhli, paudhe, bacteria (biotic) plus paani, oxygen, light, temperature (abiotic) sab hain. Bas insaan banata aur sambhalta hai, isliye* ***artificial ecosystem****.*



MNEMONIC: ABIOTIC FACTORS

"SWAT-S": Sunlight, Water, Air, Temperature, Soil: yeh paanch jaan nahi rakhte, par ecosystem inke bina nahi chalta.

REMEMBER

An ecosystem can be as small as a pond or as big as a forest.

2 PRODUCERS, CONSUMERS, DECOMPOSERS ★★★ MOST ASKED

Producers (autotrophs) ★★★

- Green plants and blue-green algae make food from simple inorganic substances using sunlight and chlorophyll (photosynthesis)
- All other organisms depend on them, directly or indirectly, for food

Consumers (heterotrophs): 4 types ★★★

Type	Eats	Examples
Herbivores	plants only	cow, deer, goat, rabbit
Carnivores	animals only	lion, tiger, frog, snake
Omnivores	plants and animals	human, crow, bear
Parasites	live on a host and feed on it	lice, tapeworm, Cuscuta

Decomposers: nature's cleaners ★★★

- **Bacteria and fungi** break down dead plants, animals and wastes into simple inorganic substances
- These go back into the soil and are used again by plants: **nutrients are recycled**
- Without decomposers, dead bodies would pile up and soil would lose its fertility



FROGS EXIST

Frog bas exist nahi karta, woh ***secondary consumer (T3)*** *hai: keede khaata hai aur saanp ka khana banta hai.*



PRODUCER POWER

Grass sunlight se khana banati hai, deer seedha le leta hai. Yahi hai ***T1 → T2*** *energy transfer.*

3 FOOD CHAINS AND TROPHIC LEVELS

★★★ MOST ASKED

A five-step food chain ★★★



Arrow = direction of energy (food) flow

T1 producer → T2 primary → T3 secondary → T4 tertiary consumer

DEFINITION

- **Food chain:** a series of organisms in which each one eats the one before it; it shows how food and energy pass from one organism to the next
- **Trophic level:** each step or level of a food chain where energy is transferred; producers are always at the first trophic level (T1)

Trophic levels at a glance ★★★

Level	Who	Example
T1	Producers	grass, phytoplankton
T2	Primary consumers (herbivores)	grasshopper, deer
T3	Secondary consumers (small carnivores)	frog, small fish
T4	Tertiary consumers (large carnivores)	snake, lion, eagle

Food chains you can write in the exam ★★★

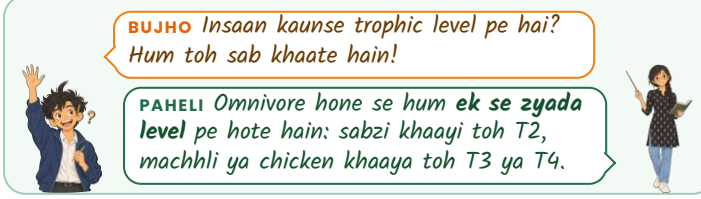
- **Grassland:** grass → deer → lion
- **Pond:** phytoplankton → zooplankton → small fish → large fish
- **Forest:** plants → insects → frog → snake → eagle
- **Field:** grains → mouse → snake → hawk

NITESH BHAIYA KI GUARANTEE

★★★ Food chain likho toh **arrow khane se khane wale ki taraf** (grass → deer), ulta arrow = seedha minus. Board mein yeh har saal aata hai!

NCERT thinker: remove one trophic level? ★★☆

- Kill all herbivores: producers grow unchecked and carnivores starve and die
- No trophic level can be removed without harm; the whole balance of the ecosystem is upset



MNEMONIC: ORDER OF A FOOD CHAIN

"Pehle Paudha, Phir Pet-bhar": Producer → Primary → Secondary → Tertiary; yaad karo "P-P-S-T: Pani Puri Sabko Tasty".

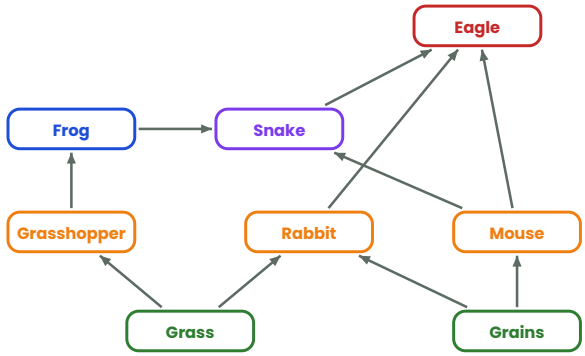
4 FOOD WEB

★★★ MOST ASKED

DEFINITION

- **Food web:** a network of many interconnected food chains in an ecosystem, because one organism can be eaten by several others
- A food web is more stable: if one food source is lost, consumers can switch to another

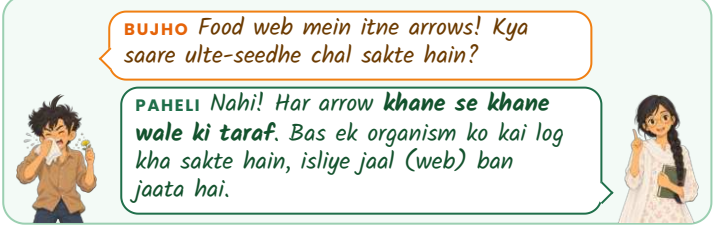
Food web in a grassland ★★★



Arrows show the direction of energy flow, from the food to the eater.

Food chain vs food web ★★☆

Food chain	Food web
A single straight path	Many interlinked chains
Each organism has one food source	An organism may have many food sources
Removal of one link breaks it	More stable; has alternative paths





SUCH ECOSYSTEM

*Sunlight se producer, producer se consumer, aur har step pe sirf **10% aage**. Doge bhi bolta: wow, such loss!*

EXAM TIP

A food web diagram must have at least 5 organisms, labelled, with every arrow from the food to the eater.

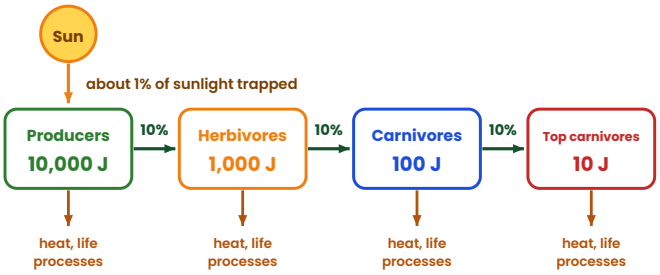
5 ENERGY FLOW AND THE 10% LAW

★★★ MOST ASKED

10% LAW

- **Ten percent law (Lindeman):** only about **10%** of the energy at one trophic level is passed on to the next; the rest is used in life processes and lost as heat
- Green plants trap only about **1%** of the sunlight falling on their leaves and change it into food (chemical energy)

Energy flow through a food chain ★★★



Worked numericals (verified) ★★★

1 Producers trap 10,000 J

- T2 = 10,000 × 10/100 = 1,000 J; T3 = 100 J; T4 = **10 J**

2 Grass → deer → lion, grass has 20,000 J

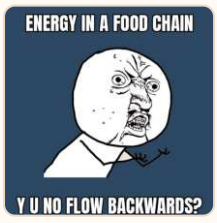
- Deer = 2,000 J; lion = 2,000 × 10/100 = **200 J**

3 A snake at T4 gets 5 J. Energy at T1?

- Go back ×10 per step: 5 → 50 (T3) → 500 (T2) → **5,000 J** (T1)

MNEMONIC: 10% LAW

"Das ka dum, baaki garam": har step pe 10% aage jaata hai, baaki 90% life processes aur heat mein chala jaata.



ONE WAY

*Energy ka rasta **one-way** hai: Sun se producer, producer se consumer. Wapas nahi aata, chahe kitna bhi Y U No bolo.*

6 PRACTICE MCQS ★★★ MOST ASKED

1. Which of these is an artificial ecosystem? NCERT ★★★

(a) forest (b) pond
(c) crop field (d) lake

2. Which of the following is an abiotic component of an ecosystem? ★★☆

(a) bacteria (b) temperature
(c) fungi (d) grass

3. Decomposers in an ecosystem are mainly: ★★★

(a) green plants (b) bacteria and fungi
(c) herbivores (d) parasites

Energy flow is unidirectional ★★★

- Energy moves only one way: Sun → producers → herbivores → carnivores
- Energy passed to the next level never comes back to the previous level; heat lost cannot be reused by plants
- So every ecosystem needs a constant input of energy from the Sun

Why only 3–4 steps in a food chain? ★★★

- About 90% of the energy is lost at every step
- After 4 levels very little energy is left (10,000 J → 10 J), not enough to support another level of organisms
- So there are fewer organisms at higher levels, and top carnivores are few

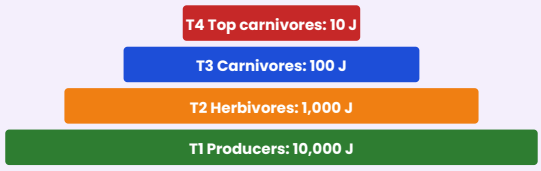


TOP PAR BHI GHAATA

*Top carnivore bano aur mile sirf **10 J out of 10,000 J**. 10% law ne Brian ko phir haraaya.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The **pyramid of energy** stacks trophic levels by the energy they hold. Because energy falls ten-fold at each step, it is always upright, never inverted.



Common mistakes that cost marks

× WRONG	✓ RIGHT
Arrow from lion to deer	Arrow from food to eater: deer → lion
Plants trap 10% of sunlight	About 1% of sunlight; 10% is between trophic levels
Decomposers are producers	They break down dead matter; plants are producers
Energy flow is cyclic	Energy is unidirectional; nutrients cycle

Day 58 hit-list: Eduspark PYQ analysis

Topic	Rating
10% law and energy numericals	★★★
Food chain and trophic levels	★★★
Role of decomposers	★★★
Unidirectional flow of energy	★★☆
Food web (diagram)	★★☆
Natural vs artificial ecosystems	★☆☆

4. Which of the following constitute a food chain? NCERT ★★★
(a) grass, wheat and mango (b) grass, goat and human
(c) goat, cow and elephant (d) grass, fish and goat

5. In the chain grass → deer → lion, the lion is at trophic level: ★★☆
(a) T1 (b) T2
(c) T3 (d) T4

6. If producers trap 10,000 J, the energy available to secondary consumers is: ★★★
(a) 1,000 J (b) 100 J
(c) 10 J (d) 1 J

7. The flow of energy in an ecosystem is: ★★★
(a) cyclic (b) unidirectional
(c) bidirectional (d) random

8. Green plants convert about what percentage of the sunlight falling on their leaves into food energy? ★★☆
(a) 1% (b) 10%
(c) 50% (d) 100%

9. **Assertion (A):** Food chains generally have only three or four trophic levels.
Reason (R): Only about 10% of the energy passes to the next level, so very little energy is left after four levels. ★★★
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

10. **Assertion (A):** An aquarium is an artificial ecosystem.
Reason (R): An aquarium has only biotic components. ★★☆
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

Short and long answer questions

1. What are decomposers? What will happen if there are no decomposers in an ecosystem? NCERT **(2 marks)**
2. Why are food chains generally limited to three or four trophic levels? **(2 marks)**
3. What are trophic levels? Give an example of a food chain and state the different trophic levels in it. NCERT **(3 marks)**
4. Draw a food web with at least five organisms. In the chain grass → deer → lion, grass has 20,000 J. Find the energy reaching the lion and explain why energy flow is unidirectional. **(5 marks)**



NITESH SIR SAYS
Food chain ka arrow aur 10% ka hisaab, yeh do cheezein pakki karo toh is topic ke marks pakke. ***Kal milte hai Day 59 pe!***

★ How we starred (Eduspark PYQ analysis): we counted how often each topic has come in board and sample-paper questions on this chapter. Year tags appear only where the paper year is known.

Answer key: MCQs

Q	Ans	Why
1	(c)	Crop fields are made and kept by humans.
2	(b)	Temperature is non-living.
3	(b)	They break down dead matter.
4	(b)	Producer → herbivore → omnivore.
5	(c)	Grass T1, deer T2, lion T3.
6	(b)	10,000 → 1,000 (T2) → 100 J (T3).
7	(b)	Energy never flows back.
8	(a)	About 1% of the sunlight.
9	(a)	R explains A.
10	(c)	R false: it has abiotic parts too.

Model answers

1 Decomposers
- Bacteria and fungi that break down dead remains into simple substances (1); without them dead matter piles up and nutrients are not returned to the soil (1)

2 3–4 trophic levels
- 90% energy lost at each level (1); after 4 levels too little is left to support another level (1)

3 Trophic levels
- Each step of a food chain where energy is transferred (1)
- Grass → grasshopper → frog → snake (1)
- Grass T1 producer, grasshopper T2, frog T3, snake T4 (1)

4 Food web and energy
- Food web diagram with five or more organisms and correct arrows (2)
- Deer 2,000 J; lion 200 J (1)
- Energy flows Sun → producers → consumers and never returns; heat lost cannot be reused (2)

Day 58 checklist: tick before you sleep

- ☐ I can define an ecosystem and list its biotic and abiotic components
- ☐ I can classify consumers and explain the role of decomposers
- ☐ I can draw a food chain and name its trophic levels
- ☐ I can draw a food web and say why it is more stable
- ☐ I can solve 10% law numericals and explain unidirectional energy flow

My score: ____ / 5

LEVEL 58 COMPLETE +125 XP earned. Next: Level 59



BUJHO *Samajh gaya! Arrow khane wale ki taraf, aur har step pe sirf 10%.*

PAHELI *Shabaash! Kal Level 59: biological magnification, ozone aur garbage.*

DAY 59 LEVEL 59

BIOLOGY | CHAPTER 13 | PART 2 ASTRA SERIES

OUR ENVIRONMENT & BOSS LEVEL

Goal: DDT, ozone and garbage, then the Chapter 13 Boss.



by Nitesh Prakash

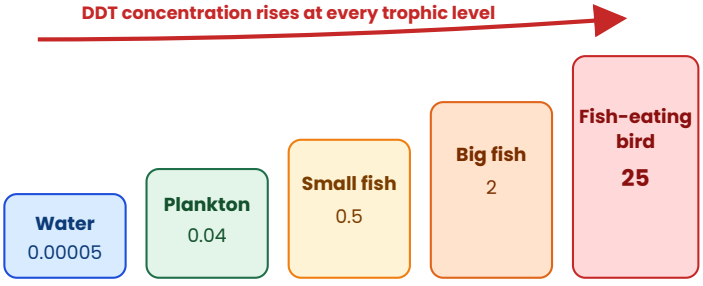
01 Biomagnification | 02 Ozone Layer | 03 CFCs and UNEP | 04 Garbage | 05 Boss Level | +175 XP today | Level 59 / 60

STAR TAGS: ★★★ **MOST ASKED** = Most frequently asked (Eduspark PYQ analysis) | ★★☆ = Often asked | ★☆☆ = Asked

1 BIOLOGICAL MAGNIFICATION ★★★ MOST ASKED

DEFINITION

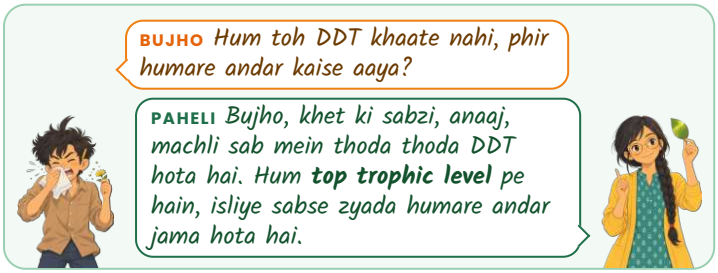
- **Biological magnification:** the progressive **increase in concentration** of harmful, non-biodegradable chemicals (like pesticides) at each higher trophic level of a food chain
- Pesticides sprayed on crops are washed into the **soil and water bodies**, are absorbed by plants with water and minerals, and so **enter the food chain**
- Such chemicals are **not degraded** and not excreted, so they **accumulate** in the body (mostly in fat)



DDT in ppm (parts per million), classic lake study

Why does the amount keep rising? ★★☆

- Each organism eats a **large amount** of the food of the lower level, so it collects the chemical from **many** organisms
- The chemical stays in its body for life, so the dose **adds up** step by step
- Result: the **top consumer** carries the highest concentration



WHY LEAST IN PRODUCERS?

Plants take the pesticide only from soil and water, so their level is the **lowest**. Every consumer above eats many organisms, so its level jumps.

Humans: the top trophic level ★★★

- Humans eat plants, animals and fish, so they occupy the **top level** in many food chains
- Hence the **maximum concentration** of pesticides builds up in humans (found even in food grains, vegetables, fish and milk)
- Common example: **DDT**, a pesticide that is non-biodegradable and fat-soluble



MEME BREAK

Top of the food chain hone ka "reward": sabse zyada DDT! Biological magnification ka matlab yahi hai.



★★★ *"Concentration of pesticides is maximum in humans or the top carnivore" ye one-liner har saal aata hai. Nitesh Bhaiya ki guarantee!*

MNEMONIC: BIOLOGICAL MAGNIFICATION

"Neeche kam, upar dam": neeche wale level mein chemical kam, upar wale level pe sabse zyada jama.

2 OZONE LAYER ★★★ MOST ASKED

DEFINITION

- **Ozone (O_3):** a molecule of **three oxygen atoms**; found in the upper atmosphere (stratosphere) as the **ozone layer**
- It **shields** the surface of the Earth from harmful **ultraviolet (UV)** radiation of the Sun
- At ground level ozone is a **deadly poison**; only the high-altitude layer helps us

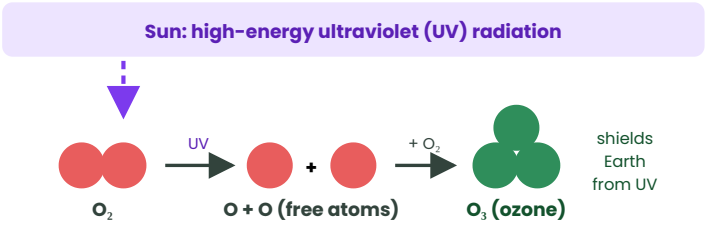
How ozone is formed ★★★

- UV radiation splits an oxygen molecule into free atoms:

$$O_2 \xrightarrow{(UV)} O + O$$

- Each free atom combines with another oxygen molecule:

$$O + O_2 \rightarrow O_3 \text{ (ozone)}$$

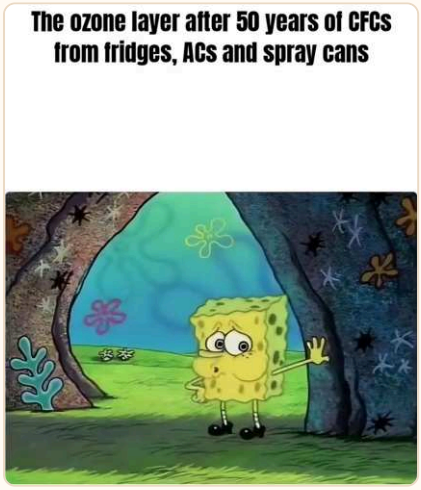


Depletion of the ozone layer ★★★

- The amount of ozone began to **fall sharply in the 1980s** (the "ozone hole")
- Main cause: **CFCs** (chlorofluorocarbons), used as **refrigerants** in fridges and ACs and in **fire extinguishers** and aerosol sprays
- Less ozone means more UV reaches us: **skin cancer**, cataract, weaker immunity and harm to plants and animals
- **1987:** the **UNEP** (United Nations Environment Programme) got an agreement to **freeze CFC production at 1986 levels**; now CFC-free fridges are made

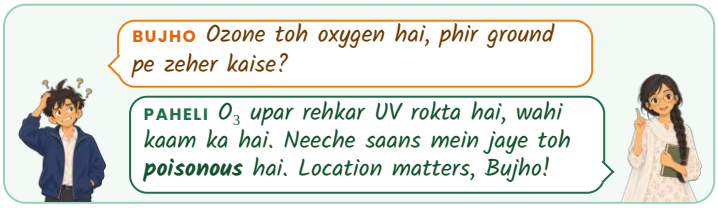
MNEMONIC: OZONE FACTS

"3-O, UV ka shield; CFC villain; 86 pe freeze, 87 ka deal": O_3 teen oxygen, CFC se depletion, UNEP 1987 ne production 1986 level pe freeze kiya.



MEME BREAK

Ozone layer bhi thak gayi hai yaar! Fridge aur AC CFC-free lo, warna UV seedha skin pe.



FACT FILE

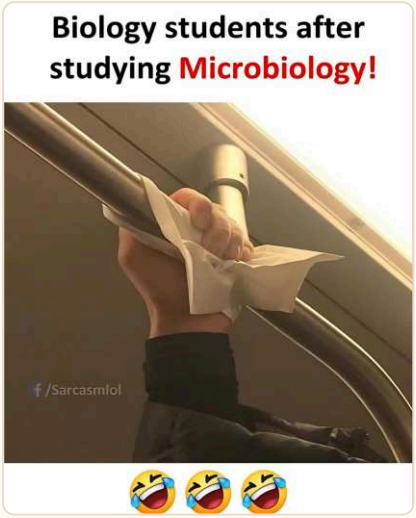
The thinning is worst over **Antarctica**, where an "ozone hole" forms every spring. Thanks to the CFC ban, the layer is now slowly healing.

3 MANAGING THE GARBAGE ★★★ MOST ASKED

Biodegradable vs non-biodegradable ★★★

Basis	Biodegradable	Non-biodegradable
Meaning	broken down by **biological processes** (bacteria, fungi)	not broken down by biological processes
Effect	return to nature, no lasting harm	persist for long, harm the ecosystem
Examples	vegetable and fruit peels, paper, cotton, wood, leaves, cow dung	plastics, polythene, glass, DDT, metal cans, synthetic fibres

Enzymes act on substances in a **specific** way, which is why they cannot break down plastics. Inert substances simply **persist**.



MEME BREAK

Decomposers ka asli kaam: bacteria aur fungi biodegradable kachre ko tod kar mitti mein wapas bhejte hain.

NCERT debate: disposable cups and kulhads ★★☆

- Trains once served tea in **plastic cups** that had to be returned; later **disposable plastic cups** came, creating huge waste
- **Kulhads** (clay cups) were suggested as a substitute, but making them on a large scale would use up **fertile top-soil**
- Now **disposable paper cups** are used: paper is biodegradable, but it still uses trees, so **reuse** is the real answer

Ways to dispose of waste ★★☆

Method	What happens
Segregation	separate wet (green bin) and dry (blue bin) waste at home
Composting	kitchen and garden waste turn into manure; earthworms speed it up (vermicomposting)
Recycling	paper, plastic, glass, metal are made into new products
Landfill	waste is dumped in low-lying areas and covered with soil
Incineration	burning at high temperature; reduces bulk but pollutes air



MEME BREAK

*Green bin mein chips ka packet? Arthur ki tarah gussa control karo aur use **dry waste** wale bin mein daalo.*

EXTRA KNOWLEDGE (HIGH LEVEL) not in current CBSE syllabus

The 3 Rs to save the environment: **Reduce** (use less, e.g. switch off fans and taps), **Reuse** (use things again, e.g. jars, envelopes), **Recycle** (send plastic, paper, glass, metal for remaking).
Reuse is better than recycling because recycling itself uses energy.

Plastics: the big problem

- Plastic is **non-biodegradable**: it stays in the soil for hundreds of years and blocks drains
- Animals like cows eat polythene bags and fall ill; burning plastic releases **toxic gases**
- Carry a **cloth bag**, refuse single-use plastic, and segregate

MNEMONIC: BIODEGRADABLE TEST

"Jo sadta hai, wo bachta hai": jo cheez gal-sad jaati hai (peels, paper, leaves) wo biodegradable; plastic, glass, DDT kabhi nahi sadte.



MEME BREAK

Squid Game ka glass bridge: kulhad wala panel mitti mein mil jaata hai, plastic wala 500 saal tak toot ke bhi nahi jaata. Sahi chuno!

Exam tips and common mistakes

- ✔ Write "concentration **increases** at each trophic level" for biological magnification, not "energy increases"
- ✔ Ozone equations: write **both steps** with UV over the first arrow
- ✔ UNEP agreement: **1987**, freeze at **1986** levels; do not swap the years
- ✔ Kulhad answer must mention **loss of fertile top-soil**



NITESH SIR SAYS

*Environment ke 5 marks pakke hain: DDT, ozone aur garbage ka table yaad karo. Ab Boss Level, poora Chapter 13, ek saath. **Chalo, jalwa dikhao!***

4 BOSS LEVEL: OUR ENVIRONMENT

★★★ MOST ASKED

Boss rules: 60 minutes, 30 marks

1. Section A: 9 MCQs incl. 2 Assertion–Reason (1 mark each) = **9**
2. Section B: 3 × 2 marks = **6**; Section C: 2 × 3 marks = **6**
3. Section D: 1 × 5 marks = **5**; Section E: 1 case-based × 4 marks = **4**
4. Covers the whole chapter: Day 58 (ecosystem, food chains, 10% law) and Day 59

SECTION A (1 mark each)

1. Which of these is an artificial ecosystem? ★★☆

(a) pond (b) forest
(c) crop field (d) lake

2. Which of the following constitutes a food chain? ★★★

(a) grass, wheat, mango (b) grass, goat, human
(c) goat, cow, elephant (d) grass, fish, goat

3. Green plants in a food chain form the: ★★☆

(a) first trophic level (b) second trophic level
(c) top trophic level (d) decomposer level

4. If 20,000 J of energy is trapped by producers, the energy reaching secondary consumers is: ★★★

(a) 2000 J (b) 200 J
(c) 20 J (d) 2 J

5. The flow of energy in an ecosystem is: ★★☆

(a) cyclic (b) bidirectional
(c) unidirectional (d) reversible

6. Which group contains only non-biodegradable items? ★★★

(a) paper, glass, DDT (b) polythene, glass, DDT
(c) peels, wood, plastic (d) leaves, cotton, cans

7. Ozone in the upper atmosphere is formed when: ★★★

(a) UV splits O_2 and O joins O_2 (b) CFCs react with O_2
(c) CO_2 breaks down (d) water vapour splits

8. **Assertion (A):** A food chain usually has only three or four trophic levels.
Reason (R): Only about 10% of the energy of one trophic level is passed on to the next. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

9. **Assertion (A):** The concentration of DDT is highest in humans.
Reason (R): DDT is a biodegradable chemical. ★★★

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B (2 marks each)

10. What are decomposers? What would happen if all the decomposers of an ecosystem were removed? **(2)**

11. What will happen if we kill all the organisms of one trophic level, for example all herbivores? **(2)**

12. Why is damage to the ozone layer a cause of concern? What step was taken in 1987 to limit it? **(2)**

SECTION C (3 marks each)

13. **Diagram-based:** Draw a food chain of four trophic levels in a grassland and label each trophic level. If the producers fix 15,000 J, how much energy reaches the top level? **(3)**

14. What is biological magnification? Will its level be different at different trophic levels? Explain with an example. **(3)**

SECTION D (5 marks)

15. (a) Give two differences between biodegradable and non-biodegradable substances, with two examples of each. (b) Suggest three ways to reduce the garbage produced at home. (c) Why are kulhads not a good large-scale substitute for disposable cups? **(2 + 2 + 1)**

SECTION E (case-based, 4 marks)

Q16. Case: the lake near the cotton fields

A lake beside cotton fields sprayed with a pesticide was tested. Pesticide level (ppm): water 0.02, algae 5, small fish 40, large fish 300, fish-eating bird 1500.

- (i) Name the phenomenon shown by the data. **(1)**
- (ii) Write the food chain and the trophic level of the bird. **(1)**
- (iii) Why does the pesticide not get broken down or excreted? **(1)**
- (iv) Suggest one way farmers can reduce this problem. **(1)**

5 ANSWER KEY ★★★

Answer key: Section A

Q	Ans	Why
1	(c)	Crop fields, gardens, aquaria are made by humans.
2	(b)	Producer → herbivore → omnivore.
3	(a)	Producers (autotrophs) are level 1.
4	(b)	20,000 → 2000 → 200 J (10% law).
5	(c)	Energy does not return to the Sun.
6	(b)	Paper is biodegradable, so (a) is out.
7	(a)	$O_2 \rightarrow O + O$ (UV); $O + O_2 \rightarrow O_3$.
8	(a)	90% loss per step; R explains A.
9	(c)	A true; DDT is non-biodegradable.

BOSS STRATEGY

Section A in 10 min, B and C in 25 min, D and E in 20 min, last 5 min to check units and labels. Food-chain arrows always point **towards the eater**.

6 MARKING POINTS ★★★

Marking points: Sections B to E

10 Decomposers
- Bacteria and fungi that break down dead remains and wastes into simple substances (1)
- Without them, dead matter piles up and nutrients are not returned to the soil (1)

11 Killing one level
- The level above (carnivores) starves and declines (1)
- The level below (plants) grows unchecked; the whole ecosystem is upset (1)

12 Ozone concern
- Less ozone lets UV in: skin cancer, cataract, harm to crops (1)
- UNEP, 1987: CFC production frozen at 1986 levels (1)

13 Food chain diagram
- Grass → grasshopper → frog → snake with arrows (1); levels labelled T1 to T4 (1)
- 15,000 → 1500 → 150 → **15 J** at the top level (1)

14 Biological magnification
- Definition: rise in concentration of harmful chemicals at each higher level (1)
- Yes, highest at the top (1); DDT: water → plankton → fish → birds or humans (1)

15 Garbage (5 marks)
- (a) Breakdown by microbes or not; harmless or persistent (1); examples: peels, paper / plastic, glass (1)
- (b) Segregate wet and dry, compost kitchen waste, carry cloth bags (2)
- (c) Making them on a large scale wastes fertile top-soil (1)

16 Case answers
- (i) Biological magnification (ii) water → algae → small fish → large fish → bird; bird is at the top (fifth) level
- (iii) It is non-biodegradable and stored in fat (iv) Use less pesticide, or biological pest control

Day 59 checklist: tick before you sleep

- [ ] I can explain biological magnification with the DDT example
- [ ] I can write both ozone-formation equations and the CFC story
- [ ] I know the 1987 UNEP step and the 1986 freeze level
- [ ] I can classify waste and argue the kulhad debate
- [ ] I finished the Boss Level and checked it with the key

My score: ____ / 5

LEVEL 59 COMPLETE +175 XP earned. Next: Level 60



BUJHO *Chapter 13 ka Boss bhi gaya! DDT upar jama hota hai, ozone UV rokta hai, kachra alag karo.*

PAHELI *Shabaash! Kal Level 60: Science Grand Mock, poori book.*

DAY 60
LEVEL 60

SCIENCE | ALL 13 CHAPTERS | GRAND MOCK

SCIENCE GRAND MOCK TEST

Goal: write one full 80-mark board paper in 3 hours.



by Nitesh Prakash

01 Instructions | 02 Sections A–E | 03 39 Questions | 04 Answer Key | 05 Marking Hints | +200 XP Boss Level | Level 60 / 60

SCIENCE (086) GRAND MOCK

Class X | Time: 3 hours | Max. Marks: 80

Based on the CBSE 2026-27 syllabus and sample-paper pattern

GENERAL INSTRUCTIONS

1. This question paper has 39 questions in 5 sections, A to E. All questions are compulsory; internal choices are given in some questions.
2. Section A: Q1–20 are multiple choice questions of 1 mark each (Q17–20 are Assertion–Reason).
3. Section B: Q21–26 are very short answer questions of 2 marks each (30 to 50 words).
4. Section C: Q27–33 are short answer questions of 3 marks each (50 to 80 words).
5. Section D: Q34–36 are long answer questions of 5 marks each (80 to 120 words).
6. Section E: Q37–39 are case-based units of 4 marks each, with sub-parts of 1, 1 and 2 marks.

Unit tally: Chemistry 25 | Biology (World of Living) 25 | Light and Eye 12 | Electricity and Magnetism 13 | Our Environment 5 | Total 80

SECTION A — 20 × 1 = 20 marks

1. Which of the following is a decomposition reaction? 1

(a) $CaO + H_2O \rightarrow Ca(OH)_2$

(b) $2AgCl \rightarrow 2Ag + Cl_2$ (in sunlight)

(c) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

(d) $NaOH + HCl \rightarrow NaCl + H_2O$

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 1 (b) 4

(c) 5 (d) 10

3. Which metal is kept immersed in kerosene to stop it catching fire? 1

(a) sodium (b) magnesium

(c) aluminium (d) zinc

4. The functional group present in ethanoic acid is: 1

(a) –OH (b) –CHO

(c) –COOH (d) >C=O

5. Pyruvate is broken down completely into CO_2 and water, releasing the most energy, in: 1

(a) cytoplasm (b) mitochondria

(c) chloroplast (d) nucleus

6. The blood vessel that brings oxygenated blood from the lungs to the heart is: 1

(a) pulmonary artery (b) pulmonary vein

(c) aorta (d) vena cava

7. The plant hormone that promotes cell division is: 1

(a) auxin (b) abscisic acid

(c) cytokinin (d) gibberellin

8. Plasmodium reproduces asexually by: 1

(a) budding (b) binary fission

(c) fragmentation (d) multiple fission

9. In the F_2 generation of Mendel's monohybrid cross (TT × tt), the percentage of dwarf plants is: 1

(a) 75% (b) 50%

(c) 25% (d) 0%

10. The power of a lens is −2 D. The lens and its focal length are: 1

(a) convex, +50 cm (b) concave, −50 cm

(c) concave, −2 m (d) convex, +0.5 cm

11. The focal length of the eye lens is adjusted by the: 1

(a) iris (b) pupil

(c) ciliary muscles (d) retina

12. A wire of resistance R is stretched uniformly to double its length. Its new resistance is: 1

(a) R/2 (b) R

(c) 2R (d) 4R

13. One kilowatt hour (1 kWh) is equal to: 1

(a) 3.6×10^3 J (b) 3.6×10^5 J
(c) 3.6×10^6 J (d) 3.6×10^9 J

14. The direction of the force on a current-carrying conductor placed in a magnetic field is given by: 1

(a) right-hand thumb rule (b) Fleming's left-hand rule
(c) Ohm's law (d) Snell's law

15. Which of the following does NOT depend on the length of a wire? 1

(a) resistance (b) resistivity
(c) both (d) neither

16. Which group contains only biodegradable items? 1

(a) grass, flowers, leather
(b) grass, wood, plastic
(c) fruit peels, cake, polythene
(d) cake, wood, glass

Q17–20: ASSERTION–REASON

Two statements are given, Assertion (A) and Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

17. **Assertion (A):** Silver chloride kept in sunlight turns grey. 1
Reason (R): Sunlight decomposes silver chloride into silver and chlorine.

18. **Assertion (A):** Lactic acid is formed in our muscle cells when there is a lack of oxygen. 1
Reason (R): In yeast, pyruvate is converted into lactic acid in the cytoplasm.

19. **Assertion (A):** The image formed by a concave lens is always real. 1
Reason (R): A concave lens always forms a virtual, erect and diminished image.

20. **Assertion (A):** The flow of energy in an ecosystem is unidirectional. 1
Reason (R): The energy that passes to the herbivores does not come back to the autotrophs.

SECTION B — 6 × 2 = 12 marks

21. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, identify the substance oxidised and the substance reduced. Why is it called a redox reaction? 2

OR

Why should a magnesium ribbon be cleaned before burning it in air? Write the balanced equation for its burning.

22. An electric iron draws a current of 4 A from the 220 V mains. Find its power and the energy it uses in 2 hours, in kWh. 2

23. State the role of (a) salivary amylase and (b) bile juice in the digestion of food. 2

24. What is a reflex arc? Write, in the correct order, the components through which the impulse travels in a reflex action. 2

25. A concave mirror forms a real image three times the size of an object placed 10 cm in front of it. Where is the image located? 2

26. Draw the pattern of magnetic field lines around a straight conductor carrying current. What happens to the field when the current is increased? 2

OR

State Fleming's left-hand rule. What happens to the force on the conductor if the direction of the current is reversed?

SECTION C — 7 × 3 = 21 marks

27. A shiny brown metal X, used in electric wires, turns black when heated in air. Name X and the black compound. Write the balanced equation and name the type of reaction involved. 3

28. Write the chemical name and formula of (a) bleaching powder, (b) baking soda and (c) washing soda. Give one use of each. 3

29. Explain with a labelled diagram how soap molecules form micelles and remove oily dirt from clothes. 3

30. What is double circulation? Why is the separation of oxygenated and deoxygenated blood necessary in mammals and birds? 3

31. (a) What is the role of the placenta? (b) Name one barrier method and one surgical method of contraception. (c) Why is female foeticide a threat to society? 3

32. A 2 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, size and nature of the image. 3

33. What is biological magnification? Will its level be the same at every trophic level? Explain with the example of a pesticide such as DDT. 3

SECTION D — 3 × 5 = 15 marks

34. Answer the following about ionic compounds. 5

(a) Show the formation of NaCl and MgO by the transfer of electrons. **(2)**

(b) Why do ionic compounds have high melting points? **(1)**

(c) Why do they conduct electricity in the molten state but not in the solid state? **(2)**

OR

Answer the following about metallurgy.

(a) Distinguish between roasting and calcination with one equation each. **(2)**

(b) Write the thermit reaction and one of its uses. **(2)**

(c) Why are sulphide and carbonate ores changed into oxides before reduction? **(1)**

35. Answer the following about excretion in humans. 5

(a) Draw a neat labelled diagram of a nephron. **(2)**

(b) Explain how urine is formed in it. **(2)**

(c) How is the amount of urine produced regulated? **(1)**

OR

Answer the following about sexual reproduction in plants.

(a) Draw a labelled diagram of the germination of pollen on the stigma. **(2)**

(b) Describe fertilisation in flowering plants. What do the ovule and the ovary develop into? **(3)**

36. Resistors of 2 Ω, 3 Ω and 6 Ω are given. 5

(a) Find their equivalent resistance in series and in parallel. **(2)**

(b) The parallel combination is connected to a 6 V battery. Find the total current and the current through the 2 Ω resistor. **(2)**

(c) Why are household appliances connected in parallel? **(1)**

OR

Answer the following about electric power.

(a) Define electric power and show that $P = I^2R$. **(2)**

(b) A bulb is rated 100 W, 220 V. Find the current it draws and its resistance. **(2)**

(c) Find the energy it uses in 10 hours, in kWh. **(1)**

SECTION E — 3 × 4 = 12 marks

37. Five solutions A to E were tested with universal indicator paper. Their pH values are given below. 4

Solution	A	B	C	D	E
pH	1	4	7	10	13

(i) Which solution is strongly alkaline? **(1)**

(ii) Which solution is neutral? Could it be pure water? **(1)**

(iii) Arrange A to E in increasing order of hydrogen ion concentration. **(2)**

OR

(iii) Name the reaction when A is mixed with E, and give its general word equation. **(2)**

38. Mendel crossed pure pea plants with round yellow seeds (RRYY) with plants having wrinkled green seeds (rryy). All F_1 seeds were round yellow. Selfing F_1 gave 556 seeds: 315 round yellow, 108 round green, 101 wrinkled yellow and 32 wrinkled green. 4

(i) Which traits are dominant? **(1)**

(ii) Write the genotype of the F_1 plants and the gametes they form. **(1)**

(iii) Find the F_2 phenotypic ratio and name the law it proves. **(2)**

OR

(iii) Name the new combinations that appear in F_2. What fraction of the F_2 plants is RRYY? **(2)**

39. Riya, a Class 10 student, can read her book clearly but cannot see the writing on the board clearly from the last bench. The doctor finds that her far point is 2 m. 4

(i) Name the defect of vision she has. **(1)**

(ii) State one cause of this defect. **(1)**

(iii) Find the power of the lens she needs and name the lens. **(2)**

OR

(iii) Draw ray diagrams of her eye before and after correction. **(2)**

BLUEPRINT: MARKS BY UNIT AND SECTION

Unit	A	B	C	D	E	Total
Chemical Substances	5	2	9	5	4	**25**
World of Living	6	4	6	5	4	**25**
Light and Eye	3	2	3	0	4	**12**
Electricity, Magnetism	4	4	0	5	0	**13**
Our Environment	2	0	3	0	0	**5**
Total	**20**	**12**	**21**	**15**	**12**	**80**

END OF QUESTION PAPER

2 ANSWER KEY AND MARKING HINTS

Section A: answers — 20 marks

1 (b)	2 (d)	3 (a)	4 (c)	5 (b)
6 (b)	7 (c)	8 (d)	9 (c)	10 (b)
11 (c)	12 (d)	13 (c)	14 (b)	15 (b)
16 (a)	17 (a)	18 (c)	19 (d)	20 (a)

A–R: 17 (a) sunlight splits AgCl | 18 (c) yeast makes ethanol + CO_2 | 19 (d) concave lens image is virtual | 20 (a) energy never returns

Section B: value points — 2 each

- **21** H_2 is oxidised (gains O), CuO is reduced (loses O); both together = redox. OR: removes the oxide layer; $2Mg + O_2 \rightarrow 2MgO$
- **22** P = VI = 220 × 4 = **880 W**; E = 0.88 kW × 2 h = **1.76 kWh**
- **23** (a) starch → sugar (maltose) in the mouth; (b) emulsifies fats and makes the medium alkaline for lipase
- **24** Nerve pathway of a reflex: receptor → sensory neuron → relay neuron (spinal cord) → motor neuron → effector (muscle/gland)
- **25** m = −v/u = −3, u = −10 cm, so **v = −30 cm**: 30 cm in front of the mirror
- **26** Concentric circles around the wire (diagram); stronger field when the current is increased. OR: forefinger field, middle finger current, thumb force (all at right angles); force reverses

Section C: value points — 3 each

- **27** X = copper, black CuO; $2Cu + O_2 \rightarrow 2CuO$; combination (oxidation) reaction
- **28** Calcium oxychloride $CaOCl_2$ (disinfecting water); sodium hydrogencarbonate $NaHCO_3$ (baking, antacid); sodium carbonate decahydrate $Na_2CO_3.10H_2O$ (removes hardness of water)
- **29** Ionic head in water, carbon tail in oil; tails trap the oil inside a micelle; the micelles are rinsed away with water (diagram 1 mark)
- **30** Blood passes through the heart twice per cycle (body → right side → lungs → left side → body); separation gives the high O_2 supply needed by warm-blooded animals
- **31** (a) supplies food and O_2 to the embryo and removes wastes; (b) condom; vasectomy or tubectomy; (c) lowers the sex ratio and is illegal and unjust
- **32** 1/v = 1/10 − 1/15 = 1/30, **v = +30 cm**; m = v/u = −2; h′ = **−4 cm**: real, inverted, magnified
- **33** Non-biodegradable chemicals build up at each trophic level; highest at the top (humans, birds of prey); e.g. DDT in water → plankton → fish → birds

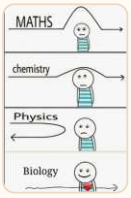
Score tracker

Section	Max.	My marks
A (MCQ, A–R)	20	____
B + C	33	____
D + E	27	____
Total	**80**	____

70+ Topper zone | 60–69 Strong | 45–59 Revise weak chapters | below 45 Redo the chapter Boss Levels

Sections D and E: marking points — 5 and 4 each

- **34** Na(2,8,1) gives $1e^-$ to Cl(2,8,7); Mg(2,8,2) gives $2e^-$ to O(2,6); strong forces of attraction between ions; ions move freely only in the molten state. OR: roasting $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; calcination $ZnCO_3 \rightarrow ZnO + CO_2$; $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat (joining rail tracks); oxides are easier to reduce
- **35** Labels: glomerulus, Bowman's capsule, tubule, collecting duct; filtration in the glomerulus, selective reabsorption of glucose, amino acids, salts and water in the tubule; amount depends on excess water and dissolved wastes. OR: pollen tube grows through the style to the ovule; male gamete fuses with the egg to form the zygote; ovule → seed, ovary → fruit
- **36** Series 11 Ω; parallel 1/R = 1/2 + 1/3 + 1/6, **R = 1 Ω**; I = 6/1 = **6 A**; $I_{2\Omega}$ = 6/2 = **3 A**; same voltage, separate switches, one fault does not stop the rest. OR: P = VI = I^2R; I = 100/220 = **0.45 A**; R = $220^2/100$ = **484 Ω**; E = 0.1 kW × 10 h = **1 kWh**
- **37** (i) E; (ii) C, yes; (iii) E < D < C < B < A. OR: neutralisation, acid + base → salt + water
- **38** (i) round, yellow; (ii) RrYy; gametes RY, Ry, rY, ry; (iii) 9 : 3 : 3 : 1, law of independent assortment. OR: round green, wrinkled yellow; 1/16
- **39** (i) myopia; (ii) excessive curvature of the eye lens or elongated eyeball; (iii) f = −2 m, P = 1/f = **−0.5 D**, concave lens. OR: image in front of the retina; concave lens shifts it on to the retina



BOOK COMPLETE
*Maths, Chemistry, Physics, Biology: graph jaisa bhi ho, **Day 60 tak aa gaye** toh tum already topper ho!*

Day 60 checklist: tick before you sleep

- ☐ I wrote the full paper in 3 hours
- ☐ I attempted every OR choice once
- ☐ I checked my answers with the key
- ☐ I redrew every diagram I got wrong
- ☐ I listed my three weakest chapters



My score: ____ / 5

5/5 = Board ready! Weak chapters ek baar aur.

LEVEL 60 COMPLETE

+200 XP earned. BOOK COMPLETE. Book total: 8000 XP



BUJHO *60 din, 60 levels, 8000 XP! Chemistry, Physics, Biology: sab clear!*

PAHELI *Ab bas revision aur sample papers. Boards mein dikhana hai jalwa!*



NITESH SIR SAYS
*Book khatam, safar nahi. Har hafte ek full paper, time lagake. Diagram saaf, units sahi, steps poore. **All the best, champions!***

BIOLOGY | CHAPTER 13 COMPLETE | BOOK COMPLETE

CHAPTER 13 SWAHAA!

Our Environment | Days 58–60



BUJHO *60 din, 60 levels, 8000 XP! Humne poori Science book khatam kar di, Swahaa!*

PAHELI *Proud of you, Bujho! Ab bas revision aur boards mein full marks. All the best!*

CHAPTER XP EARNED
+500 XP

LEVELS CLEARED
3 levels (Days 58–60)

BOOK PROGRESS
Level 60 / 60 8000 XP total

WHAT YOU HAVE MASTERED

- ✔ Ecosystem and its components
- ✔ Ten percent law of energy flow
- ✔ Ozone layer depletion
- ✔ Food chains, food webs and trophic levels
- ✔ Biological magnification
- ✔ Biodegradable and non-biodegradable waste

BOOK STATUS BOOK COMPLETE: Level 60 / 60, 8000 XP

All 60 levels cleared!

EDUSPARK

BOOK COMPLETE

LEVEL 60 / 60 CLEARED

SCIENCE SWAHAA!

60 days, 13 chapters, 8000 XP. You did not just read this book, you **finished** it. Revise every topic you marked R or RA, re-attempt the Boss Levels, and walk into the board exam like a topper.



LEVELS CLEARED
60 / 60
Chemistry, Physics, Biology

TOTAL XP EARNED
8000 XP
2500 + 2500 + 3000

NEXT MISSION
BOARD EXAM
You are ready

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